

Eigenvalues of Tetrahedral sums of Hermitian matrices

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Abstract

Horn's problem is concerned with characterizing the eigenvalues (α, β, γ) of a triple of Hermitian matrices (A, B, C) satisfying the constraint $A + B = C$. The development of a complete solution to Horn's problem, due in large part to breakthroughs by A. Klyachko, A. Knutson, and T. Tao, has led to deep connections between eigenvalues of Hermitian matrices, tensor product invariants of matrix groups & Littlewood-Richardson coefficients, geometric invariant theory and the intersection theory of Schubert varieties via cohomology.

Here, we define and investigate a generalization of Horn's problem which we refer to as the *tetrahedral Horn's problem*. The tetrahedral Horn's problem is concerned with characterizing the eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of a sextuple of Hermitian matrices (A, B, C, D, E, F) satisfying the constraints $A + B = C$, $B + D = F$, $D + C = E$ and $A + F = E$. In this paper, we describe the scaling, shifting and tetrahedral symmetries of the tetrahedral Horn's problem, as well as conjecture the existence of additional symmetries generalizing the Regge symmetries present in the case of 2×2 Hermitian matrices. Furthermore, unlike the solution to Horn's problem, we show that the space of admissible sextuples of eigenvalues is neither convex nor a polytope. We demonstrate this qualitative difference by solving the tetrahedral Horn's problem for 2×2 matrices explicitly.

For the general case of $n \times n$ Hermitian matrices, we collect a handful of previously known matrix inequalities for sums of positive semidefinite Hermitian matrices and reinterpret them as necessary conditions for the tetrahedral Horn's problem. For a complete characterization, we provide a direct proof, using spectrum estimation techniques from quantum information theory, that the admissibility of a given sextuple of eigenvalues is encoded in the semiclassical asymptotics of the norm of the generalized Wigner $6j$ operator for the unitary group $U(n)$. Our result generalizes the well-known correspondence between admissible eigenvalues for Horn's problem and Littlewood-Richardson coefficients, and partially generalizes the well-known semiclassical analysis of J. Roberts for the case of $n = 2$.

Moreover, we provide a new elementary proof of an upper bound on the Hilbert-Schmidt norms of $6j$ operators using the Cauchy-Schwarz inequality. Combining this upper bound with estimates coming from the large $n \rightarrow \infty$ asymptotic limit, we obtain a new proof of a previously known submodular inequality involving unnormalized entropies of sums of positive semidefinite matrices.

Finally, we show how it is possible to embed positive semidefinite Hermitian matrices and their sums into multipartite density matrices such that further generalizations of the tetrahedral Horn's problem can be viewed as a special case of the spectral quantum marginals problem. From this perspective, we obtain an alternative new proof of the aforementioned submodular unnormalized entropy inequality. Motivated by this connection to multipartite quantum states, we consider the set of possible allocations of unnormalized entropies of sums of positive semidefinite matrices and prove it forms a convex cone. This convex cone, which we call the additive quantum entropy cone, serves as an analogue of Pippenger's quantum entropy cone; indeed we prove that every balanced inequality that is valid for the quantum entropy cone is also valid for additive quantum entropy cone, thus extending an open conjecture of J. Cadney, N. Linden, and A. Winter.

TC: This is the main document for the project "TETRASPECTRA".
Previous drafts are located in the folder titled "old" and will not necessarily compile properly.

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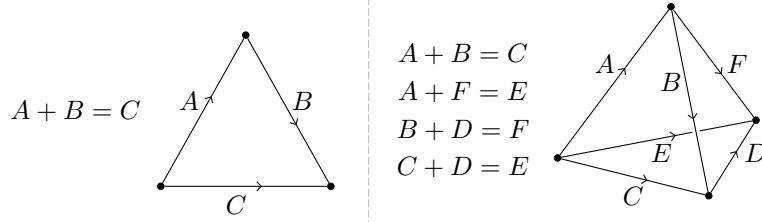


Figure 1: (Left) The *triangular* Horn's problem subjects three Hermitian matrices, (A, B, C) , to the constraint $A + B = C$ forming a triangle in the vector space of Hermitian matrices. (Right) The *tetrahedral* Horn's problem subjects six Hermitian matrices, (A, B, C, D, E, F) , to a quadruple of constraints forming a tetrahedron in the vector space of Hermitian matrices. fig:tetrahedron_labelling_scheme

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1 Introduction

There are a number of operations involving the transformation of Hermitian matrices under which it is possible to track the effect had of the corresponding sets of eigenvalues. For example, the eigenvalues, $\text{eig}(X)$, of a Hermitian matrix, X , are invariant under conjugation by unitary matrices, $\text{eig}(X) = \text{eig}(UXU^\dagger)$, covariant under taking positive powers $\text{eig}(X^p) = \text{eig}(X)^p$, covariant under multiplication by real numbers, $\text{eig}(rX) = r \text{eig}(X)$, and covariant under the addition of scalar matrices, $\text{eig}(X + rI) = \text{eig}(X) + (r, \dots, r)$. The operation of adding together Hermitian matrices is not an operation of this type. Specifically, the eigenvalues $\gamma = \text{eig}(A + B)$ of the sum of two matrices cannot be determined from the eigenvalues $\alpha = \text{eig}(A)$ and $\beta = \text{eig}(B)$ of the summands *alone*. Stated differently, to determine the eigenvalues involved, additional knowledge about the eigenvectors involved is typically required. Nevertheless, there are numerous mathematical situations wherein knowledge about the eigenvectors involved is lacking, yet it remains desirable to determine the set of admissible eigenvalues of a sum given the eigenvalues of the summands.

The problem of determining the set (α, β, γ) of admissible eigenvalues has been investigated for over a century yet has come to be known as *Horn's problem* after Horn who, in 1962, conjectured a solution to the problem in the form of an inductively defined, finite set of homogeneous linear inequalities. These inequalities, now known as the *Horn inequalities* describe a polytope of admissible values for (α, β, γ) [Hor62]. In 1998, Klyachko provided a solution to Horn's problem [Kly98] which, when combined with a proof of the so-called saturation conjecture, due to Knutson and Tao [KT99], constitutes a proof of Horn's original conjecture [Ful98].

For applications of Horn's problem to quantum information theory and Nielsen's theorem [Nie99], see the article due to Daftuar and Hayden [DH05].

In parallel to Horn's problem, numerous variations and extensions of Horn's problem have been explored by various authors. For example, numerous authors have considered the admissible eigenvalues of unitary matrices and their product [AW97; Bel08; BK16], or the admissible singular values of matrices and their product [Spe05; TT71]. In 2000, it was shown by [Kly00] that *logarithms* of the admissible singular values of non-singular matrices and their product are identical to the admissible eigenvalues of Hermitian matrices and their sum, thus proving a conjecture due to Thompson [Kly00]. This correspondence between the singular values of a product and the eigenvalues of sum has been generalized substantially by Alekseev, Meinrenken, and Woodward [AMW01]. For a pedagogical and historical introduction to Horn's problem, we recommend the review articles of Bhatia [Bha01], Knutson [Knu00], and Fulton [Ful00].

Recent developments surrounding Horn's problem not covered by the aforementioned reviews include the development of the geometric method by Sherman [She15] and Berline, Vergne, and Walter [BVW18] originally pioneered by the geometric proofs due to Belkale [Bel02; Bel05], as well the characterization and computation of the induced distribution over the eigenvalues of the sum given random matrices with fixed spectra in a series of papers by Coquereaux and Zuber [CZ18; CZ18; CMZ19; Zub18].

In this paper, we consider a generalization of Horn’s problem previously considered only briefly in a paper due to Christandl, Şahinoğlu, and Walter [CŞW18]. While Horn’s problem aims to characterize the eigenvalues (α, β, γ) of a triple of $n \times n$ Hermitian matrices (A, B, C) which are subjected to the constraint $A + B = C$, the generalization consider here aims to characterize the eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of a *sextuple* of $n \times n$ matrices (A, B, C, D, E, F) subject to four constraints: $A + B = C$, $B + D = F$, $D + C = E$ and $A + F = E$. These constraints arise naturally from building a kind of tetrahedral arrangement of Hermitian matrices inside the vector space of Hermitian matrices according to Fig. 1. Consequently, throughout the paper, we refer to this generalization of Horn’s problem as the *tetrahedral Horn’s problem* and, by contrast, Horn’s original problem, involving the sum of just two Hermitian matrices, as the *triangular Horn’s problem*.

Beyond the superficial connection between the imposed constraints and the resulting shape formed in the vector space of Hermitian matrices (as in Fig. 1), in the special case of 2×2 Hermitian matrices, this connection is much deeper. It is well-known that the space of admissible eigenvalues for a triangular arrangement of 2×2 traceless¹ Hermitian matrices is in correspondence with the moduli space of Euclidean *triangles* in $\mathbb{R}^3$ [HK96; FKL10; Knu00]. As we shall see in Example 5, this correspondence generalizes in the sense that the admissible eigenvalues for a tetrahedral arrangements of 2×2 traceless Hermitian matrices is in correspondence with the moduli space of Euclidean *tetrahedra* in $\mathbb{R}^3$. While this latter result is straightforward to prove and also well-known, indeed it forms the basis of the semiclassical asymptotics of the $SU(2)$ Wigner $6j$ symbols [Rob99], it leads to a complete solution to the tetrahedral Horn’s problem for 2×2 matrices. In Example 5, we will see that the moduli space of Euclidean tetrahedra in $\mathbb{R}^3$ is provided by the positivity of so-called Cayley-Menger determinant [SJS18].

Although we are ultimately interested in tackling the tetrahedral Horn’s problem for $n \times n$ Hermitian matrices with $n > 2$, the $n = 2$ case already demonstrates a strong qualitative difference between the triangular and tetrahedral variants of Horn’s problem. In particular, while the triangular Horn’s problem is completely characterized by a *finite* number of *linear* inequalities in the eigenvalues (indeed the solution space is an instance of a moment polytope known as *Horn’s polytope*), the solution space for the tetrahedral Horn’s problem is in general *not a polytope*.

TODO: Finish the introduction

- Talk about connections between Horn’s problem & Schubert calculus
- talk about semiclassical asymptotics of $6j$ symbols
- talk about entropy cones and open problems regarding entropy cones
- talk about known submodular inequalities for PSD matrices
- describe the scope & results of the paper (more detailed abstract)
- describe structure of the paper

¹One can always assume, without loss of generality, that the Hermitian matrices involved are traceless due to a simple shifting symmetry which is described in Section 2.3.

2 Tetrahedra of Hermitian matrices

2.1 The tetrahedral Horn problem

`prob:tri_problem`

Problem 1 (Trigonal Horn problem). A triple $(\alpha, \beta, \gamma) \in (\mathbb{R}_{\text{dec}}^n)^{\times 3}$ of candidate eigenvalues are said to be *trigonal* if there exists a corresponding triple (A, B, C) of $d \times d$ Hermitian matrices satisfying (i) the linear constraint

`{eq:tri_linear_constraints}`

$$A + B = C, \quad (1)$$

and (ii) the three eigenvalue constraints

`{eq:tri_eigenvalue_constraints}`

$$\text{eig}(A) = \alpha, \quad \text{eig}(B) = \beta, \quad \text{eig}(C) = \gamma. \quad (2)$$

The problem of determining the set of trigonal eigenvalues (α, β, γ) will be referred to as the *trigonal Horn problem*.

`prob:tet_problem`

Problem 2 (Tetrahedral Horn problem). A sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \in (\mathbb{R}_{\text{dec}}^n)^{\times 6}$ of candidate eigenvalues are said to be *tetrahedral* if there exists a corresponding sextuple (A, B, C, D, E, F) of $n \times n$ Hermitian matrices satisfying (i) the four linear constraints

`{eq:tet_linear_constraints}`

$$A + B = C, \quad B + D = F, \quad C + D = E, \quad A + F = E, \quad (3)$$

and (ii) the six eigenvalue constraints

`{eq:tet_eigenvalue_constraints}`

$$\text{eig}(A) = \alpha, \quad \text{eig}(B) = \beta, \quad \text{eig}(C) = \gamma, \quad \text{eig}(D) = \delta, \quad \text{eig}(E) = \epsilon, \quad \text{eig}(F) = \phi. \quad (4)$$

The problem of determining the set of tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ will be referred to as the *tetrahedral Horn problem*.

`exam:one_dim_soln`

Example 3 (Tetrahedral Horn's problem for $n = 1$). Perhaps the simplest example of the tetrahedral Horn's problem occurs when $n = 1$; i.e., when Hermitian matrices and their eigenvalues are identified with real numbers. In this case, the tetrahedral Horn's problem reduces to the problem of which sextuples of real numbers (a, b, c, d, e, f) simultaneously satisfy

`{eq:one_dim_tet_conds}`

$$a + b = c, \quad b + d = f, \quad d + c = e, \quad a + f = e, \quad (5)$$

or equivalently, are such that $(a, b, c, d, e, f) \in \ker \Gamma$ where Γ is the 4×6 matrix of coefficients encoding the linear system from Eq. (5):

`{eq:tet_coeff_matrix}`

$$\Gamma := \begin{pmatrix} 1 & 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 & -1 & 0 \\ 1 & 0 & 0 & 0 & -1 & 1 \end{pmatrix}. \quad (6)$$

It can be shown that the above coefficient matrix has a three dimensional kernel and, in particular, one can parameterize the solution space by $(x, y, z) \in \mathbb{R}^3$ according to the formula

`{eq:one_dim_soln}`

$$(a, b, c, d, e, f) = (x, y, x + y, z, x + y + z, y + z). \quad (7)$$

Remark 4 (Traces are tetrahedral). In the original, trigonal version of Horn's problem, the eigenvalues (α, β, γ) of Hermitian matrices (A, B, C) satisfying $A + B = C$ necessarily must satisfy a basic equality condition known as the *trace condition*:

`{eq:tri_trace_cond}`

$$|\alpha| + |\beta| = |\gamma|, \quad (8)$$

where $|\lambda| := \sum_i \lambda_i$ is shorthand for the sum of the eigenvalues $\lambda = (\lambda_1 \geq \dots \geq \lambda_n)$. This condition must hold because $A + B = C$ implies $\text{Tr}(C) = \text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$ and $\text{Tr}(X) = |\lambda|$ for any Hermitian matrix X with eigenvalues λ . For the tetrahedral Horn's problem, a set of tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ necessarily satisfy four *trace conditions*:

`{eq:tet_trace_conds}`

$$|\alpha| + |\beta| = |\gamma|, \quad |\beta| + |\delta| = |\phi|, \quad |\delta| + |\gamma| = |\epsilon|, \quad |\alpha| + |\phi| = |\epsilon|. \quad (9)$$

Comparing the tetrahedral trace conditions (Eq. (9)) with the $n = 1$ case of the tetrahedral Horn's problem (Eq. (3)), we see that a sextuple of eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ can only be tetrahedral if their traces, $(|\alpha|, |\beta|, |\gamma|, |\delta|, |\epsilon|, |\phi|)$, are tetrahedral.

Example 5 (Tetrahedral Horn's problem for $d = 2$). Here we articulate the solution to the tetrahedral Horn's problem for 2×2 Hermitian matrices. According to Corollary 17, one can restrict their focus to the case of *traceless* 2×2 Hermitian matrices without loss of generality. In this context, the sextuple of eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ can be identified with a sextuple of non-negative real numbers $(a, b, c, d, e, f) \in \mathbb{R}_{\geq 0}^6$ such that

$$\alpha = (a, -a), \quad \beta = (b, -b), \quad \gamma = (c, -c), \quad \delta = (d, -d), \quad \epsilon = (e, -e), \quad \phi = (f, -f). \quad (10)$$

Moreover, each traceless 2×2 Hermitian matrix, X , may be identified with a three-dimensional vector $(x_1, x_2, x_3) \in \mathbb{R}^3$ using the standard Pauli-matrix basis $\{\sigma_1, \sigma_2, \sigma_3\}$ according to the formula:

$$X = x_1\sigma_1 + x_2\sigma_2 + x_3\sigma_3 = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}. \quad (11)$$

With respect to this parameterization, one sees that the eigenvalues of X are determined by the Euclidean length of the vector (x_1, x_2, x_3) :

$$\text{eig}(X) = \text{eig} \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix} = (\|x\|, -\|x\|), \quad \text{where,} \quad \|x\| = \sqrt{x_1^2 + x_2^2 + x_3^2}. \quad (12)$$

Furthermore, if A is determined by the coordinates (a_1, a_2, a_3) and B by the coordinates (b_1, b_2, b_3) , then their sum $C = A + B$ is determined by the coordinates $(c_1, c_2, c_3) = (a_1 + b_1, a_2 + b_2, a_3 + b_3)$. Consequently, in this setting, the tetrahedral Horn's problem reduces to problem of describing the space of all possible edge-lengths (a, b, c, d, e, f) of a Euclidean tetrahedron in $\mathbb{R}^3$. This problem admits of a well-studied solution [Ric02; WD09], namely, the conjunction of a quadruplet of triangle-inequalities

$$|a - b| \leq c \leq a + b, \quad |b - d| \leq f \leq b + d, \quad |d - c| \leq e \leq d + c, \quad |a - f| \leq e \leq a + f, \quad (13)$$

(which ensure the possibility of forming a Euclidean triangle for each of the four faces), together with a single non-linear inequality of the form

$$D(a, b, c, d, e, f) = \frac{1}{288} \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & a^2 & e^2 & c^2 \\ 1 & a^2 & 0 & f^2 & b^2 \\ 1 & e^2 & f^2 & 0 & d^2 \\ 1 & c^2 & b^2 & d^2 & 0 \end{pmatrix} \geq 0. \quad (14)$$

The quantity $D(a, b, c, d, e, f)$ is known as a Cayley-Menger determinant and, when $D(a, b, c, d, e, f)$ is positive and when the triangle inequalities above are also satisfied, $D(a, b, c, d, e, f)$ is always equal to *square* of the volume of the corresponding tetrahedron [SJS18]. Figure 2 depicts a three-dimensional projection of the solution space described by Eq. (13) and Eq. (14).

The tetrahedral Horn problem can be viewed as a generalization of the trigonal Horn problem in a variety of ways. First and foremost, the trigonal Horn problem is a special case of the tetrahedral Horn problem.

Proposition 6. *A triplet (α, β, γ) of eigenvalues is trigonal if and only if the sextuple $(\alpha, \beta, \gamma, 0, \gamma, \beta)$ is tetrahedral.*

Proof. If (α, β, γ) are the eigenvalues of (A, B, C) satisfying $A+B = C$, then one can see that $(A, B, C, 0, C, B)$ satisfy Eq. (3) and have eigenvalues $(\alpha, \beta, \gamma, 0, \gamma, \beta)$. Conversely, if $(\alpha, \beta, \gamma, 0, \gamma, \beta)$ are tetrahedral then it is straightforward to see that (α, β, γ) are trigonal. $\square$

More generally, it is possible to view the tetrahedral Horn problem as the conjunction of four separate instances of the trigonal Horn problem subject to additional matching constraints.

Lemma 7. *A sextuple of eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ is tetrahedral only if each of*

$$(\alpha, \beta, \gamma), \quad (\beta, \delta, \phi), \quad (\gamma, \delta, \epsilon), \quad (\alpha', \phi', \epsilon'), \quad (15)$$

are trigonal.

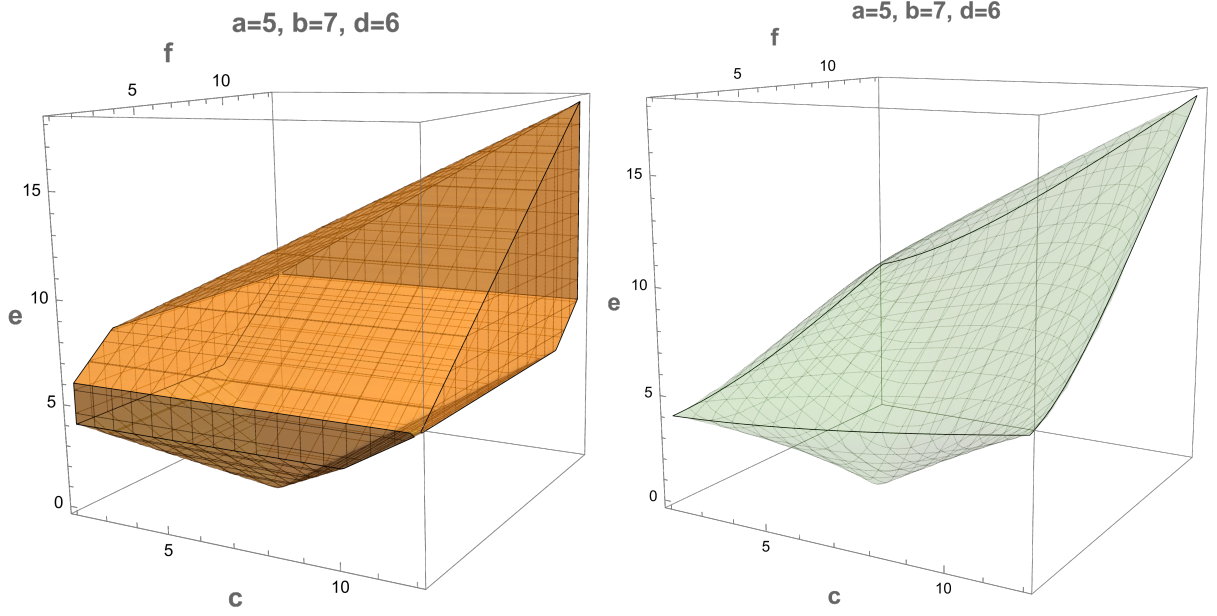


Figure 2: A slice of the set of admissible side-lengths (a, b, c, d, e, f) for Euclidean tetrahedron along the coordinates $(a, b, d) = (5, 7, 6)$ (Left) The polyhedron of values (c, e, f) satisfying the triangle inequalities for each face Eq. (13). (Right) The set of all values (c, e, f) satisfying both Eq. (13) and Eq. (14).

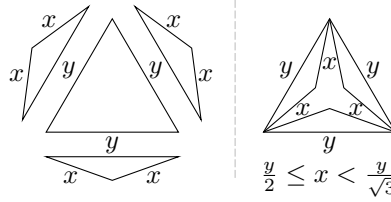


fig:frustrated-tet

Figure 3: A example of a sextuple of edge lengths, $(a, b, c, d, e, f) = (y, y, y, x, x, x)$, which satisfies all triangle inequality constraints whenever $y \leq 2x$, but cannot be the edge lengths of any Euclidean tetrahedron when $\sqrt{3}x < y$.

Proof. The claim follows from a relaxation of the conditions defining the tetrahedral Horn problem. In particular, there exists a sextuple (A, B, C, D, E, F) of Hermitian matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ satisfying Eq. (3) *only if* there exists twelve Hermitian matrices

$$(A, A', B, B', C, C', D, D', E, E', F, F') \quad \text{\texttt{eq:twelve_fold}} \quad (16)$$

with eigenvalues $(\alpha, \alpha, \beta, \beta, \gamma, \gamma, \delta, \delta, \epsilon, \epsilon, \phi, \phi)$ satisfying

$$A + B = C, \quad B' + D = F, \quad C' + D' = E, \quad A' + F' = E'. \quad (17)$$

Now notice that each of the four linear constraints above depend on a disjoint subset of the twelve matrices in Eq. (16) and therefore the respective eigenvalues are independently constrained. In other words, this relaxation of the tetrahedral Horn problem is equivalent to the statement that $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral eigenvalues *only if* each of four triplets in Eq. (15) are trigonal. $\square$

2.2 Eigenvalues of blocks of positive semidefinite matrices

\texttt{sec:blocks_of_psd_matrices}

The purpose of this section is to reformulate both the trigonal and tetrahedral Horn's problem as a instance of a slightly more general kind of eigenvalue problem involving positive semidefinite matrices. Throughout this section, we will restrict our focus to the case of positive semidefinite Hermitian matrices and their eigenvalues. As a reminder we note that this can be done without loss of generality due to Corollary 18.

Given a positive semidefinite Hermitian matrix, X , with eigenvalues $\text{eig}(X) = \chi = (\chi_1 \geq \dots \geq \chi_n \geq 0)$ and eigendecomposition $X = \sum_{i=1}^n \chi_i \Pi_i$, one can define $X^{\frac{1}{2}}$ by $X^{\frac{1}{2}} := \sum_{i=1}^n \chi_i^{\frac{1}{2}} \Pi_i$. Inspired by [Bha97, Theorem IV.2.13], we obtain the following fundamental lemma relating the eigenvalues γ of a sum $C = A + B$ of $n \times n$ Hermitian matrices to the eigenvalues of a larger $2n \times 2n$ matrix M built from A and B .

\texttt{lem:trigonal_block}

Lemma 8. *Let (A, B, C) be a triple of $n \times n$ Hermitian matrices with eigenvalues (α, β, γ) such that $C = A + B$. Let $\Sigma_{A,B}$ be the $2n \times 2n$ Hermitian matrix*

$$\Sigma_{A,B} = \begin{bmatrix} A & A^{\frac{1}{2}} B^{\frac{1}{2}} \\ B^{\frac{1}{2}} A^{\frac{1}{2}} & B \end{bmatrix}. \quad (18)$$

Then $\Sigma_{A,B} \geq 0$ is positive semidefinite, has rank at most n and has eigenvalues given by γ

$$\text{eig}(\Sigma_{A,B}) \simeq \text{eig}(A + B) = \text{eig}(C) = \gamma. \quad (19)$$

Proof. That $\Sigma_{A,B}$ is positive semidefinite and has rank at most n follows because $\Sigma_{A,B}$ can be decomposed as $\Sigma_{A,B} = T^* T$ where T is the $n \times 2n$ block matrix

$$T = \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} \end{bmatrix}. \quad (20)$$

That $\Sigma_{A,B}$ and $A + B$ share the same eigenvalues (up to padding by n zeros) follows because for any rectangular matrix T , the eigenvalues of $T^* T$ and $T T^*$ are equal (up to padding by zeros) and in this instance

$$T^* T = \begin{bmatrix} A^{\frac{1}{2}} \\ B^{\frac{1}{2}} \end{bmatrix} \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} \end{bmatrix} = \begin{bmatrix} A & A^{\frac{1}{2}} B^{\frac{1}{2}} \\ B^{\frac{1}{2}} A^{\frac{1}{2}} & B \end{bmatrix} = \Sigma_{A,B} \quad T T^* = \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} \end{bmatrix} \begin{bmatrix} A^{\frac{1}{2}} \\ B^{\frac{1}{2}} \end{bmatrix} = A + B. \quad (21)$$

$\square$

From the perspective of Lemma 8, one can view the trigonal Horn problem involving the eigenvalues (α, β, γ) of $n \times n$ Hermitian matrices (A, B, C) satisfying $A + B = C$ as a special case of a more general eigenvalue problem concerned with the relations between the eigenvalues of $2n \times 2n$ Hermitian matrices Σ of the form

$$\Sigma = \begin{bmatrix} A & X \\ X^* & B \end{bmatrix}, \quad \text{\texttt{eq:block_ABX}} \quad (22)$$

and the eigenvalues of its two $n \times n$ diagonal blocks A and B . It is known, by [HJ12, Theorem 7.7.9] or more generally [MKX19, Theorem 5.11] that if Σ is a 2×2 block matrix such that $A, B \geq 0$, then $\Sigma \geq 0$ if and only if the off-diagonal block X is of the form

$$X = A^{\frac{1}{2}} K B^{\frac{1}{2}} \quad (23)$$

for some contraction matrix K (largest eigenvalue of $(K^* K)^{\frac{1}{2}}$ is at most unity). Additionally, when $X = A^{\frac{1}{2}} K B^{\frac{1}{2}}$, one has the decomposition

$$\Sigma = \begin{bmatrix} A & X \\ X^* & B \end{bmatrix} = \begin{bmatrix} A^{\frac{1}{2}} & 0 \\ 0 & B^{\frac{1}{2}} \end{bmatrix} \begin{bmatrix} I & K \\ K^* & I \end{bmatrix} \begin{bmatrix} A^{\frac{1}{2}} & 0 \\ 0 & B^{\frac{1}{2}} \end{bmatrix}, \quad (24)$$

noting that K is a contraction if and only if the middle block matrix above is positive semidefinite [MKX19, Lemma 5.10].

Therefore, the special case covered by Lemma 8 corresponds to the case where the corresponding contraction matrix K is the $n \times n$ identity matrix $K = I$ (or more generally any contractive matrix satisfying $A^{\frac{1}{2}} K B^{\frac{1}{2}} = A^{\frac{1}{2}} B^{\frac{1}{2}}$). A simple observation which distinguishes between eigenvalue problem concerning diagonal blocks and the eigenvalue problem concerning sums is simply that the rank of Σ can in general be up to $2n$ whereas the rank of the sum $C = A + B$ is at most n . Despite this additional freedom, there remains non-trivial relations constraining the eigenvalues of Σ and $A + B$ in terms of properties of X (see [Gum+18; BL22]). Nevertheless, when the rank of Σ is constrained to be at most n , we obtain the following theorem which provides an equivalent formulation of the trigonal Horn problem. thm:trigonal_block

Theorem 9. *Let (α, β, γ) be a triple of positive semidefinite eigenvalues of length at most n . The following are equivalent:*

- (i) *there exists a triple of Hermitian matrices (A, B, C) with eigenvalues (α, β, γ) satisfying $A + B = C$,*
- (ii) *there exists an $2n \times 2n$ Hermitian matrix,*

$$\Sigma = \begin{bmatrix} A' & X \\ X^* & B' \end{bmatrix}, \quad (25)$$

such that $\text{eig}(A') = \alpha$, $\text{eig}(B') = \beta$, and $\text{eig}(\Sigma) = \gamma$.

Proof. Since we explicitly assume all of the eigenvalues in the triple (α, β, γ) are non-negative, the Hermitian matrices A, B, C , and Σ, A', B' involved in the statements above are necessarily positive semidefinite. The claim (i) $\implies$ (ii) follows from Lemma 8 by setting $(\Sigma, A', B') = (\Sigma_{A,B}, A, B)$. That (ii) $\implies$ (i) follows by noticing that if Σ has eigenvalues γ , then it has rank at most n . Therefore, as $\Sigma \geq 0$, there exists a $n \times 2n$ block matrix T of the form

$$T = \begin{bmatrix} T_A & T_B \end{bmatrix} \quad (26)$$

such that $\Sigma = T^* T$. The diagonal blocks of Σ are therefore $A' = T_A^* T_A$, and $B' = T_B^* T_B$. As in the proof of Lemma 8, we note that $T^* T$ and $T T^*$ have the same eigenvalues up to padding by zeros. Therefore, by letting $A = T_A T_A^*$ and $B = T_B T_B^*$ we see that

$$\begin{aligned} \text{eig}(A) &= \text{eig}(T_A T_A^*) \simeq \text{eig}(T_A^* T_A) = \text{eig}(A') = \alpha, \\ \text{eig}(B) &= \text{eig}(T_B T_B^*) \simeq \text{eig}(T_B^* T_B) = \text{eig}(B') = \beta, \end{aligned} \quad (27)$$

while the eigenvalues of $C = A + B = T_A T_A^* + T_B T_B^*$ match the eigenvalues of γ because

$$\text{eig}(C) = \text{eig}(A + B) = \text{eig}(T_A T_A^* + T_B T_B^*) = \text{eig}(T T^*) \simeq \text{eig}(T^* T) = \text{eig}(\Sigma) = \gamma. \quad (28)$$

□

As a corollary of this correspondence between the trigonal Horn problem and the eigenvalues of blocks of 2×2 matrices, we obtain an equivalent formulation of the tetrahedral Horn problem.

Corollary 10. Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of positive semidefinite eigenvalues of length at most n . Then $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral if and only if there exists a $3n \times 3n$ matrix Σ of the form

$$\Sigma = \begin{bmatrix} A' & Z & Y \\ Z^* & B' & X \\ Y^* & X^* & D' \end{bmatrix}, \quad \text{\texttt{eq:block_3by3}} \quad (29)$$

such that

$$\text{eig}(A') = \alpha, \quad \text{eig}(B') = \beta, \quad \text{eig}\begin{pmatrix} A' & Z \\ Z^* & B' \end{pmatrix} = \gamma, \quad \text{eig}(D') = \delta, \quad \text{eig}(\Sigma) = \epsilon, \quad \text{eig}\begin{pmatrix} B' & X \\ X^* & D' \end{pmatrix} = \phi. \quad \text{\texttt{eq:block_eigenvalue_tet}} \quad (30)$$

Proof. The proof is analogous to the proofs of Lemma 8 and Theorem 9. First we prove the “only if” portion of the claim which is a corollary of Lemma 8. Suppose (A, B, C, D, E, F) is a tetrahedral sextuple of Hermitian matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \gamma)$. Let $\Sigma = \Sigma_{A,B,D}$ be given by

$$\Sigma_{A,B,D} = \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} & D^{\frac{1}{2}} \end{bmatrix}^* \begin{bmatrix} A^{\frac{1}{2}} & B^{\frac{1}{2}} & D^{\frac{1}{2}} \end{bmatrix} = \begin{bmatrix} A & A^{\frac{1}{2}}B^{\frac{1}{2}} & A^{\frac{1}{2}}D^{\frac{1}{2}} \\ B^{\frac{1}{2}}A^{\frac{1}{2}} & B & B^{\frac{1}{2}}D^{\frac{1}{2}} \\ D^{\frac{1}{2}}A^{\frac{1}{2}} & D^{\frac{1}{2}}B^{\frac{1}{2}} & D \end{bmatrix} \quad (31)$$

Then the eigenvalues of the corresponding principal submatrices of Σ satisfy Eq. (30) by Lemma 8. Now we prove the “if” portion of the claim which is a corollary of Theorem 9. If Σ is a matrix of form given by Eq. (29) with eigenvalues ϵ with length at most n , then Σ is positive semidefinite with rank at most n . Therefore, there exists an $n \times 3n$ block matrix

$$T = \begin{bmatrix} T_A & T_B & T_D \end{bmatrix} \quad (32)$$

such that $\Sigma = T^*T$ and thus

$$\begin{aligned} A' &= T_A^*T_A, & B' &= T_B^*T_B, & D' &= T_D^*T_D, \\ X &= T_B^*T_D, & Y &= T_A^*T_D, & Z &= T_A^*T_B. \end{aligned} \quad (33)$$

Now let (A, B, C, D, E, F) be given by

$$\begin{aligned} A &= T_A T_A^*, & B &= T_B T_B^*, & C &= T_A T_A^* + T_B T_B^*, \\ D &= T_D T_D^*, & E &= T_A T_A^* + T_B T_B^* + T_D T_D^*, & F &= T_B T_B^* + T_D T_D^*. \end{aligned} \quad (34)$$

From here, one may directly verify that these $n \times n$ Hermitian matrices satisfy Eq. (3) and have eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ (using a similar argument presented in the proof of Theorem 9); i.e., one concludes that $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral. $\square$

Remark 11. The correspondence between the tetrahedral Horn problem and the eigenvalues of principal submatrices of 3×3 matrices provided by Corollary 10 enables us to import a myriad of known results concerning the eigenvalues of principal submatrices. Specifically, the results due to Audenaert, Hiai, and Petz [AHP10], Lami et al. [Lam+17], Niculescu [Nic25], Paksoy, Turkmen, and Zhang [PTZ14], Lewin and Sabin [LS14].

Many of the results concerning the eigenvalues of principal submatrices are born out of the correspondence between PSD matrices and covariance matrices of Gaussian multivariate random variables [Lam+17]. From this perspective, the Gaussian entropy region has been partially characterized by Shadbakht and Hassibi [SH11]. From this perspective, one may view Horn problems are kinds of “spectral” classical marginal problems for blocks of Gaussian random variables.

2.3 Symmetries of tetrahedral eigenvalues

There are numerous types of symmetries present in the statement of Problem 2. The purpose of this section is to describe the scaling, inversion, tetrahedral, and shifting symmetries of the problem. These symmetries can, in many cases, be used to simplify the problem by selecting a canonical representative of the corresponding equivalence class; in other cases, it will be useful to leave Problem 2 as stated. Later we will comment on the possibility of additional hidden symmetries not previously mentioned.

Proposition 12 (Scaling symmetry). *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of candidate eigenvalues and let $s \in (0, \infty)$ be a positive real number. Then the scaled eigenvalues, $(s\alpha, s\beta, s\gamma, s\delta, s\epsilon, s\phi)$, are tetrahedral if and only if the original, unscaled eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral.*

Proof. The statement follows from the fact that $\text{eig}(sX) = s \text{eig}(X)$ holds for all Hermitian matrices X and positive scalars s and that each of the constraints in Eq. (3) are homogeneous in the six Hermitian matrices (A, B, C, D, E, F) ; for example, we have $sA + sB - sC = 0$ if and only if $s(A + B - C) = 0$ if and only if $A + B - C = 0$. $\square$

One may wonder with or not it is possible to extend the the scaling symmetry to consider scaling by negative $s < 0$. Indeed, this can be accomplished, provided one accounts for the fact that negating a Hermitian matrices both negates and reverses the order of its eigenvalues. prop:inversion

Proposition 13 (Inversion symmetry). *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of candidate eigenvalues. Given any list of non-increasing eigenvalues $\lambda = (\lambda_1 \geq \dots \geq \lambda_n)$, let λ^* denote the inverted list of resorted and negated eigenvalues: $\lambda^* = (-\lambda_n \geq \dots \geq -\lambda_1)$. Then $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral if and only if $(\alpha^*, \beta^*, \gamma^*, \delta^*, \epsilon^*, \phi^*)$ are tetrahedral.*

Proof. The statement holds because (i) (A, B, C, D, E, F) satisfy Eq. (3) if and only if $(A', B', C', D', E', F') := (-A, -B, -C, -D, -E, -F)$ satisfy Eq. (3), and (ii) $\text{eig}(X) = \lambda$ if and only if $\text{eig}(-X) = \lambda^*$. $\square$

The inversion symmetry Proposition 13 is just one element belonging to a group of 48 symmetries consisting of inversions of eigenvalues, $\lambda \rightarrow \lambda^*$, or permutations of elements in the sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$. prop:tet_symmetry

Proposition 14 (Oriented Tetrahedral symmetry). *The sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of eigenvalues are tetrahedral if and only if any (and thus all) of the following sextuples of eigenvalues are tetrahedral:*

$$(\alpha \ \phi \ \epsilon \ \delta^* \ \gamma \ \beta) \quad (\gamma \ \epsilon^* \ \delta^* \ \phi \ \beta \ \alpha^*) \quad \text{eq:gens_of_tet_sym} \quad (35)$$

Moreover, these two operations generate a finite group of order 48 whose action, grouped by conjugacy classes, produces:

$$\begin{array}{cccc} (\alpha \ \beta \ \gamma \ \delta \ \epsilon \ \phi) & (\alpha^* \ \gamma \ \beta \ \delta \ \phi \ \epsilon) & (\alpha \ \gamma^* \ \beta^* \ \delta^* \ \phi^* \ \epsilon^*) & (\alpha^* \ \beta^* \ \gamma^* \ \delta^* \ \epsilon^* \ \phi^*) \\ (\delta \ \beta \ \phi \ \alpha \ \epsilon \ \gamma) & (\beta^* \ \alpha^* \ \gamma^* \ \epsilon \ \delta \ \phi) & (\beta \ \alpha \ \gamma \ \epsilon^* \ \delta^* \ \phi^*) & (\delta^* \ \beta^* \ \phi^* \ \alpha^* \ \epsilon^* \ \gamma^*) \\ (\delta^* \ \epsilon \ \gamma \ \alpha^* \ \beta \ \phi) & (\phi^* \ \beta \ \delta^* \ \gamma^* \ \epsilon^* \ \alpha^*) & (\phi \ \beta^* \ \delta \ \gamma \ \epsilon \ \alpha) & (\delta \ \epsilon^* \ \gamma^* \ \alpha \ \beta^* \ \phi^*) \\ (\alpha \ \epsilon^* \ \phi^* \ \delta \ \beta^* \ \gamma^*) & (\epsilon \ \delta^* \ \gamma \ \beta^* \ \alpha \ \phi^*) & (\epsilon^* \ \delta \ \gamma^* \ \beta \ \alpha^* \ \phi) & (\alpha^* \ \epsilon \ \phi \ \delta^* \ \beta \ \gamma) \\ (\beta \ \phi^* \ \delta^* \ \epsilon \ \gamma \ \alpha) & (\alpha \ \phi \ \epsilon \ \delta^* \ \gamma \ \beta) & (\alpha^* \ \phi^* \ \epsilon^* \ \delta \ \gamma^* \ \beta^*) & (\beta^* \ \phi \ \delta \ \epsilon^* \ \gamma \ \alpha^*) \\ (\gamma \ \alpha^* \ \beta \ \phi^* \ \delta^* \ \epsilon^*) & & & (\gamma^* \ \alpha \ \beta^* \ \phi \ \delta \ \epsilon) \\ (\epsilon \ \gamma^* \ \delta \ \beta \ \phi \ \alpha^*) & (\beta \ \delta \ \phi \ \epsilon^* \ \alpha^* \ \gamma^*) & (\beta^* \ \delta^* \ \phi^* \ \epsilon \ \alpha \ \gamma) & (\epsilon^* \ \gamma \ \delta^* \ \beta^* \ \phi^* \ \alpha) \\ (\phi \ \alpha \ \epsilon \ \gamma^* \ \delta \ \beta^*) & (\delta \ \phi^* \ \beta^* \ \alpha^* \ \gamma^* \ \epsilon^*) & (\delta^* \ \phi \ \beta \ \alpha \ \gamma \ \epsilon) & (\phi^* \ \alpha^* \ \epsilon^* \ \gamma \ \delta^* \ \beta) \\ (\phi^* \ \delta \ \beta^* \ \gamma \ \alpha \ \epsilon) & (\phi \ \epsilon^* \ \alpha^* \ \gamma \ \beta \ \delta^*) & (\phi^* \ \epsilon \ \alpha \ \gamma^* \ \beta^* \ \delta) & (\phi \ \delta^* \ \beta \ \gamma^* \ \alpha^* \ \epsilon^*) \\ (\epsilon^* \ \phi \ \alpha^* \ \beta^* \ \gamma^* \ \delta) & (\epsilon^* \ \alpha \ \phi^* \ \beta \ \delta^* \ \gamma) & (\epsilon \ \alpha^* \ \phi \ \beta^* \ \delta \ \gamma^*) & (\epsilon \ \phi^* \ \alpha \ \beta \ \gamma \ \delta^*) \\ (\gamma^* \ \delta^* \ \epsilon^* \ \phi \ \alpha^* \ \beta) & (\delta^* \ \gamma^* \ \epsilon^* \ \alpha \ \phi^* \ \beta^*) & (\delta \ \gamma \ \epsilon \ \alpha^* \ \phi \ \beta) & (\gamma \ \delta \ \epsilon \ \phi^* \ \alpha \ \beta^*) \\ (\beta^* \ \gamma \ \alpha \ \epsilon^* \ \phi^* \ \delta^*) & (\gamma^* \ \epsilon \ \delta \ \phi^* \ \beta^* \ \alpha) & (\gamma \ \epsilon^* \ \delta^* \ \phi \ \beta \ \alpha^*) & (\beta \ \gamma^* \ \alpha^* \ \epsilon \ \phi \ \delta) \end{array} \quad \text{eq:all_of_tet_sym} \quad (36)$$

Proof. To begin, notice that each of the operations on $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ appearing in Eq. (35) are invertible (indeed the first operation has order two, while the second operation has order four). The claim that these two operations generate a finite group of order 48 can be checked explicitly using computer software. TODO: Supply mathematica code used for this. Now suppose $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are the eigenvalues of a sextuple of Hermitian matrices (A, B, C, D, E, F) satisfying Eq. (3). Then, for example, the sextuple $(A', B', C', D', E', F') = (A, F, E, -D, C, B)$ (which has the eigenvalues $(\alpha, \phi, \epsilon, \delta^*, \gamma, \beta)$) is also tetrahedral because

$$\begin{aligned} A' + B' - C' &= (A + F - E) = 0, \\ B' + D' - F' &= -(B + D - F) = 0, \\ C' + D' - E' &= -(C + D - E) = 0, \\ A' + F' - E' &= (A + B - C) = 0. \end{aligned} \quad \text{eq:signed_perm_rows} \quad (37)$$

Moreover, as the operation mapping $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to $(\alpha^*, \gamma, \beta, \delta, \phi, \epsilon)$ is invertible, one also has that if $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ is not tetrahedral, then neither is $(\alpha^*, \gamma, \beta, \delta, \phi, \epsilon)$. An analogous argument holds for the operation mapping $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to $(\gamma, \epsilon^*, \delta^*, \phi, \beta, \alpha^*)$ with (A', B', C', D', E', F') given by $(C, -E, -D, F, B, -A)$. $\square$

Remark 15. The group of order 48 appearing in Proposition 14 may be viewed as a subgroup of the group of all $6! \times 2^6 = 46,080$ possible signed permutations of 6 elements generated by permutations and negations of the elements of the sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$. Moreover, this subgroup is maximal in the sense that it is the maximal subgroup of permutations and negations of the columns of coefficient matrix Γ (appearing in Eq. (6)) which is equivalent to a permutation and negation of the rows of Γ (as in Eq. (37)). It can also be shown that this group of symmetries is isomorphic to the signed permutation group of three elements, i.e., $(\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2) \rtimes S_3$. In order to see this, it is helpful to arrange the sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ into a 2×3 grid using the following identification:

$$(\alpha \ \beta \ \gamma \ \delta \ \epsilon \ \phi) \cong \begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix}. \quad (38)$$

Then one can see that each $\mathbb{Z}_2$ factor in the direct product $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ acts faithfully on the sextuple by both negating a certain subset of the columns and permuting the upper and lower rows leaving just one column unchanged.

$$\begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix} \leftrightarrow \begin{pmatrix} \alpha & \epsilon^* & \phi^* \\ \delta & \beta^* & \gamma^* \end{pmatrix}, \quad \begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix} \leftrightarrow \begin{pmatrix} \delta & \beta & \phi \\ \alpha & \epsilon & \gamma \end{pmatrix}, \quad \begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix} \leftrightarrow \begin{pmatrix} \delta^* & \epsilon & \gamma \\ \alpha^* & \beta & \phi \end{pmatrix}. \quad (39)$$

The symmetric group S_3 of order $3! = 6$ acts instead by permuting the columns and negating a subset of the entries in the top row, producing a subgroup of order 6 of the form:

$$\begin{pmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{pmatrix} \quad \begin{pmatrix} \beta & \gamma^* & \alpha^* \\ \epsilon & \phi & \delta \end{pmatrix} \quad \begin{pmatrix} \gamma^* & \alpha & \beta^* \\ \phi & \delta & \epsilon \end{pmatrix} \\ \begin{pmatrix} \alpha^* & \gamma & \beta \\ \delta & \phi & \epsilon \end{pmatrix} \quad \begin{pmatrix} \beta^* & \alpha^* & \gamma^* \\ \epsilon & \delta & \phi \end{pmatrix} \quad \begin{pmatrix} \gamma & \beta^* & \alpha \\ \phi & \epsilon & \delta \end{pmatrix} \quad (40)$$

Moreover, the above group of symmetries admits a finite presentation with respect to generators $g_1 \simeq (\alpha, \phi, \epsilon, \delta^*, \gamma, \beta)$ and $g_2 \simeq (\gamma, \epsilon^*, \delta^*, \phi, \beta, \alpha^*)$ as a free group over g_1, g_2 subject to the relations

$$\{g_1^2, g_2^4, (g_1 g_2)^6, (g_2 g_1 g_2 g_1 g_2)^2\}. \quad (41)$$

prop:shifting

Proposition 16 (Shifting symmetry). *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of candidate eigenvalues and let $x, y, z \in \mathbb{R}$ be fixed. Given any list of non-increasing eigenvalues $\lambda = (\lambda_1 \geq \dots \geq \lambda_n)$, let $\lambda + x := (\lambda_1 + x \geq \dots \geq \lambda_d + x)$. Then define shifted eigenvalues by the formula*

$$\begin{aligned} \tilde{\alpha} &:= \alpha + x, & \tilde{\beta} &:= \beta + y, & \tilde{\gamma} &:= \gamma + x + y, & \text{\textcolor{blue}{eq:shifted_tet_eigenvalues}} \\ \tilde{\delta} &:= \delta + z, & \tilde{\epsilon} &:= \epsilon + x + y + z, & \tilde{\phi} &:= \phi + y + z. \end{aligned} \quad (42)$$

The shifted eigenvalues $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ are tetrahedral if and only if the unshifted eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral.

Proof. Given a Hermitian matrix X with eigenvalues $\text{eig}(X) = \lambda$ and a constant $x \in \mathbb{R}$, the eigenvalues of $X + xI$ (where I is the identity matrix) are

$$\text{eig}(X + xI) = \lambda + x = (\lambda_1 + x \geq \dots \geq \lambda_d + x). \quad (43)$$

The statement then follows from Example 3 because

$$(A + xI, B + yI, C + (x + y)I, D + zI, E + (x + y + z)I, F + (y + z)I) \quad (44)$$

satisfy the tetrahedral constraints in Eq. (3) if and only if (A, B, C, D, E, F) satisfy the tetrahedral constraints in Eq. (3) $\square$

As a corollary of the shifting symmetries (Proposition 16) one concludes that to solve the tetrahedral Horn's problem, in a certain sense, it suffices to solve the tetrahedral Horn's problem merely for traceless matrices (Corollary 17) or, alternatively, for positive definite or positive semidefinite matrices (Corollary 18). cor:traceless

Corollary 17 (Traceless eigenvalues). *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of candidate eigenvalues and for any list of n eigenvalues $\lambda = (\lambda_1 \geq \dots \geq \lambda_n)$, let $\lambda^0 := \lambda - |\lambda|/n$ denote the traceless list of eigenvalues obtained by shifting λ by $-|\lambda|/n$. Then $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral if and only if (i) their traces $(|\alpha|, |\beta|, |\gamma|, |\delta|, |\epsilon|, |\phi|)$ are tetrahedral (they satisfy Eq. (9)) and (ii) $(\alpha^0, \beta^0, \gamma^0, \delta^0, \epsilon^0, \phi^0)$ are tetrahedral.*

Proof. This “only if” portion of the statement follows from both the tetrahedral trace conditions (Eq. (9)) and applications of the shifting symmetry Proposition 16 with (x, y, z) values

$$x = -n^{-1} \text{Tr}(A), \quad y = -n^{-1} \text{Tr}(B) \quad z = -n^{-1} \text{Tr}(D). \quad (45)$$

The “if” portion of the statement follows from the shifting symmetry applied to the negated (x, y, z) values. $\square$

The next corollary of the shifting symmetry proves that to solve the tetrahedral Horn's problem it suffices to solve the tetrahedral Horn's problem for positive definite spectra only. Indeed, one can simply take the shifting parameters (x, y, z) appearing in Proposition 16 to be sufficiently large. The following corollary establishes, in a certain sense, the minimum shifting required to ensure strictly positive eigenvalues for all six Hermitian matrices (assuming some other basic inequalities hold). Note that in subsequent sections we will often explicitly assume the candidate eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are either non-negative or positive for the sake of technical convenience or sometimes to establish a connection with density matrices from quantum theory (which are positive semidefinite Hermitian matrices with unit trace). cor:psd

Corollary 18 (Positive eigenvalues). *If $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are a tetrahedral sextuple of eigenvalues, then necessarily the smallest eigenvalues satisfy*

$$\gamma_d \geq \alpha_d + \beta_d, \quad \phi_d \geq \beta_d + \delta_d, \quad \epsilon_d \geq \delta_d + \gamma_d, \quad \epsilon_d \geq \alpha_d + \phi_d. \quad (46) \quad \text{\textcolor{blue}{eq:min_eig_ineqs}}$$

Moreover, if $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are a sextuple of candidate eigenvalues satisfying Eq. (46), then for any $(x, y, z) \in \mathbb{R}^3$ satisfying eq:positive_shift

$$x > -\alpha_d, \quad y > -\beta_d, \quad z > -\delta_d, \quad (47)$$

the resulting shifted eigenvalues $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ (given by Eq. (42)) are all positive.

Proof. That tetrahedral eigenvalues satisfy Eq. (46) follows from four applications of the well-known inequality $\gamma_d \geq \alpha_d + \beta_d$ for the trigonal Horn's problem where $C = A + B$ [Bha01]. As a reminder, the smallest eigenvalue λ_n of a matrix X is characterized by a minimization: $\lambda_d = \min_{\|v\|=1} \langle v, Xv \rangle$ which produces

$$\gamma_d = \min_{\|v\|=1} \langle v, Cv \rangle = \min_{\|v\|=1} (\langle v, Av \rangle + \langle v, Bv \rangle) \geq \min_{\|v\|=1} \langle v, Av \rangle + \min_{\|w\|=1} \langle w, Bw \rangle = \alpha_d + \beta_d. \quad (48)$$

Four applications of this inequality produces Eq. (46). Now if (x, y, z) satisfy Eq. (47), we have for all j

$$\alpha_j + x \geq \alpha_d + x > 0, \quad (49)$$

meaning $\alpha + x = (\alpha_1 + x \geq \dots \geq \alpha_d + x)$ is strictly positive. Similar statements hold for $\beta, \gamma, \delta, \epsilon$ and ϕ ; for instance, for all j ,

$$\epsilon_j + x + y + z \geq \epsilon_d + x + y + z > \epsilon_d - \alpha_d - \beta_d - \delta_d \geq 0, \quad (50)$$

meaning $\epsilon + x + y + z$ is strictly positive also. $\square$

Remark 19. Although this section listed a handful of symmetries of the tetrahedral Horn's problem, there remains the possibility of additional symmetries which are unknown at this time. For example, in the case 2×2 Hermitian matrices, an additional class of symmetries on the set of edge-lengths (a, b, c, e, d, f) of tetrahedra, known as the *Regge symmetries*, hold [AI19; MS18; Boa07; PR68, Theorem 1.1]. If there exists a tetrahedron with edge lengths (a, b, c, d, e, f) , then there also exists a tetrahedron with edge lengths

$$(s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \quad \text{where} \quad s_1 = \frac{a + b + d + e}{2}. \quad (51)$$

Combining this symmetry with the permutational symmetries given by Proposition 14 (and noting that for the eigenvalues λ of traceless 2×2 Hermitian matrices, $\lambda = (x, -x)$, we have $\lambda^* = \lambda$) one concludes the co-existence of a family of six tetrahedra with edge lengths given by [VMK88, Chapter 9]:

$$\begin{aligned} (a, b, c, d, e, f) & \quad (s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \\ (a, s_3 - b, s_3 - c, d, s_3 - e, s_3 - f) & \quad (s_2 - a, b, s_2 - c, s_2 - d, e, s_2 - f) \\ (s_2 - d, s_1 - e, s_3 - f, s_2 - a, s_1 - b, s_3 - c) & \quad (s_1 - d, s_3 - e, s_2 - f, s_1 - a, s_3 - b, s_2 - c) \end{aligned} \quad (52)$$

where

$$s_1 = \frac{a+b+d+e}{2}, \quad s_2 = \frac{a+c+d+f}{2}, \quad s_3 = \frac{b+c+e+f}{2}. \quad (53)$$

Whether there exists analogous types of symmetries for the tetrahedral Horn's problem when $d > 2$ remains an open problem. **TODO:** Describe how Regge symmetries emerge from $SU(2)$ 6j symbol symmetries [Boa07], how they can be viewed as a scissors-congruence of Euclidean tetrahedra perserving both the volume invariants and Dhen invariants [AI19], and how higher rank Regge symmetries are connected to the eigenvalue conjecture [MS18].

3 Inequalities for tetrahedral eigenvalues

- In [AZ99], Ando and Zhan proved a subadditivity inequality for non-negative operator monotone functions $f(t)$ on $[0, \infty)$. Specifically, for any PSD $A, B \geq 0$, and an unitarily invariant norm,

$$\|f(A) + f(B)\| \geq \|f(A+B)\|. \quad (54)$$

- Later in [Kos06], Kosem considered a generalization of the Ando-Zhan theorem which asks if for every non-negative operator monotone function $f(t)$ and every unitarily invariant norm and $A, B, C \geq 0$ PSD whether

$$\|f(A+B) + f(B+C) - f(B)\| \geq \|f(A+B+C)\|? \quad (55)$$

Kosem showed that this statement fails to hold for 2×2 PSD matrices when $f(t) = \frac{t}{t+1}$ and the norm $\|\cdot\|$ is the Ky Fan 2-norm (which for PSD matrices $X = (X^\dagger X)^{\frac{1}{2}}$ has the form $\|X\|_{(k)} = \sum_{i=1}^k \text{eig}_i(X)$).

- See also the inequalities presented by Aujla and Bourin [AB07], namely that if f is monotone concave on $[0, \infty)$ then there exists unitaries U, V such that as operators

$$f(A+B) \leq Uf(A)U^* + Vf(B)V^*. \quad (56)$$

- In [AHP10], Audenaert, Hiai, and Petz investigate the set of so-called *strongly subadditive functions* $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ such that

$$\text{Tr } f(P_{123}XP_{123}) + \text{Tr } f(P_2XP_2) \leq \text{Tr } f(P_{12}XP_{12}) + \text{Tr } f(P_{23}XP_{23}), \quad (57)$$

holds for all positive operators X and orthogonal projection operators P_1, P_2, P_3 where $P_{12} = P_1 + P_2$, $P_{23} = P_2 + P_3$ and $P_{123} = P_1 + P_2 + P_3$.

- It seems to be that analogous results were also derived by Friedland and Gaubert [FG13].
- Later, in [Lam+17], Lami et al. is concerned with the option $f(x) = \ln x$ which pertains to log-determinant inequalities and its connection to entropy inequalities.
- It seems probable that could be a unifying submodularity inequality (maybe related to [GJL17]?) prop:simple_quadratic

Proposition 20. Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be positive semidefinite tetrahedral eigenvalues. Then

$$\sum_j \alpha_j \delta_{d-j+1} \leq \frac{1}{2} \left(\sum_j \beta_j^2 + \sum_j \epsilon_j^2 - \sum_j \gamma_j^2 - \sum_j \phi_j^2 \right) \leq \sum_j \alpha_j \delta_j \quad (58)$$

Proof. Let (A, B, C, D, E, F) be positive semidefinite Hermitian matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ satisfying $A + B = C$, $B + D = F$, and $A + F = C + D = E$. The following identity holds:

$$\begin{aligned}
& \text{Tr}[B^2] + \text{Tr}[E^2] - \text{Tr}[C^2] - \text{Tr}[F^2] \\
&= \text{Tr}[(B)^2] + \text{Tr}[(A + B + D)^2] - \text{Tr}[(A + B)^2] - \text{Tr}[(B + D)^2] \\
&= (\text{Tr}[B^2]) + (\text{Tr}[A^2] + \text{Tr}[B^2] + \text{Tr}[D^2] + 2 \text{Tr}[AB] + 2 \text{Tr}[BD] + 2 \text{Tr}[AD]) \\
&\quad - (\text{Tr}[A^2] + \text{Tr}[B^2] + 2 \text{Tr}[AB]) - (\text{Tr}[B^2] + \text{Tr}[D^2] + 2 \text{Tr}[BD]) \\
&= 2 \text{Tr}[AD].
\end{aligned} \tag{59}$$

While the value of $\text{Tr}(AD)$ is not determined the eigenvalues $\alpha = \text{eig}(A)$ and $\delta = \text{eig}(D)$, one has the following bounds [MOA10, H.1.g & H.1.h]:

$$\sum_j \alpha_j \delta_{d-j+1} \leq \text{Tr}[AD] \leq \sum_j \alpha_j \delta_j, \tag{60}$$

where $\delta_{d-j+1} = \delta_j^*$. The claim then follows because $\text{Tr}[X^2] = \sum_j \lambda_j^2$ for any Hermitian operator X with eigenvalues $\text{eig}(X) = \lambda$. $\square$

3.1 Trigonal Horn inequalities

sec:horn_ineqs

There are a wealth of inequalities constraining the set of tetrahedral eigenvalues which may be viewed as generalization of the triangle inequalities which hold for case where $d = 2$ (see Eq. (13)). From Lemma 7 we see that $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral only if each of (α, β, γ) , (β, γ, ϕ) , $(\gamma, \delta, \epsilon)$, and (α, ϕ, ϵ) are trigonal. Consequently, every inequality constraining trigonal eigenvalues can be lifted to four inequalities constraining tetrahedral eigenvalues. The purpose of this section, therefore, is to briefly summarize some known inequalities which hold for triangular eigenvalues (α, β, γ) . Of particular interest are Horn's inequalities [Hor62] which are known to be necessary and sufficient [Kly98; KT99]. For a more complete historical account, the reader is directed to the survey articles [Bha01; Ful98].

In addition to the aforementioned trace condition, $\sum_i \alpha_i + \sum_j \beta_j = \sum_k \gamma_k$, one always has the following pair of elementary inequalities,

$$\alpha_1 + \beta_1 \geq \gamma_1, \quad \text{and} \quad \alpha_d + \beta_d \leq \gamma_d, \tag{61}$$

which constrain the largest and smallest eigenvalues respectively. The first somewhat non-trivial inequality, derived by Weyl in 1912, is now known as *Weyl's inequality* and states [Wey12]

$$\alpha_i + \beta_j \geq \gamma_{i+j-1}, \tag{62}$$

provided that $i + j - 1 \leq d$. For 2×2 matrices ($d = 2$) one can see that these necessary inequalities are also sufficient to ensure that (α, β, γ) are trigonal.²

In 1949, Fan derived an *majorization* inequality constraining the first r largest eigenvalues [Fan49]:

$$\sum_{i=1}^r \alpha_i + \sum_{j=1}^r \beta_j \geq \sum_{k=1}^r \gamma_k, \tag{63}$$

and shortly thereafter Lidskii & Wielandt found proofs of another family of inequalities which hold for any non-empty subset $I = \{i_1, \dots, i_r\} \subseteq \{1, \dots, d\}$ of indices of size r of the form

$$\sum_{j=1}^r \alpha_{i_j} + \sum_{j=1}^r \beta_j \geq \sum_{j=1}^r \gamma_{i_j}, \tag{64}$$

along with the inequality obtained by exchanging α and β . Finally, in 1962, Horn proved that these inequalities together with

$$\alpha_1 + \alpha_3 + \beta_1 + \beta_3 \geq \gamma_2 + \gamma_3 \tag{65}$$

²After shifting to traceless eigenvalues using Corollary 17, $\alpha = (a, -a)$, $\beta = (b, -b)$ and $\gamma = (c, -c)$, these inequalities become equivalent to the triangle inequalities $c \leq a + b$ and $c \geq \max\{b - a, a - b\}$.

are sufficient to ensure that (α, β, γ) are trigonal for the case where $d = 3$ [Hor62]. Moreover, also in [Hor62], Horn conjectured that in addition to the trace condition, a set of inequalities of the form

$$\sum_{i \in I} \alpha_i + \sum_{j \in J} \beta_j \geq \sum_{k \in K} \gamma_k \quad (66)$$

(for some inductively defined subsets $I, J, K \subset \{1, \dots, d\}$ of equal cardinality) was both necessary and sufficient to ensure that (α, β, γ) are trigonal.

TODO: Articulate how the four sets of Horn inequalities are necessary but insufficient conditions for the tetrahedral Horn problem. **TODO:** Articulate how the Horn inequalities can be inductively defined but also defined in terms of non-vanishing LR coefficients.

3.2 Submodular inequalities

DISCLAIMER: In this section I've switched to the A, B, C convention instead of the A, B, D convention. When discussing the tetrahedral Horn problem and its relation to the ordinary Horn problem or when matching the problem to the properties of the $6j$ symbols, the A, B, D convention is preferred. However when talking about submodular/strong subadditivity inequalities, the A, B, C convention is preferred. Before completing the paper, it will be desirable to establish a globally consistent convention.

Throughout this section, we will be in inequalities constraining the eigenvalues of $B, A + B, B + C$, and $A + B + C$ for PSD matrices $A, B, C \geq 0$. Note that under this formulation, we are not interested in the eigenvalues of either A or C . An equivalent formulation is to consider eigenvalues of four PSD matrices W, X, Y, Z such that $X + Y = W + Z$ and $W \geq X \geq Z$ and $W \geq Y \geq Z$. That these two formulations are the same follows from the identification $X = B + C, Y = A + B, Z = B$ and $W = A + B + C$.

Bouhtou, Gaubert, and Sagnol claim to prove the following family of inequalities [BGS08, Lemma 5.3]. lem:bouhtou_ineq

Lemma 21. *Let $A, B, C \geq 0$ be positive semidefinite Hermitian matrices and let $p \in [0, 1]$. Then*

$$\sum_j \lambda_j (A + B)^p + \sum_j \lambda_j (B + C)^p \geq \sum_j \lambda_j (B)^p + \sum_j \lambda_j (A + B + C)^p. \quad (67)$$

The following operator inequality can be used to prove a number of inequalities [TCL11, Lemma 2.2]. lem:mme_operator_ineq

Lemma 22. *Let $A, B, C \geq 0$ be positive semidefinite matrices and let k be a positive integer. Then*

$$(A + B + C)^{\otimes k} + B^{\otimes k} \geq (A + B)^{\otimes k} + (B + C)^{\otimes k}. \quad (68)$$

Proof. The result is obtained from applying a non-commutative multinomial expansion and noting that every summand on the right-hand-side above also appears on the left-hand-side above. $\square$

To obtain an inequality which depends only on the eigenvalues of $B, A + B, B + C$, and $A + B + C$, one can simply pair the positive semidefinite operator in Lemma 22 with an $SU(n)$ -equivariant and positive semidefinite operator:

Corollary 23. *Let $A, B, C \geq 0$ be positive semidefinite matrices and let $\lambda \vdash k$ be a Young diagram with k boxes. Then* cor:isotypic_ineq

$$\text{Tr}[P_\lambda (A + B + C)^{\otimes k}] + \text{Tr}[P_\lambda B^{\otimes k}] \geq \text{Tr}[P_\lambda (A + B)^{\otimes k}] + \text{Tr}[P_\lambda (B + C)^{\otimes k}]. \quad (69)$$

Moreover, as $\text{Tr}[P_\lambda X^{\otimes k}] \propto E_\lambda(X)^3$, where $E_\lambda(X) = s_\lambda(\text{eig}(X))$ is the Schur polynomial of the eigenvalues of X , we equivalently have

$$s_\lambda(A + B + C) + s_\lambda(B) \geq s_\lambda(A + B) + s_\lambda(B + C). \quad (70)$$

As a special case of the previous corollary, we obtain an inequality which complements Proposition 20.

³Specifically the proportionality constant here is equal to $\dim W_\lambda$ where W_λ is the irreducible representation of S_k labelled by λ .

Corollary 24. Let $A, B, C \geq 0$ be positive semidefinite Hermitian matrices, then

$$\mathrm{Tr}[A+B]^2 - \mathrm{Tr}[(A+B)^2] + \mathrm{Tr}[B+C]^2 - \mathrm{Tr}[(B+C)^2] \leq \mathrm{Tr}[B]^2 - \mathrm{Tr}[B^2] + \mathrm{Tr}[A+B+C]^2 - \mathrm{Tr}[(A+B+C)^2]. \quad (71)$$

Proof. This result is a special case of Corollary 23 below when $\lambda = (1, 1)$ because

$$\mathrm{Tr}[P_{(1,1)} X^{\otimes 2}] = \frac{1}{2} (\mathrm{Tr}(X)^2 - \mathrm{Tr}(X^2)). \quad (72)$$

□

Paksoy, Turkmen, and Zhang proved a family of polynomial inequalities which are generally not functions of just the eigenvalues [PTZ14]. These inequalities are in terms a so-called generalized matrix function first defined by Marcus and Minc [MM65].

Definition 25 (Generalized matrix function). Let H be a subgroup of S_n and let χ be an irreducible character of H . The *generalized matrix function* for the character χ of H is

$$d_\chi^H(X) = \sum_{\sigma \in H} \chi(\sigma) \prod_i X_{i, \sigma(i)} \quad (73)$$

The notion of a *generalized matrix function* generalizes the notion of an *immanant* (when $H = S_n$) which, in turn, contains as special cases the notion of a permanent and determinant [HM21].

thm:ptz_ineqs

Theorem 26. Let $A, B, C \geq 0$ be positive semidefinite $n \times n$ matrices and let H be a subgroup of S_n with irreducible character χ . Then

$$d_\chi^H(A+B+C) + d_\chi^H(B) \geq d_\chi^H(A+B) + d_\chi^H(B+C). \quad (74)$$

Proof. See [PTZ14].

□

As a corollary of Theorem 26, we obtain the following determinant and permanent inequalities.

Corollary 27. Let $A, B, C \geq 0$ be positive semidefinite matrices. Then

$$\det(A+B+C) + \det(B) \geq \det(A+B) + \det(B+C), \quad (75)$$

$$\mathrm{per}(A+B+C) + \mathrm{per}(B) \geq \mathrm{per}(A+B) + \mathrm{per}(B+C). \quad (76)$$

Note that with the exception of the above determinant inequality, the inequalities provided by Theorem 26 are generally not equivalent to inequalities involving only eigenvalues.

One also has another determinant inequality which is non-linear in determinants [TCL11, Lemma 2.3].

lem:det_prod_ineq

Lemma 28. Let $A, B, C \geq 0$ be positive semidefinite matrices. Then

$$\det(A+B+C) \det(B) \leq \det(A+B) \det(B+C). \quad (77)$$

Note that Lemma 28 can be seen as a special of Lemma 21 in the limit as $p \rightarrow 0$; the derivative of Lemma 21 in p at $p = 0$ yields

$$\sum_j \log \lambda_j(A+B) + \sum_j \log \lambda_j(B+C) \geq \sum_j \log \lambda_j(B) + \sum_j \log \lambda_j(A+B+C), \quad (78)$$

{eq:bouhtou_ineq_p_zero}

which is equivalent to Lemma 28.

thm:ssa_ineq

Theorem 29. Let $A, B, C \geq 0$ be positive semidefinite Hermitian matrices, then

$$\sum_j \lambda_j(A+B) \log \lambda_j(A+B) + \sum_j \lambda_j(B+C) \log \lambda_j(B+C) \quad (79)$$

{eq:ssa_like_ineq}

$$\leq \sum_j \lambda_j(B) \log \lambda_j(B) + \sum_j \lambda_j(A+B+C) \log \lambda_j(A+B+C). \quad (80)$$

The most comprehensive result concerning the eigenvalues of partial sums of Hermitian matrices we have encountered thus far is [Sag13, Proposition 2.3] which provides the following result.

thm:sagnol_ineq

Theorem 30. *Let $A, B, C \geq 0$ be positive semidefinite Hermitian matrices and let f be a real valued and differentiable function over the non-negative real numbers $\mathbb{R}_{\geq 0}$ such that the derivative of f is an operator monotone function over the positive real numbers $\mathbb{R}_{> 0}$. Then the following inequality holds*

{eq:sagnol_ineq}

$$\mathrm{Tr} f(A + B + C) + \mathrm{Tr} f(B) \geq \mathrm{Tr} f(A + B) + \mathrm{Tr} f(B + C). \quad (81)$$

Proof. The following proof is a rough summary of the proof found in [Sag13, Section 2.1]. Let X, Y be positive semidefinite and consider following difference:

$$F(Y|X) := \mathrm{Tr} f(X + Y) - \mathrm{Tr} f(Y). \quad (82)$$

The inequality in Eq. (81) is thus equivalent to $F(C|B) \leq F(A + C|B)$. To prove Eq. (81) it suffices to prove that the value of $F(Y|X)$ is a non-decreasing function under $Y \mapsto Y + Z$ for arbitrary positive semidefinite X, Y , and Z . To establish the claim, let $\epsilon > 0$ be small and let $Z = \epsilon W$ for non-zero positive semidefinite W . It can be shown⁴ that

$$(\partial_Y F(Y|X))(W) := \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (F(Y + \epsilon W|X) - F(Y|X)) \quad (83)$$

$$= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (\mathrm{Tr} f(X + Y + \epsilon W) - \mathrm{Tr} f(Y + \epsilon W) - \mathrm{Tr} f(X + Y) + \mathrm{Tr} f(Y)) \quad (84)$$

$$= \mathrm{Tr} [(f'(X + Y) - f'(Y))W]. \quad (85)$$

Using the assumption that f' is an operator monotone together with $X + Y \geq Y$ (because $X \geq 0$), we have $f'(X + Y) \geq f'(Y)$ and thus $f'(X + Y) - f'(Y) \geq 0$. Therefore $(\partial_Y F(Y|X))(W) \geq 0$ for all W and thus $F(Y|X)$ is non-decreasing with respect to $Y \mapsto Y + Z$ as desired. $\square$

As corollaries one obtains some of the previously mentioned inequalities for various functions f satisfying the conditions of the theorem; for example, choosing $f(x) = x \log x$ yields Theorem 29, while choosing $f(x) = -x^p$ for $p \in (0, 1]$ yields Lemma 21, and choosing $f(x) = x^p$ for $p \in (1, 2]$ yields the following.

Corollary 31. *Let $A, B, C \geq 0$ be positive semidefinite Hermitian matrices and let $p \in (1, 2]$. Then*

$$\sum_j \lambda_j(A + B)^p + \sum_j \lambda_j(B + C)^p \leq \sum_j \lambda_j(B)^p + \sum_j \lambda_j(A + B + C)^p. \quad (86)$$

TC: On Sept. 18th, I came across an interesting discussion on [mathoverflow](https://mathoverflow.net/questions/257562/a-determinant-inequality) <https://mathoverflow.net/questions/257562/a-determinant-inequality> which appears deeply related to the topic of submodular inequalities. In that post the following result is stated without explicit proof and is based on the theory of so-called “hyperbolic polynomials” p and their real roots λ_i . There are a handful of conjectured inequalities on the page. For example, it seems to prove

$$E_k(B)E_k(A + B + C) \leq E_k(A + B)E_k(B + C) \quad (87)$$

where $E_k(X) = e_k(\mathrm{eig}(X))$ where E_k is the k th elementary symmetric polynomial.

Remark 32. For 2×2 Hermitian matrices it is possible to visualize the set of eigenvalues (in the same (a, b, c, d, e, f) coordinates used above) which are known to be non-tetrahedral on the basis that they violate Eq. (79). Specifically, consider the *minimal* shifting of the traceless eigenvalues in Eq. (10), i.e.,

$$\begin{aligned} \alpha &= (2a, 0), & \beta &= (2b, 0), & \gamma &= (a + b + c, a + b - c), \\ \delta &= (2d, 0), & \epsilon &= (a + b + d + e, a + b + d - e), & \phi &= (b + d + f, b + d - f). \end{aligned} \quad (88)$$

⁴The statement follows, without loss of generality, by assuming that all the matrices involved are positive *definite*. This assumption is permitted because f is assumed continuous and the eigenvalues of a matrix are also continuous functions (see [Sag13] for a detailed explanation).

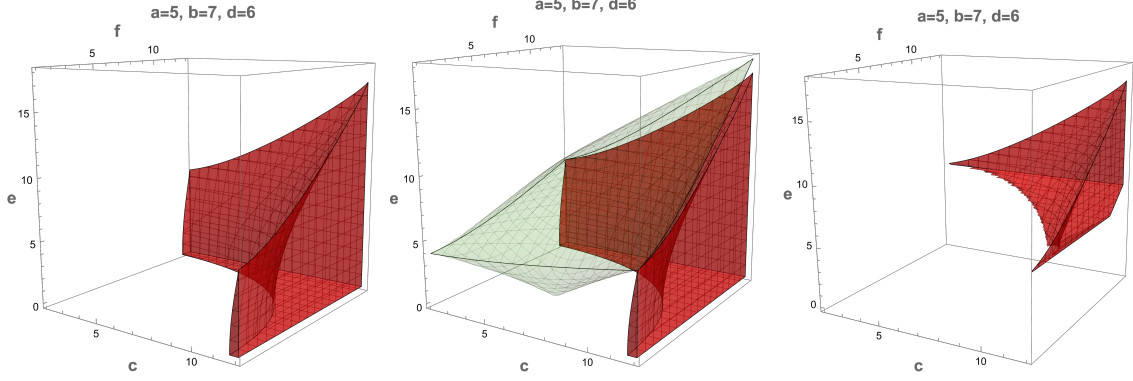


Figure 4: (Left) The region for which (c, e, f) violates Eq. (89) with $(a, b, c) = (5, 7, 6)$. (Middle) Comparing the region of tetrahedral (c, e, f) with those which violate Eq. (89). (Right) The values of (c, e, f) satisfying Eq. (13) but violating Eq. (89). fig:ssa_extra_mileage

With respect to these coordinates, Eq. (79) becomes

$$\begin{aligned}
 F(a, b, c, d, e, f) := & -(a + b + c) \log(a + b + c) - (a + b - c) \log(a + b - c) \\
 & - (b + d + f) \log(b + d + f) - (b + d - f) \log(b + d - f) \quad \text{\textcolor{blue}{eq:ssa_like_d_two}} \\
 & + (a + b + d + e) \log(a + b + d + e) + (a + b + d - e) \log(a + b + d - e) \\
 & + (2b) \log(2b) \geq 0
 \end{aligned} \tag{89}$$

See Fig. 4 for a depiction of the region violating Eq. (89).

Here I attempt to generalize the derivation of Sagnol:

$$\lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (\text{Tr } f(X + Y + \epsilon W) - \text{Tr } f(Y + \epsilon W) - \text{Tr } f(X + Y) + \text{Tr } f(Y)) = \text{Tr}[(f'(X + Y) - f'(Y))W] \tag{90}$$

$$\lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (f(\text{Tr}(X + Y + \epsilon W)) - f(\text{Tr}(Y + \epsilon W)) - f(\text{Tr}(X + Y)) + f(\text{Tr}(Y))) \tag{91}$$

$$= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (f(x + y + \epsilon w) - f(y + \epsilon w) - f(x + y) + f(y)) \tag{92}$$

$$= w(f'(x + y) - f'(y)) \tag{93}$$

$$= \text{Tr}(W)(f'(\text{Tr}(X + Y)) - f'(\text{Tr}(Y))). \tag{94}$$

If m is an operator monotone, then $X \geq 0$ implies $m(X) \geq 0$. Moreover, $X \geq 0$ implies $\text{Tr}(m(X)W) \geq 0$ for all PSD W . Under which additional conditions can we expect $\text{Tr}(m(X)W) \leq m(\text{Tr}(X)) \text{Tr}(W)$? If $X = \sum_i \lambda_i \Pi_i$ then $m(X) = \sum_i m(\lambda_i) \Pi_i$ and

$$\text{Tr}(m(X)W) = \sum_i m(\lambda_i) \text{Tr}(\Pi_i W) \leq \sum_i m(\lambda_i) \text{Tr}(W) = \text{Tr}(m(X)) \text{Tr}(W) \tag{95}$$

If $\sum_i m(\lambda_i) \leq m(\sum_i \lambda_i)$ then we also conclude $\text{Tr}(m(X)W) \leq m(\text{Tr}(X)) \text{Tr}(W)$.

Also consider if $X \geq Y$ then $f(X) \geq f(Y)$ but also $X \mapsto X' := \text{Tr}(X)I - X$ sends PSD matrices to PSD matrices so the inequality concerning $(\text{Tr} \circ f)(X)$ above also holds for $(\text{Tr} \circ f)(\text{Tr}(X)I - X)$.

3.3 The CSW inequality

At this stage, it is worth comparing the inequality in Eq. (79) to a similar inequality derived by Christandl, Şahinoğlu, and Walter [CSW18], which holds for tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ which are non-negative and sufficiently small in the sense that $\lambda_1 + |\lambda| \leq 1$:

$$h(|\gamma|) + |\gamma| S(\bar{\gamma}) + h(|\phi|) + |\phi| S(\bar{\phi}) - h(|\beta|) - |\beta| S(\bar{\beta}) - h(|\epsilon|) - |\epsilon| S(\bar{\epsilon}) \geq 0. \quad \text{\textcolor{blue}{eq:ssa_like_CSW}} \tag{96}$$

The inequality appearing in Eq. (79) is equivalent to the following by Eq. (219):

$$-|\gamma| \log |\gamma| + |\gamma| S(\bar{\gamma}) - |\phi| \log |\phi| + |\phi| S(\bar{\phi}) + |\beta| \log |\beta| - |\beta| S(\bar{\beta}) + |\epsilon| \log |\epsilon| - |\epsilon| S(\bar{\epsilon}) \geq 0. \quad \text{\textcolor{blue}{(eq:SSA_like_ineq_equiv)}} \quad (97)$$

The difference between Eq. (96) and Eq. (97) is captured by a formal substitution replacing the binary entropy $h(|\lambda|)$ with the expression $-|\lambda| \log |\lambda|$. We note that it is possible to derive Eq. (97) from Eq. (96) by a down-scaling argument. Specifically, while the set of tetrahedral eigenvalues is closed under scaling all eigenvalues by a non-negative r ,

$$(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \mapsto (r\alpha, r\beta, r\gamma, r\delta, r\epsilon, r\phi), \quad \text{\textcolor{blue}{(eq:tet_scaling)}} \quad (98)$$

the form and positivity of Eq. (96) is *not* invariant under scaling by r . Define q_r by the formula

$$q_r := r^{-1} (h(|r\gamma|) + h(|r\phi|) - h(|r\beta|) - h(|r\epsilon|)). \quad (99)$$

Letting $r \in (0, 1)$ be positive and small, we see that

$$r^{-1} h(|r\lambda|) = |\lambda| (1 - \log r) - |\lambda| \log |\lambda| + O(r^2) \quad (100)$$

which implies the following limit holds when $|\gamma| + |\phi| = |\beta| + |\epsilon|$:

$$q_0 := \lim_{r \rightarrow 0} q_r = -|\gamma| \log |\gamma| - |\phi| \log |\phi| + |\beta| \log |\beta| + |\epsilon| \log |\epsilon|. \quad (101)$$

In summary, by scaling down the eigenvalues in Eq. (96) by $r \in (0, 1)$ according to Eq. (98) and then scaling up Eq. (96) by multiplication by r^{-1} and taking the limit as $r \rightarrow \infty$, one recovers Eq. (97). Moreover, it can be shown that $q_r \geq q_0$ for all r and thus Eq. (97) is strictly stronger than Eq. (96). To prove this, we see that the trace equalities imply

$$\begin{aligned} q_r - q_0 &= r^{-1} (1 - r|\beta|) \log(1 - r|\beta|) + r^{-1} (1 - r|\epsilon|) \log(1 - r|\epsilon|) \\ &\quad - r^{-1} (1 - r|\gamma|) \log(1 - r|\gamma|) - r^{-1} (1 - r|\phi|) \log(1 - r|\phi|). \end{aligned} \quad (102)$$

Approximating $-(1-x) \log(1-x)$ by a Taylor series for $x \approx 0$,

$$-(1-x) \log(1-x) = x - \frac{x^2}{2} + O(x^3), \quad (103)$$

produces

$$q_r - q_0 = \frac{r}{2} (|\beta|^2 + |\epsilon|^2 - |\gamma|^2 - |\phi|^2) + O(r^2) = r |\alpha| |\delta| + O(r^2) \quad (104)$$

which implies $q_r > q_0$ and least for sufficiently small r . To prove $q_r > q_0$ holds for all r , by continuity, it therefore suffices to show that the gradient of $q_r - q_0$ with respect to r never changes sign for $r > 0$.

$$\begin{aligned} \partial_r (q_r - q_0) &= \frac{1}{r^2} (\log(1 - r|\gamma|) + \log(1 - r|\phi|) - \log(1 - r|\beta|) - \log(1 - r|\epsilon|)) \\ &= \frac{1}{r^2} \log \left(\frac{(1 - r|\gamma|)(1 - r|\phi|)}{(1 - r|\beta|)(1 - r|\epsilon|)} \right). \end{aligned} \quad (105)$$

Thus the gradient $\partial_r (q_r - q_0)$ vanishes only if the argument of the above logarithm is one, or equivalently when

$$0 = (1 - r|\gamma|)(1 - r|\phi|) - (1 - r|\beta|)(1 - r|\epsilon|) = r^2 (|\gamma| |\phi| - |\epsilon| |\beta|) = r^2 |\alpha| |\delta|. \quad (106)$$

As we are implicitly assuming $|\alpha| \neq 0$ and $|\delta| \neq 0$, we see that the gradient of $q_r - q_0$ never changes when $r > 0$ and thus $q_r - q_0$ is increasing with increasing $r > 0$.

4 Asymptotics of $SU(n)$ $6j$ operators

The purpose of this section is to derive some upper and lower bounds on generalized $6j$ symbol for $SU(n)$ in relation to the existence or non-existence of Hermitian operators satisfying the tetrahedral relations Eq. (3). This section essentially constitutes a generalization of the proof techniques in the “quantum-information-theoretic” approach taken in [Chr08]. Specifically, the results of this section are derived from considering a type of pseudo-commutative binomial expansion of tensor powers of sums of Hermitian operators and applying spectrum estimation to the resulting terms.

TC: TODO: There is a lot of work left to do to clean up this section. In particular, it would be very useful to present a brief pedagogical account of the connection between the Littlewood-Richardson coefficients and the trigonal Horn problem and then use that intuition as a basis for extrapolating to the case of the tetrahedral Horn problem and the need for a representation-theory object which captures the additional consistency constraints imposed by the tetrahedral Horn problem.

4.1 Schur-Weyl decomposition & spectrum estimation

Throughout this section, we make use of the following decomposition of a tensor-power of a n -dimensional complex vector space.

Proposition 33 (Schur-Weyl decomposition). *Let k and n be positive integers. Then the decomposition of $(\mathbb{C}^n)^{\otimes k}$ into irreducible representations of $S_k \times SU(n)$, known as the Schur-Weyl decomposition is given by*

$$(\mathbb{C}^n)^{\otimes k} \cong \bigoplus_{\lambda \vdash_n k} W_\lambda \otimes V_\lambda^n \quad \text{\texttt{eq:schur_weyl_decomp}} \quad (107)$$

where the direct summands are indexed by Young diagrams λ with k boxes and at most $n - 1$ rows.

The following result constrains the associated Schur-Weyl decomposition of the k -th tensor power, $X^{\otimes k}$, of a positive semidefinite Hermitian matrix X [Chr08; KW01].

Proposition 34 (Spectrum estimation). *Let P_λ denote the Hermitian projection operator associated to the isotypic subspace indexed by $\lambda \vdash_n k$ in Eq. (107) and let X be a positive semidefinite matrix with eigenvalues χ . Then*

$$\text{Tr}(P_\lambda X^{\otimes k}) \leq |\chi|^k \dim(V_\lambda^n) \exp(-kH(\bar{\lambda}|\bar{\chi})), \quad \text{\texttt{prop:spec_est}} \quad (108)$$

where $H(\bar{\lambda}|\bar{\chi})$ denotes the Kullback-Leibler divergence of $\bar{\lambda}$ from $\bar{\chi}$.

Proof. See [KW01] for a proof (noting that the $|\chi|^k$ term simply accounts for the possibility that $\text{Tr}(X) \neq 1$ because $\text{Tr}(X) = |\chi|$). $\square$

4.2 Littlewood-Richardson coefficients & recoupling operators

Definition 35 (Littlewood-Richardson coefficients). Let $\alpha, \beta, \gamma \in \mathbb{Z}_{\text{dec}}^n$ be a triple of dominant weights for $GL(n)$ and let $V_\alpha, V_\beta, V_\gamma$ be the triple of irreducible representations of $GL(n)$ with corresponding highest weights. Let $C_{\alpha, \beta}^\gamma$ denote the multiplicity space of $GL(n)$ -intertwiners from V_γ to $V_\alpha \otimes V_\beta$:

$$C_{\alpha, \beta}^\gamma := \text{Hom}_{GL(n)}(V_\gamma, V_\alpha \otimes V_\beta) \simeq \text{Inv}_{GL(n)}(V_{\gamma^*} \otimes V_\alpha \otimes V_\beta). \quad (109)$$

The *Littlewood-Richardson coefficient* is defined as the dimension of this multiplicity space:

$$c_{\alpha, \beta}^\gamma := \dim(C_{\alpha, \beta}^\gamma). \quad (110)$$

Definition 36 (Clebsch-Gordon isometries). Given α, β, γ as in Definition 35, a *Clebsch-Gordon isometry* is a $GL(n)$ -equivariant isometric linear map sending $f \otimes v \in C_{\alpha, \beta}^\gamma \otimes V_\gamma$ to $f(v) \in V_\alpha \otimes V_\beta$ ⁵. Let

$$Y_{\alpha, \beta}^\gamma : C_{\alpha, \beta}^\gamma \otimes V_\gamma \rightarrow V_\alpha \otimes V_\beta, \quad (111)$$

⁵If $C_{\alpha, \beta}^\gamma = 0$, then $Y_{\alpha, \beta}^\gamma$ is identified with the origin in $V_\alpha \otimes V_\beta$.

be a Clebsch-Gordon isometry and depict it diagrammatically by

$$\begin{array}{c} V_\alpha \\ \nearrow \\ \circ \\ \searrow \\ V_\beta \end{array} \begin{array}{c} V_\gamma \\ \nearrow \\ \circ \\ \searrow \\ V_\gamma \end{array} = \begin{array}{c} V_\alpha \\ \nearrow \\ \boxed{Y_{\alpha,\beta}^\gamma} \\ \searrow \\ V_\beta \end{array} \begin{array}{c} V_\gamma \\ \nearrow \\ \circ \\ \searrow \\ V_\gamma \end{array}. \quad (112)$$

The conjugate transpose of the Clebsch-Gordon isometry is

$$Y_{\alpha,\beta}^{\gamma\dagger} : V_\alpha \otimes V_\beta \rightarrow C_{\alpha,\beta}^\gamma \otimes V_\gamma, \quad (113)$$

will be depicted diagrammatically by

$$\begin{array}{c} C_{\alpha,\beta}^\gamma \\ \nearrow \\ \circ \\ \searrow \\ V_\gamma \end{array} \begin{array}{c} V_\alpha \\ \nearrow \\ \circ \\ \searrow \\ V_\beta \end{array} = \begin{array}{c} C_{\alpha,\beta}^\gamma \\ \nearrow \\ \boxed{Y_{\alpha,\beta}^{\gamma\dagger}} \\ \searrow \\ V_\gamma \end{array} \begin{array}{c} V_\alpha \\ \nearrow \\ \circ \\ \searrow \\ V_\beta \end{array}. \quad (114)$$

`defn:ud_recoupling_operator`

Definition 37 (Recoupling operators). Let x, y, z and n be positive integers and let

$$\alpha \in \Upsilon_{x,n}, \delta \in \Upsilon_{z,n}, \beta \in \Upsilon_{y,n}, \epsilon \in \Upsilon_{x+y+z,n}, \gamma \in \Upsilon_{x+y,n}, \quad \phi \in \Upsilon_{y+z,n}. \quad (115)$$

The $U(n)$ -recoupling operator is the operator

$$\left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} : C_{\alpha,\beta}^\gamma \otimes C_{\delta,\gamma}^\epsilon \rightarrow C_{\alpha,\phi}^\epsilon \otimes C_{\beta,\delta}^\phi, \quad (116)$$

defined by

$$\left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} := \frac{\text{Tr}_{V_\epsilon} [Y_{\alpha,\phi}^{\epsilon\dagger} (\text{id}_{V_\alpha} \otimes Y_{\beta,\delta}^{\phi\dagger}) (Y_{\alpha,\beta}^\gamma \otimes \text{id}_{V_\delta}) Y_{\gamma,\delta}^\epsilon]}{\dim(V_\epsilon)}, \quad (117)$$

which is depicted diagrammatically by

$$\begin{array}{c} C_{\alpha,\phi}^\epsilon \\ \nearrow \\ \boxed{\left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\}} \\ \searrow \\ C_{\gamma,\delta}^\epsilon \end{array} \begin{array}{c} C_{\beta,\delta}^\phi \\ \nearrow \\ \circ \\ \searrow \\ C_{\gamma,\delta}^\epsilon \end{array} := \frac{1}{\dim(V_\epsilon)} \begin{array}{c} C_{\alpha,\phi}^\epsilon \\ \nearrow \\ \boxed{Y_{\alpha,\phi}^{\epsilon\dagger}} \\ \searrow \\ V_\phi \end{array} \begin{array}{c} C_{\beta,\delta}^\phi \\ \nearrow \\ \boxed{Y_{\beta,\delta}^{\phi\dagger}} \\ \searrow \\ V_\delta \end{array} \begin{array}{c} V_\alpha \\ \nearrow \\ \boxed{Y_{\alpha,\beta}^\gamma} \\ \searrow \\ V_\gamma \end{array} \begin{array}{c} V_\delta \\ \nearrow \\ \boxed{Y_{\gamma,\delta}^\epsilon} \\ \searrow \\ C_{\gamma,\delta}^\epsilon \end{array} = \frac{1}{\dim(V_\epsilon)} \begin{array}{c} C_{\alpha,\phi}^\epsilon \\ \nearrow \\ \circ \\ \searrow \\ C_{\gamma,\delta}^\epsilon \end{array} \begin{array}{c} C_{\beta,\delta}^\phi \\ \nearrow \\ \circ \\ \searrow \\ C_{\gamma,\delta}^\epsilon \end{array} \begin{array}{c} V_\alpha \\ \nearrow \\ \circ \\ \searrow \\ V_\gamma \end{array} \begin{array}{c} V_\delta \\ \nearrow \\ \circ \\ \searrow \\ V_\epsilon \end{array}. \quad (118)$$

WARNING: The convention I am using for the curly braces seems to differ from many of the uses of the curly braces to define the Wigner 6j coefficient which differs in that I think the Wigner 6j coefficient does not include the additional factor of $\dim(V_\epsilon^n)$. This lack of common convention seems to be echoed by Coquereaux [Coq23].

Subsequent sections will primarily be interested in the square of the Hilbert-Schmidt norm of the recoupling operator in Definition 37. While the individual coefficients of the recoupling operator are dependent on a choice of basis for each of the multiplicity spaces, its Hilbert-Schmidt and operator norms are not dependent on such a choice of basis. The following lemma provides a formula for the square of the Hilbert-Schmidt norm as an integral over $SU(n)^{\times 4}$.

Lemma 38 (Character formula). *Let $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} : C_{\alpha,\beta}^\mu \otimes C_{\mu,\gamma}^\lambda \rightarrow C_{\alpha,\nu}^\lambda \otimes C_{\beta,\gamma}^\nu$ be the recoupling operator defined in Definition 37. Then*

$$R_{\delta\epsilon\phi}^{\alpha\beta\gamma} := \dim(V_\epsilon)^2 \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_{\text{HS}}^2 = \int_{\text{SU}(n)^4} du_1 du_2 du_3 du_4 X_{\delta,\epsilon,\phi}^{\alpha,\beta,\gamma}(u_1, u_2, u_3, u_4). \quad (119)$$

where

$$X_{\delta,\epsilon,\phi}^{\alpha,\beta,\gamma}(u_1, u_2, u_3, u_4) := \chi_\alpha(u_1^{-1}u_2)\chi_\beta(u_4^{-1}u_2)\chi_\gamma(u_2^{-1}u_3)\chi_\delta(u_4^{-1}u_3)\chi_\epsilon(u_3^{-1}u_1)\chi_\phi(u_1^{-1}u_4) \quad (120)$$

Note that when $n = 2$, Eq. (119) recovers the well-known formula for the square of the $6j$ -symbol for $\text{SU}(n)$ (see [FL03, Equation (1)] and also [Bar98; FK00]).

Proof of Lemma 38. The proof follows from (i) the following identities concerning the character $\chi_\lambda(u) = \text{Tr}(\pi_\lambda(u))$ and duality $\pi_\lambda^*(u) = (\pi_\lambda(u^{-1}))^T$ relations

$$\chi_\lambda(u) = \boxed{\pi_\lambda(u)} \quad \pi_\lambda^*(u) = \boxed{\pi_\gamma(u^{-1})} \quad (121)$$

and (ii) the integral formula for the projection operator $\Pi_{\alpha,\beta}^\gamma$ onto the subspace of invariant vectors in $V_\alpha \otimes V_\beta \otimes V_\gamma^*$:

$$\Pi_{\alpha,\beta}^\gamma = \begin{array}{c} \alpha \quad \beta \quad \gamma \\ \uparrow \quad \uparrow \quad \uparrow \\ \boxed{\Pi_{\alpha,\beta}^\gamma} \\ \downarrow \quad \downarrow \quad \downarrow \\ \alpha \quad \beta \quad \gamma \end{array} = \begin{array}{c} \alpha \quad \beta \quad \gamma \\ \uparrow \quad \uparrow \quad \uparrow \\ \circ \\ \downarrow \quad \downarrow \quad \downarrow \\ \alpha \quad \beta \quad \gamma \end{array} C_{\alpha,\beta}^\gamma = \int du \begin{array}{c} \alpha \quad \beta \quad \gamma \\ \uparrow \quad \uparrow \quad \uparrow \\ \boxed{\pi_\alpha(u)} \quad \boxed{\pi_\beta(u)} \quad \boxed{\pi_\gamma^*(u)} \\ \downarrow \quad \downarrow \quad \downarrow \\ \alpha \quad \beta \quad \gamma \end{array}. \quad (122)$$

Applied four times, we obtain

$$\begin{aligned} & \left(\bigcirc_{L_\epsilon} \right)^2 C_{\alpha,\phi}^\epsilon C_{\alpha,\beta}^\gamma C_{\gamma,\delta}^\epsilon C_{\beta,\delta}^\phi = \\ & \begin{array}{c} \phi \quad \epsilon \quad \alpha \quad \alpha \quad \beta \quad \gamma \quad \gamma \quad \delta \quad \epsilon \quad \delta \quad \beta \quad \phi \\ \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ C_{\alpha,\phi}^\epsilon \quad C_{\alpha,\beta}^\gamma \quad C_{\gamma,\delta}^\epsilon \quad C_{\beta,\delta}^\phi \end{array} \\ & = \Pi_i \int du_i \begin{array}{c} \bar{u}_1 \quad u_1 \quad \bar{u}_1 \quad u_2 \quad u_2 \quad \bar{u}_2 \quad u_3 \quad u_3 \quad \bar{u}_3 \quad u_4 \quad u_4 \quad \bar{u}_4 \quad u_4 \\ \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ \phi \quad \epsilon \quad \alpha \quad \alpha \quad \beta \quad \gamma \quad \gamma \quad \delta \quad \epsilon \quad \delta \quad \beta \quad \phi \end{array} \\ & = \Pi_i \int du_i \begin{array}{c} \bar{u}_1 \quad u_2 \quad \bar{u}_2 \quad u_3 \quad u_3 \quad \bar{u}_4 \quad u_4 \quad \bar{u}_1 \quad u_3 \quad \bar{u}_1 \quad u_4 \\ \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ \alpha \quad \gamma \quad \delta \quad \beta \quad \epsilon \quad \phi \end{array}. \quad (123) \end{aligned}$$

□

Remark 39. With the integral character formula $R_{\delta\epsilon\phi}^{\alpha\beta\gamma}$ established, it is interesting to compare it against the character formula for the heat kernel of a compact Lie group G [Feg78], denoted by $K_G(g_1, g_2; t)$ in [McS21]:

$$K_G(g_1, g_2; t) = \sum_{\lambda} \dim V_{\lambda}^n \chi_{\lambda}(g_1 g_2^{-1}) e^{-t(|\lambda+\rho|^2 - |\rho|^2)/2}, \quad (124)$$

where the sum is over all dominant integral weights λ for G and ρ the associated Weyl vector. It is conceivable that there might be a method for expressing $R_{\delta\epsilon\phi}^{\alpha\beta\gamma}$ as a low temperature limit, $t \rightarrow 0$, of an integral over products of heat kernels.

4.3 Trigonal & polygonal spectra

sec:trigonal_spectra

TODO: This section also needs a lot of work, there are about half a dozen or so ways of building up the connection between the Littlewood-Richardson coefficients and the trigonal Horn problem. In the end, the approach we need to take here should mirror the approach in [Chr08] as closely as possible because it is this approach that we make use of when generalizing from the trigonal to the tetrahedral Horn problem. For example, one common strategy for simplifying the presentation is to shift from the asymmetric variant of the Horn problem where $A + B = C$ to the symmetric variant of the Horn problem where $A + B + C = 0$. While this makes things conceptually much clearer, when generalizing to the tetrahedral Horn problem, the more “symmetric” approach becomes a slight obstacle and forces one to consider artificial matching conditions of the form $C + C' = 0$ and therefore to consider a total of twelve matrices.

It is also worth mentioning that one can describe the Horn inequalities themselves in terms of the LR coefficients. This discussion should be included in Section 3.1.

Our first result can be viewed as a simple condition which is sufficient to conclude that a given triple of integer eigenvalues are trigonal.

Lemma 40. *If $c_{\alpha,\beta}^{\gamma} > 0$, then there exists Hermitian matrices $A \in \mathcal{O}_{\alpha}$, $B \in \mathcal{O}_{\beta}$, and $C \in \mathcal{O}_{\gamma}$, satisfying $A + B - C = 0$.*

Proof. Our proof proceeds in the style of [BVW18, Proposition 2.3]. Given any dominant weight $\lambda \in \mathbb{Z}_{\text{dec}}^n$, let P_{λ} denote the Hermitian projection operator onto the invariant □

Using the theory of spectrum estimation developed in the context of quantum information theory [ARS88; KW01; HM02], the following pair of theorems were proven in [Chr08]. Recall that $(\alpha, \beta, \gamma) \in (\mathbb{R}_{\text{dec}}^n)^{\times 3}$ are said to be *trigonal* if there exists Hermitian matrices $A, B, C \in \mathcal{H}(n)$ satisfying (i) $A + B + C = 0$ and (ii)

$$(\text{eig}(A), \text{eig}(B), \text{eig}(C)) = (\alpha, \beta, \gamma). \quad (125)$$

thm:LR_pos_implies_herm

Theorem 41. *If $c_{\alpha,\beta,\gamma} > 0$ for some $\alpha, \beta, \gamma \in \mathbb{Z}_{\text{dec}}^n$, then (α, β, γ) are trigonal.*

Theorem 42. *If $(\alpha, \beta, \gamma) \in (\mathbb{R}_{\text{dec}}^n)^{\times 3}$ are triangular, then there exists a graded sequence $(\tilde{\alpha}(n), \tilde{\beta}(n), \tilde{\gamma}(n)) \in (\mathbb{Z}_{\text{dec}}^n)^{\times 3}$ such that (i) for all $n \in \mathbb{N}_{>0}$,*

$$c_{\tilde{\alpha}(n), \tilde{\beta}(n), \tilde{\gamma}(n)} > 0, \quad (126)$$

and (ii)

$$\lim_{n \rightarrow \infty} \frac{\tilde{\alpha}(n)}{n} = \alpha, \quad \lim_{n \rightarrow \infty} \frac{\tilde{\beta}(n)}{n} = \beta, \quad \lim_{n \rightarrow \infty} \frac{\tilde{\gamma}(n)}{n} = \gamma. \quad (127)$$

prop:kempf-ness-abc

Proposition 43 (Kempf-Ness). *Let $(A, B, C) \in \mathcal{O}_{\alpha} \times \mathcal{O}_{\beta} \times \mathcal{O}_{\gamma}$ be a triple of $d \times d$ Hermitian matrices with integer-valued eigenvalues and let*

$$w_{A,B,C} := \varphi(A) \otimes \varphi(B) \otimes \varphi(C) = v_{\alpha}^{u_A} \otimes v_{\beta}^{u_B} \otimes v_{\gamma}^{u_C} \quad (128)$$

be the associated coherent state of unit norm in $V_\alpha \otimes V_\beta \otimes V_\gamma$. Define the capacity of (A, B, C) as

$$\text{cap}(A, B, C) := \inf_{g \in \text{GL}(n)} \left\| [\pi_\alpha(g) \otimes \pi_\beta(g) \otimes \pi_\gamma(g)] w_{A,B,C} \right\|^2, \quad (129)$$

$$= \inf_{g \in \text{GL}(n)} \left\| v_\alpha^{gu_A} \otimes v_\beta^{gu_B} \otimes v_\gamma^{gu_C} \right\|^2. \quad (130)$$

Then $\text{cap}(A, B, C) \in [0, 1]$ with

$$\text{cap}(A, B, C) = 1 \iff \mu(A, B, C) = A + B + C = 0. \quad (131)$$

Proof. Summarize the direct and pedagogical Kempf-Ness style proof in [BVW18, Proposition 2.3]. $\square$

The following is a combination of [FW20, Theorem 1 & Proposition 3.1] (using the convention where capacities and constant portions are defined in terms of squared norms instead of norms directly).

Theorem 44 (Strong-duality & Occasionality). *Let everything be as in Proposition 43 and let k be a positive integer. Define the constant portion of A, B, C of degree k to be*

$$\text{cst}_k(A, B, C) = \left\| \Gamma_{\alpha, \beta, \gamma}^k w_{A, B, C}^{\otimes k} \right\|^{\frac{2}{k}}, \quad (132)$$

where $\Gamma_{\alpha, \beta, \gamma}^k$ is the projection operator onto the subspace of $\text{GL}(n)$ -invariant vectors in $(V_\alpha \otimes V_\beta \otimes V_\gamma)^{\otimes k}$. Then {eq:strong-duality}

$$\text{cap}(A, B, C) = \sup_k \text{cst}_k(A, B, C) \quad (133)$$

Moreover, when $\text{cap}(A, B, C) = 1$ and $\mu(A, B, C) = 0$, there exists an integer $m > 0$ such that {eq:occasionality}

$$\lim_{k \rightarrow \infty} (mk)^s \text{cst}_{mk}(A, B, C) > 0 \quad (134)$$

where $s = \dim U(n) - \dim U(n)_{[w_{A, B, C}]}$.

Corollary 45. Consider the quantity $\tilde{c}_{\alpha, \beta, \gamma}$ which measures the fraction of vectors $V_\alpha \otimes V_\beta \otimes V_\gamma$ which are $\text{GL}(n)$ -invariant:

$$\tilde{c}_{\alpha, \beta, \gamma} = \frac{\dim[(V_\alpha \otimes V_\beta \otimes V_\gamma)^{\text{GL}(n)}]}{\dim(V_\alpha \otimes V_\beta \otimes V_\gamma)}. \quad (135)$$

If there exists $(A, B, C) \in \mathcal{O}_\alpha \times \mathcal{O}_\beta \times \mathcal{O}_\gamma$ such that $A + B + C = 0$, then there exists an integer $K > 0$ such that $c_{K\alpha, K\beta, K\gamma} \geq 1$ ⁶, and therefore for all integers $M > 0$

$$\tilde{c}_{MK\alpha, MK\beta, MK\gamma} \geq \frac{1}{\dim(V_{MK\alpha} \otimes V_{MK\beta} \otimes V_{MK\gamma})}. \quad (136)$$

On the other hand, if there does not exist $(A, B, C) \in \mathcal{O}_\alpha \times \mathcal{O}_\beta \times \mathcal{O}_\gamma$ such that $A + B + C = 0$, then the function $k \mapsto \tilde{c}_{k\alpha, k\beta, k\gamma}$ admits an exponential upper bound⁷

$$\tilde{c}_{k\alpha, k\beta, k\gamma} \leq \max_{A, B, C} \text{cap}(A, B, C)^k. \quad (137)$$

Proof. The lower-bound follows from the operator inequality

$$|v_\lambda^u\rangle \langle v_\lambda^u| \leq \text{id}_{V_\lambda}, \quad (138)$$

for all unit vectors $v_\lambda^u \in V_\lambda$ while the upper-bound follows from the resolution of identity over the normalized Haar measure du for $U(n)$: {eq:resolution}

$$\text{id}_{V_\lambda} = \dim(V_\lambda) \int_{u \in U(n)} du |v_\lambda^u\rangle \langle v_\lambda^u|. \quad (139)$$

⁶In fact the proof of the saturation conjecture implies $K = 1$ always suffices.

⁷The semigroup property of Littlewood-Richardson coefficients together with the at most polynomial growth in $\dim(V_{k\lambda})$ combined with the exponential upper bound on $\tilde{c}_{k\alpha, k\beta, k\gamma}$ actually implies $c_{\alpha, \beta, \gamma} = 0$ whenever $A + B + C = 0$.

Applying Eq. (139) three times to the definition of $\tilde{c}_{k\alpha,k\beta,k\gamma}$ yields

$$\begin{aligned}\tilde{c}_{k\alpha,k\beta,k\gamma} &= \frac{\dim[(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})^{\text{GL}(n)}]}{\dim(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})} \quad \text{\texttt{\{eq:tilde_c_k_integral\}}} \\ &= \int du_A du_B du_C \langle (v_{k\alpha}^{u_A} \otimes v_{k\beta}^{u_B} \otimes v_{k\gamma}^{u_C}), \Gamma_{k\alpha,k\beta,k\gamma}^1(v_{k\alpha}^{u_A} \otimes v_{k\beta}^{u_B} \otimes v_{k\gamma}^{u_C}) \rangle. \quad (140)\end{aligned}$$

While the operator $\Gamma_{k\alpha,k\beta,k\gamma}^1$ appearing in Eq. (140) above projects onto the subspace of $\text{GL}(n)$ -invariant vectors in $V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma}$, the operator $\Gamma_{\alpha,\beta,\gamma}^k$ projects onto the subspace of $\text{GL}(n)$ -invariant vectors in $(V_\alpha \otimes V_\beta \otimes V_\gamma)^{\otimes k}$. Since $V_{k\lambda}$ always embeds isometrically and equivariantly into $V_\lambda^{\otimes k}$ for any λ , we have that $V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma}$ is a subspace of $(V_\alpha \otimes V_\beta \otimes V_\gamma)^{\otimes k}$ with respect to a $\text{GL}(n)$ -equivariant isometry and therefore the projection operators satisfy

$$\Gamma_{k\alpha,k\beta,k\gamma}^1 = (\iota_\alpha^k \otimes \iota_\beta^k \otimes \iota_\gamma^k)^\dagger \Gamma_{\alpha,\beta,\gamma}^k (\iota_\alpha^k \otimes \iota_\beta^k \otimes \iota_\gamma^k). \quad (141)$$

Moreover, using the $\text{GL}(n)$ -equivariance of $\Gamma_{\alpha,\beta,\gamma}^k$ and the inequality $\Gamma_{\alpha,\beta,\gamma}^k \leq (\text{id}_{V_\alpha \otimes V_\beta \otimes V_\gamma})^{\otimes k}$ yields, for all $g \in \text{GL}(n)$, an upper bound on the integrand in Eq. (140):

$$\langle (v_{k\alpha}^{u_A} \otimes v_{k\beta}^{u_B} \otimes v_{k\gamma}^{u_C}), \Gamma_{k\alpha,k\beta,k\gamma}^1(v_{k\alpha}^{u_A} \otimes v_{k\beta}^{u_B} \otimes v_{k\gamma}^{u_C}) \rangle \quad (142)$$

$$= \langle (v_\alpha^{u_A} \otimes v_\beta^{u_B} \otimes v_\gamma^{u_C})^{\otimes k}, \Gamma_{\alpha,\beta,\gamma}^k(v_\alpha^{u_A} \otimes v_\beta^{u_B} \otimes v_\gamma^{u_C})^{\otimes k} \rangle, \quad (143)$$

$$= \langle (v_\alpha^{gu_A} \otimes v_\beta^{gu_B} \otimes v_\gamma^{gu_C})^{\otimes k}, \Gamma_{\alpha,\beta,\gamma}^k(v_\alpha^{gu_A} \otimes v_\beta^{gu_B} \otimes v_\gamma^{gu_C})^{\otimes k} \rangle, \quad (144)$$

$$\leq \langle (v_\alpha^{gu_A} \otimes v_\beta^{gu_B} \otimes v_\gamma^{gu_C})^{\otimes k}, (v_\alpha^{gu_A} \otimes v_\beta^{gu_B} \otimes v_\gamma^{gu_C})^{\otimes k} \rangle, \quad (145)$$

$$= \left\| v_\alpha^{gu_A} \otimes v_\beta^{gu_B} \otimes v_\gamma^{gu_C} \right\|^{2k}. \quad (146)$$

Finally, let (u'_A, u'_B, u'_C) be unitaries which achieve the maximum of the integrand appearing in Eq. (140) (continuity of the integrand and compactness of $\text{U}(n)$ guarantees existence) such that $\tilde{c}_{k\alpha,k\beta,k\gamma}$ admits the bound

$$\tilde{c}_{k\alpha,k\beta,k\gamma} \leq \left\| v_\alpha^{gu'_A} \otimes v_\beta^{gu'_B} \otimes v_\gamma^{gu'_C} \right\|^{2k}. \quad (147)$$

Since $g \in \text{GL}(n)$ is arbitrary and (u'_A, u'_B, u'_C) is independent of g by definition one obtains

$$\tilde{c}_{k\alpha,k\beta,k\gamma} \leq \inf_{g \in \text{GL}(n)} \left\| v_\alpha^{gu'_A} \otimes v_\beta^{gu'_B} \otimes v_\gamma^{gu'_C} \right\|^{2k} = \text{cap}(A', B', C')^k \leq \max_{A,B,C} \text{cap}(A, B, C)^k. \quad (148)$$

TODO: Prove that the capacity is attained by some Hermitian matrices A, B, C (maybe using the quantitative Kempf-Ness theorem [Bür+19, Theorem 1.17]. This establishes the claimed upper bound. To prove the claimed lower bound, let $v_\alpha^{u'_A} \otimes v_\beta^{u'_B} \otimes v_\gamma^{u'_C}$ be the coherent state associated with the Hermitian matrices (A, B, C) satisfying $A + B + C = 0$. Then using Eq. (138) and the positive demidefiniteness of $\Gamma_{k\alpha,k\beta,k\gamma}^1$ we obtain,

$$\begin{aligned}\tilde{c}_{k\alpha,k\beta,k\gamma} &= \frac{\dim[(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})^{\text{GL}(n)}]}{\dim(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})} \\ &\geq \frac{\langle (v_{k\alpha}^{u'_A} \otimes v_{k\beta}^{u'_B} \otimes v_{k\gamma}^{u'_C}), \Gamma_{k\alpha,k\beta,k\gamma}^1(v_{k\alpha}^{u'_A} \otimes v_{k\beta}^{u'_B} \otimes v_{k\gamma}^{u'_C}) \rangle}{\dim(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})}, \quad \text{\texttt{\{eq:tilde_c_k_inclusivity\}}} \\ &= \frac{\langle (v_\alpha^{u'_A} \otimes v_\beta^{u'_B} \otimes v_\gamma^{u'_C})^{\otimes k}, \Gamma_{\alpha,\beta,\gamma}^k(v_\alpha^{u'_A} \otimes v_\beta^{u'_B} \otimes v_\gamma^{u'_C})^{\otimes k} \rangle}{\dim(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})}, \quad (149) \\ &= \frac{\text{cst}_k(A, B, C)}{\dim(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})}.\end{aligned}$$

Now Eq. (134) implies there exists positive integers $m, s > 0$ such that for sufficiently large k , $(mk)^s \text{cst}_{mk}(A, B, C) > 0$ which implies $c_{mk\alpha,mk\beta,mk\gamma} \neq 0$ and thus $c_{mk\alpha,mk\beta,mk\gamma} \geq 1$. Taking $K = mk$ for sufficiently large k proves the claim. $\square$

4.4 Schur-Weyl presentation of the $6j$ -operator

lem:schur_weyl_presentation

Lemma 46. *Let x, y, z arbitrary positive integers and let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be Young diagrams of the following sizes:*

$$\begin{aligned} \alpha \vdash_n x, & \quad \beta \vdash_n y, & \quad \gamma \vdash_n x + y, \\ \delta \vdash_n z, & \quad \epsilon \vdash_n x + y + z, & \quad \phi \vdash_n y + z, \end{aligned} \quad \text{\textit{feq:xyz-sized-young-diagrams}} \quad (150)$$

and define corresponding projection operators acting on $(\mathbb{C}^n)^{\otimes x+y+z}$ by

$$\begin{aligned} \Pi_\alpha &:= P_\alpha \otimes \text{id}^{\otimes y} \otimes \text{id}^{\otimes z}, & \Pi_\beta &:= \text{id}^{\otimes x} \otimes P_\beta \otimes \text{id}^{\otimes z}, & \Pi_\gamma &:= P_\gamma \otimes \text{id}^{\otimes z}, \\ \Pi_\delta &:= \text{id}^{\otimes x} \otimes \text{id}^{\otimes y} \otimes P_\delta, & \Pi_\epsilon &:= P_\epsilon, & \Pi_\phi &:= \text{id}^{\otimes x} \otimes P_\phi. \end{aligned} \quad \text{\textit{feq:extended-projs}} \quad (151)$$

Then every pair of projection operators in the set $\{\Pi_\alpha, \Pi_\beta, \Pi_\gamma, \Pi_\delta, \Pi_\epsilon, \Pi_\phi\}$ is commuting with only possible exception being Π_γ and Π_ϕ . Then $\Pi_{\alpha\beta\delta\epsilon}^\gamma$ and $\Pi_{\alpha\beta\delta\epsilon}^\phi$, defined by

$$\begin{aligned} \Pi_{\alpha\beta\delta\epsilon}^\gamma &= \Pi_\gamma \Pi_\alpha \Pi_\beta \Pi_\delta \Pi_\epsilon, \\ \Pi_{\alpha\beta\delta\epsilon}^\phi &= \Pi_\phi \Pi_\alpha \Pi_\beta \Pi_\delta \Pi_\epsilon. \end{aligned} \quad \text{\textit{feq:commuting-extended-projs}} \quad (152)$$

are projection operators. Moreover, then $6j$ -operator $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\}$ has operator norm satisfying

$$\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_\infty = \left\| \Pi_{\alpha\beta\delta\epsilon}^\phi \Pi_{\alpha\beta\delta\epsilon}^\gamma \right\|_\infty. \quad \text{\textit{feq:ud-sixj-prod-proj-operator-norm}} \quad (153)$$

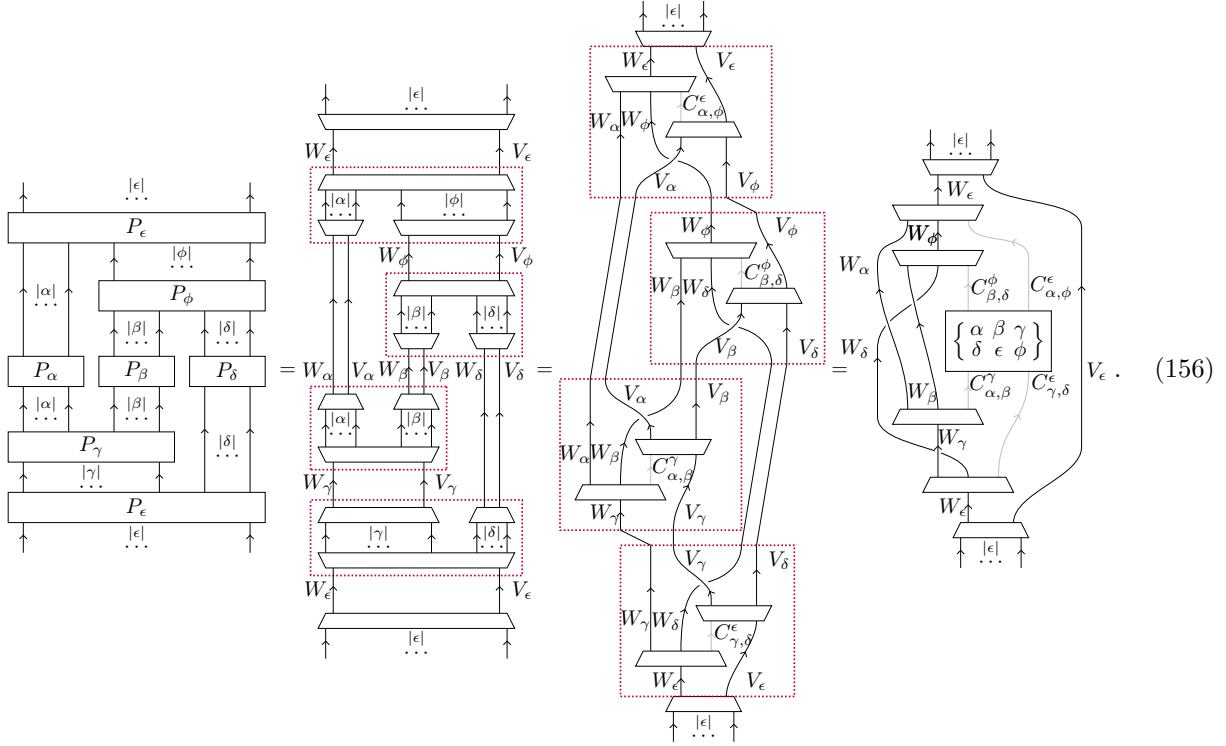
and Hilbert-Schmidt norm satisfying

$$\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_{\text{HS}}^2 = \frac{\text{Tr}(\Pi_{\alpha\beta\delta\epsilon}^\phi \Pi_{\alpha\beta\delta\epsilon}^\gamma)}{\dim(V_\epsilon^n) \dim(W_\alpha) \dim(W_\beta) \dim(W_\delta)}. \quad \text{\textit{feq:ud-sixj-prod-proj-hs-norm}} \quad (154)$$

Proof. That all projection operators except the pair (Π_γ, Π_ϕ) commute follows from elementary applications of Schur's lemma; the obstruction for the pair (Π_γ, Π_ϕ) is simply that Π_γ is not necessarily $\text{SU}(n)$ -equivariant with respect to the diagonal action on $\mathbb{C}^{\otimes y+z}$ and likewise Π_ϕ is not necessarily $\text{SU}(n)$ -equivariant with respect to the diagonal on $\mathbb{C}^{\otimes x+y}$. To prove both Eq. (153) and Eq. (154), it suffices to show there exists appropriately-typed isometries ι_I and ι_{II} such that

$$\Pi_{\alpha\beta\delta\epsilon}^\phi \Pi_{\alpha\beta\delta\epsilon}^\gamma = \iota_{II} \left(\text{id}_{V_\epsilon^n} \otimes \text{id}_{W_\alpha} \otimes \text{id}_{W_\beta} \otimes \text{id}_{W_\delta} \otimes \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right) \iota_I^\dagger \quad \text{\textit{feq:ud-sixj-prod-proj-iso}} \quad (155)$$

This follows from the follow diagrammatic calculation **TODO: Explain this.:**



defn:Q_quantity $\square$

Definition 47. Let (A, B, C, D, E, F) be a tetrahedral sextuple of non-zero positive semidefinite $n \times n$ Hermitian matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$. Let x, y, z be arbitrary positive integers and define

$$\Omega := \frac{A^{\otimes x} \otimes B^{\otimes y} \otimes D^{\otimes z}}{|\alpha|^x |\beta|^y |\delta|^z}. \quad (157)$$

Let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be Young diagrams with sizes determined by Eq. (150), and let $\Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}$ and $\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'}$ be defined according to Eq. (152). Define $Q_{A,B,C,D,E,F}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} \in \mathbb{C}$ by

$$Q_{A,B,C,D,E,F}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} := \text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} \Omega \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}). \quad (158)$$

Proposition 48. Let everything be as in Definition 47. Then

$$Q_{A,B,C,D,E,F}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} = \text{Tr}(\Pi_{\alpha'} \Pi_{\beta'} \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Pi_{\phi'} \Omega), \quad (159)$$

and therefore

$$1 = \text{Tr}(\Omega) = \sum_{\alpha' \vdash dx} \sum_{\beta' \vdash dy} \sum_{\gamma' \vdash dx+y} \sum_{\delta' \vdash dz} \sum_{\epsilon' \vdash dy+z} \sum_{\phi' \vdash dx+y+z} Q_{A,B,C,D,E,F}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} \quad (160)$$

Moreover, the magnitude of $Q_{A,B,C,D,E,F}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'}$ can be upper bounded by either

$$\left| Q_{A,B,C,D,E,F}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} \right| \leq (\text{Tr}(\Pi_{\alpha'} \Omega) \text{Tr}(\Pi_{\beta'} \Omega) \text{Tr}(\Pi_{\gamma'} \Omega) \text{Tr}(\Pi_{\delta'} \Omega) \text{Tr}(\Pi_{\epsilon'} \Omega) \text{Tr}(\Pi_{\phi'} \Omega))^{\frac{1}{6}}. \quad (161)$$

or

$$\left| Q_{A,B,C,D,E,F}^{\alpha'\beta'\gamma'\delta'\epsilon'\phi'} \right| \leq \left\| \Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'} \right\|_{\infty}. \quad (162)$$

Proof. The proof of Eq. (159) and Eq. (160) follows from elementary properties of the projection operators defined in Lemma 46. Moreover, Eq. (162) follows because Ω is positive semidefinite and trace one. To prove Eq. (161), consider an application of the Cauchy-Schwarz inequality to the Hilbert-Schmidt inner product, $\langle X, Y \rangle = \text{Tr}(X^\dagger Y)$, whenever $X = \Omega^{\frac{1}{2}} \Pi_1$ and $Y = \Omega^{\frac{1}{2}} \Pi_2$ (where Π_1 and Π_2 are generic Hermitian projectors and Ω is positive semidefinite):

$$\begin{aligned} |\text{Tr}(\Pi_1 \Omega \Pi_2)|^2 &= \left| \text{Tr}((\Omega^{\frac{1}{2}} \Pi_1)^\dagger (\Omega^{\frac{1}{2}} \Pi_2)) \right|^2 = \left| \langle \Omega^{\frac{1}{2}} \Pi_1, \Omega^{\frac{1}{2}} \Pi_2 \rangle \right|^2 \\ &\leq \langle \Omega^{\frac{1}{2}} \Pi_1, \Omega^{\frac{1}{2}} \Pi_1 \rangle \langle \Omega^{\frac{1}{2}} \Pi_2, \Omega^{\frac{1}{2}} \Pi_2 \rangle = \text{Tr}(\Pi_1 \Omega) \text{Tr}(\Pi_2 \Omega) \end{aligned} \quad \text{\textcolor{blue}{\texttt{eq:biproj_ineq}}} \quad (163)$$

Letting $\Pi_1 = \Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'}$ and $\Pi_2 = \Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'}$ and applying Eq. (163) yields

$$\begin{aligned} \left| Q_{A,B,C,D,E,F}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} \right|^6 &= |\text{Tr}[(\Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'}) \Omega (\Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'})]|^6 \\ &\leq \text{Tr}(\Pi_{\phi'} \Pi_{\alpha'} \Pi_{\beta'} \Omega)^3 \text{Tr}(\Pi_{\gamma'} \Pi_{\delta'} \Pi_{\epsilon'} \Omega)^3 \\ &\leq \text{Tr}(\Pi_{\phi'} \Omega) \text{Tr}(\Pi_{\alpha'} \Omega) \text{Tr}(\Pi_{\beta'} \Omega) \text{Tr}(\Pi_{\gamma'} \Omega) \text{Tr}(\Pi_{\delta'} \Omega) \text{Tr}(\Pi_{\epsilon'} \Omega). \end{aligned} \quad (164)$$

\text{\textcolor{blue}{\texttt{thm:Q_upper_bound}}} \quad \square

Theorem 49. *Let everything be as in Definition 47 except fix x, y, z to be*

$$x := \lceil k |\alpha| \rceil, \quad y := \lceil k |\beta| \rceil, \quad z := \lceil k |\delta| \rceil, \quad (165)$$

where k is a sufficiently large positive integer ensuring that x, y and z are non-zero. Then

$$\left| Q_{A,B,C,D,E,F}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} \right| \leq p(k) \exp(-k I_{\alpha \beta \gamma \delta \epsilon \phi}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} / 6). \quad \text{\textcolor{blue}{\texttt{eq:Q_upper_bound}}} \quad (166)$$

where $p(k)$ is a polynomial in k and where

$$I_{\alpha \beta \gamma \delta \epsilon \phi}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi'} = |\alpha| H(\bar{\alpha}' | \bar{\alpha}) + |\beta| H(\bar{\beta}' | \bar{\beta}) + |\gamma| H(\bar{\gamma}' | \bar{\gamma}) + |\delta| H(\bar{\delta}' | \bar{\delta}) + |\epsilon| H(\bar{\epsilon}' | \bar{\epsilon}) + |\phi| H(\bar{\phi}' | \bar{\phi}). \quad (167)$$

TODO: Is this relative entropy deviation measure conceptually useful or is it best to relax the deviation measure to its geometric analogue using Pinsker's inequality?

Proof. To begin the proof, we consider x, y and z to be arbitrary positive integers. For each $\lambda' \in \{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$, it is possible to upper-bound $\text{Tr}(\Pi_{\lambda'} \Omega)$ using the spectrum estimation technique Proposition 34. For a detailed example, we will consider the bound on $\text{Tr}(\Pi_{\gamma'} \Omega)$.

$$\text{Tr}(\Pi_{\gamma'} \Omega) = \frac{\text{Tr}[(P_{\gamma'} \otimes \text{id}^{\otimes z})(A^{\otimes x} \otimes B^{\otimes y} \otimes D^{\otimes z})]}{|\alpha|^x |\beta|^y |\delta|^z} = \frac{\text{Tr}[P_{\gamma'}(A^{\otimes x} \otimes B^{\otimes y})] |\delta|^z}{|\alpha|^x |\beta|^y |\delta|^z}. \quad (168)$$

Using the assumption that $C = A + B$, we have $C^{\otimes x+y} = (A + B)^{\otimes x+y}$ which implies

$$\text{Tr}(P_{\gamma'} C^{\otimes x+y}) = \sum_{k=0}^{x+y} \binom{x+y}{k} \text{Tr}(P_{\gamma'} (A^{\otimes k} \otimes B^{\otimes x+y-k})). \quad (169)$$

Here we have used the S_{x+y} -equivariance of $P_{\gamma'}$. Since each of the summands above are non-negative, we conclude for all $0 \leq k \leq x+y$ that

$$\text{Tr}(P_{\gamma'} C^{\otimes x+y}) \geq \binom{x+y}{k} \text{Tr}(P_{\gamma'} (A^{\otimes k} \otimes B^{\otimes x+y-k})). \quad (170)$$

Letting $k = x$ and rearranging yields

$$\text{Tr}(P_{\gamma'} (A^{\otimes x} \otimes B^{\otimes y})) \leq \frac{x! y!}{(x+y)!} \text{Tr}(P_{\gamma'} C^{\otimes x+y}). \quad (171)$$

Furthermore, spectrum estimation (Proposition 34) yields

$$\mathrm{Tr}(\Pi_{\gamma'}\Omega) \leq \frac{1}{|\alpha|^x |\beta|^y} \frac{x!y!}{(x+y)!} \mathrm{Tr}[P_{\gamma'} C^{\otimes x+y}], \quad (172)$$

$$\leq \frac{(|\alpha| + |\beta|)^{x+y}}{|\alpha|^x |\beta|^y} \frac{x!y!}{(x+y)!} \dim(V_{\gamma'}^n) \exp(-(x+y)H(\bar{\gamma}'|\bar{\gamma})). \quad (173)$$

Analogous calculations for $\lambda' \in \{\alpha', \beta', \gamma', \delta', \epsilon', \phi'\}$ produce the following set of upper bounds:

$$\mathrm{Tr}(\Pi_{\alpha'}\Omega) \leq \dim(V_{\alpha'}^n) \exp(-xH(\bar{\alpha}'|\bar{\alpha})), \quad (174)$$

$$\mathrm{Tr}(\Pi_{\beta'}\Omega) \leq \dim(V_{\beta'}^n) \exp(-yH(\bar{\beta}'|\bar{\beta})), \quad (175)$$

$$\mathrm{Tr}(\Pi_{\gamma'}\Omega) \leq \frac{(|\alpha| + |\beta|)^{x+y}}{|\alpha|^x |\beta|^y} \frac{x!y!}{(x+y)!} \dim(V_{\gamma'}^n) \exp(-(x+y)H(\bar{\gamma}'|\bar{\gamma})), \quad (176)$$

$$\mathrm{Tr}(\Pi_{\delta'}\Omega) \leq \dim(V_{\delta'}^n) \exp(-zH(\bar{\delta}'|\bar{\delta})), \quad (177)$$

$$\mathrm{Tr}(\Pi_{\epsilon'}\Omega) \leq \frac{(|\alpha| + |\beta| + |\delta|)^{x+y+z}}{|\alpha|^x |\beta|^y |\delta|^z} \frac{x!y!z!}{(x+y+z)!} \dim(V_{\epsilon'}^n) \exp(-(x+y+z)H(\bar{\epsilon}'|\bar{\epsilon})), \quad (178)$$

$$\mathrm{Tr}(\Pi_{\phi'}\Omega) \leq \frac{(|\beta| + |\delta|)^{y+z}}{|\beta|^y |\delta|^z} \frac{y!z!}{(y+z)!} \dim(V_{\phi'}^n) \exp(-(y+z)H(\bar{\phi}'|\bar{\phi})). \quad (179)$$

Our objective now will be to use the given values for x, y, z to prove the multinomial prefactors appearing in the bounds for γ', ϵ' , and ϕ' reasonably small compared to n . An application of Lemma 92 (when $k = 2$) implies the following bound

$$\frac{(|\alpha| + |\beta|)^{x+y}}{|\alpha|^x |\beta|^y} \frac{x!y!}{(x+y)!} \leq e^4 (n(|\alpha| + |\beta|) + 3)^2. \quad (180)$$

Moreover, using [CM06, Equation 7], we have

$$\dim(V_{\gamma'}^n) \leq (|\gamma'| + 1)^{n(n-1)/2} = (x + y + 1)^{n(n-1)/2} < (n(|\alpha| + |\beta|) + 3)^{n(n-1)/2}. \quad (181)$$

Altogether, we can relax the bound on $\mathrm{Tr}(\Pi_{\gamma'}\Omega)$ to obtain

$$\mathrm{Tr}(\Pi_{\gamma'}\Omega) \leq e^4 (n|\epsilon| + 3)^{\frac{n(n-1)}{2}+2} \exp(-n|\gamma|H(\bar{\gamma}'|\bar{\gamma})). \quad (182)$$

Similar calculations for $\mathrm{Tr}(\Pi_{\epsilon'}\Omega)$ yields (noting that $k = 3$ when applying Lemma 92):

$$\mathrm{Tr}(\Pi_{\epsilon'}\Omega) \leq e^{16} (n|\epsilon| + 4)^{\frac{n(n-1)}{2}+3} \exp(-n|\epsilon|H(\bar{\epsilon}'|\bar{\epsilon})). \quad (183)$$

It is also not too difficult to see that the polynomial in n term associated to the upper bound on $\mathrm{Tr}(\Pi_{\epsilon'}\Omega)$ above is the largest such subexponential term and thus Eq. (166) indeed holds if we take $p(n)$ to be

$$p(n) = e^{16} (n|\epsilon| + 4)^{\frac{n(n-1)}{2}+3}. \quad (184)$$

cor:seq_ud_recoupling_norm_lower_bound $\square$

Corollary 50. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be the eigenvalues of non-zero positive semidefinite tetrahedral Hermitian matrices (A, B, C, D, E, F) . Then there exists a sextuple of sequences of Young diagrams $(\alpha'_n, \beta'_n, \gamma'_n, \delta'_n, \epsilon'_n, \phi'_n)$ of sizes*

$$\begin{aligned} \alpha'_n &\vdash_n \lceil n|\alpha| \rceil, & \beta'_n &\vdash_n \lceil n|\beta| \rceil, & \gamma'_n &\vdash_n \lceil n|\alpha| \rceil + \lceil n|\beta| \rceil, \\ \delta'_n &\vdash_n \lceil n|\delta| \rceil, & \epsilon'_n &\vdash_n \lceil n|\alpha| \rceil + \lceil n|\beta| \rceil + \lceil n|\delta| \rceil, & \phi'_n &\vdash_n \lceil n|\beta| \rceil + \lceil n|\delta| \rceil, \end{aligned} \quad (185)$$

satisfying (i) the convergence conditions

$$\begin{aligned} \lim_{n \rightarrow \infty} \bar{\alpha}'_n &= \bar{\alpha}, & \lim_{n \rightarrow \infty} \bar{\beta}'_n &= \bar{\beta}, & \lim_{n \rightarrow \infty} \bar{\gamma}'_n &= \bar{\gamma}, \\ \lim_{n \rightarrow \infty} \bar{\delta}'_n &= \bar{\delta}, & \lim_{n \rightarrow \infty} \bar{\epsilon}'_n &= \bar{\epsilon}, & \lim_{n \rightarrow \infty} \bar{\phi}'_n &= \bar{\phi}, \end{aligned} \quad (186)$$

and (ii) the lower bound

$$\left\| \left\{ \frac{\alpha'_n}{\delta'_n}, \frac{\beta'_n}{\epsilon'_n}, \frac{\gamma'_n}{\phi'_n} \right\} \right\|_{\infty} \geq \frac{1}{(n|\epsilon| + 4)^{6(d-1)}}. \quad (187)$$

Proof. For each n , it suffices to pick $(\alpha'_n, \beta'_n, \gamma'_n, \delta'_n, \epsilon'_n, \phi'_n)$ be the sextuple which maximizes the value of $|Q_{A,B,C,D,E,F}^{\alpha'_n, \beta'_n, \gamma'_n, \delta'_n, \epsilon'_n, \phi'_n}|$. Such a choice will necessarily satisfy a lower bound due to the completeness relation Eq. (160). Specifically,

$$\left| Q_{A,B,C,D,E,F}^{\alpha'_n, \beta'_n, \gamma'_n, \delta'_n, \epsilon'_n, \phi'_n} \right| \geq \frac{1}{q(n)} \quad \text{\textcolor{blue}{\{eq:Q_lower_bound\}}} \quad (188)$$

where $q(n)$ is any upper bound on the number of possible summands in Eq. (160) viewed as a function of n . As there are at most $(k+1)^{d-1}$ Young diagrams with $n-1$ rows and size at most k , letting

$$q(n) = (n|\epsilon| + 4)^{6(d-1)} \quad (189)$$

ensures Eq. (188) holds because the size of ϵ'_n is at most $k = n|\epsilon| + 3$. Moreover, applying Eq. (166) yields

$$I_{\alpha\beta\gamma\delta\epsilon\phi}^{\alpha'_n, \beta'_n, \gamma'_n, \delta'_n, \epsilon'_n, \phi'_n} \leq \frac{6}{n} (\log q(n) + \log p(n)) \xrightarrow{n \rightarrow \infty} 0 \quad (190)$$

and thus Eq. (186) holds because $H(\bar{\lambda}'|\bar{\lambda}) \geq 0$ with equality if and only if $\bar{\lambda}' = \bar{\lambda}$. Then Eq. (187) follows from Eq. (188) combined with Eq. (162) and Eq. (153). $\square$

Theorem 51. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be non-negative non-zero eigenvalues which are not tetrahedral. Then for all sequences of Young diagrams $(\alpha'_n, \beta'_n, \gamma'_n, \delta'_n, \epsilon'_n, \phi'_n)$ satisfying Eq. (185) and Eq. (186), we have*

$$\left\| \left\{ \begin{array}{ccc} \alpha'_n & \beta'_n & \gamma'_n \\ \delta'_n & \epsilon'_n & \phi'_n \end{array} \right\} \right\|_{\text{HS}}^2 \leq \exp(-n I_*^{\alpha\beta\gamma\delta\epsilon\phi} / 6 + O(\log n)). \quad \text{\textcolor{blue}{\{eq:seq_ud_size_upper_bound\}}} \quad (191)$$

where $I_*^{\alpha\beta\gamma\delta\epsilon\phi} > 0$ quantifies the deviation of $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ from being tetrahedral:

$$I_*^{\alpha\beta\gamma\delta\epsilon\phi} = \min_{\substack{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi} \\ \text{tetrahedral}}} I_{\tilde{\alpha}\tilde{\beta}\tilde{\gamma}\tilde{\delta}\tilde{\epsilon}\tilde{\phi}}^{\alpha\beta\gamma\delta\epsilon\phi}. \quad (192)$$

Proof. Let dX be the normalized Hilbert-Schmidt measure over $n \times n$ density matrices and let k be a positive integer. Then define an operator Γ_k on $(\mathbb{C}^n)^{\otimes k}$ by

$$\Gamma_k := \int dX X^{\otimes k}. \quad (193)$$

As Γ_k is known to be both S_k and $\text{SU}(n)$ equivariant, by Schur's lemma, Γ_k can be expressed as

$$\Gamma_k = \sum_{\lambda' \vdash dk} r_{\lambda'} P_{\lambda'} \quad (194)$$

with coefficients $r_{\lambda'}$ that can be computed [ZS01; Wal14] to obtain

$$r_{\lambda'} = \frac{\dim(V_{\lambda'}^n)}{\dim(W_{\lambda'}) \dim(V_{(k)}^{d^2})}. \quad (195)$$

Now let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be arbitrary Young diagrams with respective sizes $(x, y, x+y, z, x+y+z, y+z)$ where x, y, z are determined by a positive integer n as in Theorem 49. Using Eq. (154), we obtain the formula

$$\left\| \left\{ \begin{array}{ccc} \alpha' & \beta' & \gamma' \\ \delta' & \epsilon' & \phi' \end{array} \right\} \right\|_{\text{HS}}^2 = \frac{\text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'})}{\dim(V_{\epsilon'}^n) \dim(W_{\alpha'}) \dim(W_{\beta'}) \dim(W_{\delta'})} \quad (196)$$

$$= R_{\alpha', \beta', \delta', \epsilon'} \text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'} (\Gamma_x \otimes \Gamma_y \otimes \Gamma_z) \Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}) \quad (197)$$

where the coefficient $R_{\alpha', \beta', \delta', \epsilon'}$ is a ratio of dimensions:

$$R_{\alpha', \beta', \delta', \epsilon'} := \frac{\dim(V_{(x)}^{d^2}) \dim(V_{(y)}^{d^2}) \dim(V_{(z)}^{d^2})}{\dim(V_{\epsilon'}^n) \dim(V_{\alpha'}^n) \dim(V_{\beta'}^n) \dim(V_{\delta'}^n)}. \quad (198)$$

Moreover, by the definition of Γ_k , we have

$$\text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'}(\Gamma_x \otimes \Gamma_y \otimes \Gamma_z)\Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}) \quad (199)$$

$$= \int dX \int dY \int dZ \text{Tr}(\Pi_{\alpha'\beta'\delta'\epsilon'}^{\phi'}(X^{\otimes x} \otimes Y^{\otimes y} \otimes Z^{\otimes z})\Pi_{\alpha'\beta'\delta'\epsilon'}^{\gamma'}), \quad (200)$$

where the integral is taken over a triple of $n \times n$ density matrices X, Y, Z . For each integrand, define a tetrahedral sextuple $(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F})$ as follows

$$\begin{aligned} \tilde{A} &:= |\alpha| X, & \tilde{B} &:= |\beta| Y, & \tilde{C} &:= |\alpha| X + |\beta| Y, \\ \tilde{D} &:= |\delta| Z, & \tilde{E} &:= |\alpha| X + |\beta| Y + |\gamma| Z, & \tilde{F} &:= |\beta| Y + |\delta| Z. \end{aligned} \quad (201)$$

Furthermore, let $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ be the eigenvalues of $(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F})$. In this notation, the integrand above is equal to

$$Q_{\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F}}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi} = \text{Tr}(\Pi_{\alpha' \beta' \delta' \epsilon'}^{\phi'}(X^{\otimes x} \otimes Y^{\otimes y} \otimes Z^{\otimes z})\Pi_{\alpha' \beta' \delta' \epsilon'}^{\gamma'}) \quad (202)$$

Moreover, by Eq. (162) we have

$$\left| Q_{\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}, \tilde{E}, \tilde{F}}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi} \right| \leq p(n) \exp(-n I_{\tilde{\alpha} \tilde{\beta} \tilde{\gamma} \tilde{\delta} \tilde{\epsilon} \tilde{\phi}}^{\alpha' \beta' \gamma' \delta' \epsilon' \phi} / 6). \quad (203)$$

Under the assumption that $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are not tetrahedral, we conclude that

$$(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \neq (\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi}) \quad (204)$$

and therefore $I_{\tilde{\alpha} \tilde{\beta} \tilde{\gamma} \tilde{\delta} \tilde{\epsilon} \tilde{\phi}}^{\alpha \beta \gamma \delta \epsilon \phi} > 0$. **TODO: Figure out how to describe the convergence, is there a triangle inequality for relative entropies? Maybe we need to relax the rate function I to its geometric analogue using Pinsker's inequality, then everything works.** $\square$

4.5 The asymptotics of irrep dimensions

It will be useful for what follows to better understand how the dimension, $\dim(V_\lambda^n)$, of the irreducible representation V_λ^n of $\text{SL}(n)$ grows as n becomes much larger $|\lambda|$ while $|\lambda|$ itself is much larger than $\ell(\lambda)$. Itzykson and Nauenberg provide the following estimate for $\dim(V_\lambda^n)$ [IN66].

lem:in_estimate

Lemma 52. *Let λ be an positive integer partition of length $\ell := \ell(\lambda)$, let $n \geq \ell$ be a positive integer and let $\rho^d := (d-1, d-2, \dots, 2, 1)$ be the Weyl-vector for $\text{SL}(n)$. The following approximation holds when $n \gg |\lambda|^2$ and $\lambda_\ell \gg \ell$:*

$$\dim(V_\lambda^n) = \dim(V_\lambda^\ell) \times \prod_{i=1}^{\ell} \frac{\rho_i^\ell!}{(\lambda_i + \rho_i^\ell)!} \times d^{|\lambda|} \times \left(1 + O\left(\frac{|\lambda|^2}{n}\right) \right). \quad (205) \quad \text{\texttt{eq:in_estimate}}$$

Proof. Weyl's character formula can be used to derive the following formula for $\dim(V_\lambda^n)$ in terms of the Young diagram λ and the dimension n [IN66]:

$$\dim(V_\lambda^n) = \frac{\Delta(\lambda + \rho^n)}{\Delta(\rho^n)} = \prod_{1 \leq i < j \leq d} \frac{(\lambda_i + \rho_i^d) - (\lambda_j + \rho_j^d)}{\rho_i^d - \rho_j^d} = \prod_{1 \leq i < j \leq d} \frac{(\lambda_i - i) - (\lambda_j - j)}{j - i}, \quad (206)$$

where $\Delta(x) = \prod_{i < j} (x_i - x_j)$ is the Vandermonde determinant. Isolating the terms above which depend on n from those that do not yields

$$\dim(V_\lambda^n) = \dim(V_\lambda^\ell) \times \prod_{i=1}^{\ell} \frac{\rho_i^\ell!}{(\lambda_i + \rho_i^\ell)!} \times \prod_{i=1}^{\ell} \frac{(\lambda_i + \rho_i^n)!}{\rho_i^n!}. \quad (207) \quad \text{\texttt{eq:dim_separation_large_d}}$$

Given positive integers a and b such that $b \gg a^2$, one obtains the following approximation:

$$\begin{aligned} \log \left(\frac{(a+b)!}{b!} \right) &= \log \prod_{j=1}^a (b+j) = \sum_{j=1}^a \log(b+j) \\ &= a \log b + \sum_{j=1}^a \log \left(1 + \frac{j}{b} \right) = a \log b + O \left(\frac{a^2}{b} \right). \end{aligned} \quad (208)$$

The assumption that $n \gg |\lambda|^2$ and $i \leq \ell \ll \lambda_\ell \leq |\lambda|$ also implies $n - i = \rho_i^n \gg \lambda_i^2$. Therefore,

$$\log \left(\frac{(\lambda_i + \rho_i^n)!}{\rho_i^n!} \right) = \lambda_i \log \rho_i^n + O \left(\frac{\lambda_i^2}{\rho_i^n} \right). \quad (209)$$

The assumptions also imply the following simplification

$$\log \left(\frac{(\lambda_i + \rho_i^n)!}{\rho_i^n!} \right) = \lambda_i \log n + O \left(\frac{|\lambda|^2}{n} \right) \quad (210)$$

which, when applied to Eq. (207), proves the claim. $\square$

Using Lemma 52, it can be seen that for fixed λ , $\dim(V_\lambda^n)$ generally grows polynomially in n with leading exponent $|\lambda|$ and leading coefficient depending only on λ . The objective of the following lemma is to characterize the asymptotics of this leading coefficient for large $|\lambda|$ and even large n . **TODO: Figure out a elegant way to present these “double asymptotics”, maybe just epsilonifying everything, like $\max\{\ell/|\lambda|, |\lambda|/d\} \leq \epsilon$.** lem:double_asymptotics

Lemma 53. *Let everything be as in Lemma 52. Then*

$$\log \dim(V_\lambda^n) = |\lambda| + |\lambda| \log(n) + s(\lambda) + O(\ell^2 \log |\lambda|) + O \left(\frac{|\lambda|^2}{n} \right) \quad (211)$$

where $s(\lambda)$ is defined by

$$s(\lambda) := - \sum_{i=1}^{\ell(\lambda)} \lambda_i \log \lambda_i. \quad \text{\textcolor{blue}{(eq:unnorm_shannon)}} \quad (212)$$

Proof. The proof proceeds by considering the multiplicands in Eq. (205) separately. First, we upper bound the dimension $\dim(V_\lambda^\ell)$; for indices $i < j \leq \ell$, one has $(\lambda_i - i) - (\lambda_j - j) \leq |\lambda| + 1$ as well as $j - i \geq 1$ which implies

$$\dim(V_\lambda^\ell) = \prod_{1 \leq i < j \leq \ell} \frac{(\lambda_i - i) - (\lambda_j - j)}{j - i} \leq (|\lambda| + 1)^{\ell(\ell-1)/2}, \quad \text{\textcolor{blue}{(eq:dim_lambda_ell_bound)}} \quad (213)$$

and thus $\log \dim(V_\lambda^\ell) = O(\ell^2 \log |\lambda|)$.

Second, we apply Stirling's approximation [Rob55] to the second term in Eq. (205) under the given assumption that $\lambda_\ell \gg \ell$ (which implies $\rho_i^\ell \ll \lambda_i$),

$$\log \left(\prod_{i=1}^{\ell} \frac{\rho_i^\ell!}{(\lambda_i + \rho_i^\ell)!} \right) = \sum_{i=1}^{\ell} \log \rho_i^\ell! - \sum_{i=1}^{\ell} \log(\lambda_i + \rho_i^\ell)!, \quad (214)$$

$$= |\lambda| + \sum_{i=1}^{\ell} (\rho_i^\ell \log \rho_i^\ell - (\lambda_i + \rho_i^\ell) \log(\lambda_i + \rho_i^\ell) + O(\log \lambda_i)), \quad (215)$$

$$= |\lambda| - \sum_{i=1}^{\ell} \lambda_i \log \lambda_i + O(\ell \log |\lambda|), \quad (216)$$

$$\text{\textcolor{blue}{(eq:unnorm_shannon_term)}} \quad (217)$$

Combining Eq. (213) and Eq. (217) with Lemma 52 proves the claim. $\square$

Remark 54. Note that the quantity $s(\lambda)$ defined in Eq. (212) resembles the Shannon entropy,

$$S(p) = - \sum_{i=1}^{\ell} p_i \log p_i \quad (218)$$

of a probability distribution $p = (p_1, \dots, p_{\ell})$. Indeed, $s(\lambda)$ is related to the Shannon entropy of the *normalization* of λ by the formula

$$s(\lambda) = |\lambda| S\left(\frac{\lambda}{|\lambda|}\right) - |\lambda| \log |\lambda|. \quad \text{\textcolor{blue}{\small{eq:little_s_relation}}} \quad (219)$$

TODO: Perhaps a more direct way of relating the asymptotics of $\dim(V_{\lambda}^n)$ to $\dim(W_{\lambda})$ can be found in Harrow's paper [Har23, Lemma 2]:

Lemma 55. *Let λ be a Young diagram with at most n parts. Then*

$$\frac{|\lambda|!}{d^{|\lambda|}} \frac{\dim(V_{\lambda}^n)}{\dim(W_{\lambda})} = \prod_{(i,j) \in \lambda} \left(1 + \frac{j-i}{n}\right) \quad (220)$$

4.6 An entropic inequality for tetrahedral spectra

thm:ud_recouple_bound

Theorem 56. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \in \mathbb{Z}_{\text{dec}}^n$ be a sextuple of dominant weights for $\text{GL}(n)$ and let $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\}$ be the corresponding $\text{GL}(n)$ -recoupling operator. Then*

$$\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_{HS}^2 \leq \left(c_{\alpha, \phi}^{\epsilon} c_{\beta, \delta}^{\phi} c_{\alpha, \beta}^{\gamma} c_{\gamma, \delta}^{\epsilon} \right)^{\frac{1}{2}} \frac{\dim(V_{\gamma}) \dim(V_{\phi})}{\dim(V_{\beta}) \dim(V_{\epsilon})} \quad (221)$$

Proof. TODO: Explain what this all means carefully. Applying both the Cauchy-Schwarz inequality with

respect to the contraction annotated below, and Schur's lemma multiple times yields the following inequality:

$$\begin{aligned}
& \left(\epsilon \right)^2 C_{\alpha,\beta}^\gamma C_{\alpha,\phi}^\epsilon C_{\beta,\delta}^\phi C_{\gamma,\delta}^\epsilon = \\
& \left[\text{Diagram with two squares and Young diagrams} \right] = \\
& \leq \left[\text{Diagram with two columns of four Young diagrams} \right]^{\frac{1}{2}} = \frac{1}{(\beta \epsilon)} \left[\text{Diagram with two columns of four Young diagrams} \right]^{\frac{1}{2}} \\
& = \frac{\left(\epsilon \right)^2 \left(\gamma \right) \left(\phi \right)}{\left(\beta \right) \left(\epsilon \right)} \left[C_{\alpha,\phi}^\epsilon C_{\beta,\delta}^\phi C_{\alpha,\beta}^\gamma C_{\gamma,\delta}^\epsilon \right]^{\frac{1}{2}}
\end{aligned} \tag{222}$$

lem:asymptotic_dimension_ratio $\square$

Lemma 57. Let $\gamma, \phi, \beta, \epsilon$ be a quadruple of Young diagrams such that

$$|\gamma| + |\phi| = |\beta| + |\epsilon| \tag{223}$$

and that $|\epsilon| = \max\{|\gamma|, |\phi|, |\beta|, |\epsilon|\}$. Let $\ell := \max\{\ell(\gamma), \ell(\phi), \ell(\beta), \ell(\epsilon)\}$ and let n be a positive integer such that $n \geq \ell$. Then

$$\log \left(\frac{\dim(V_\gamma^n) \dim(V_\phi^n)}{\dim(V_\beta^n) \dim(V_\epsilon^n)} \right) = I(\gamma, \phi; \beta, \epsilon) + O(\ell^2 \log |\epsilon|) + O \left(\frac{\epsilon^2}{n} \right) \tag{224}$$

where $I(\gamma, \phi; \beta, \epsilon)$ is the function

$$I(\gamma, \phi; \beta, \epsilon) := s(\gamma) + s(\phi) - s(\beta) - s(\epsilon). \tag{225}$$

Moreover, $I(k\gamma, k\phi; k\beta, k\epsilon) = kI(\gamma, \phi; \beta, \epsilon)$ for any $k > 0$.

Proof. The proof follows from four applications of Lemma 53. The assumption that the sizes of the Young diagrams are balanced in the sense of Eq. (223) implies that the $\log(n)$ terms appearing in Lemma 53 cancel each other and one obtains Eq. (224).

Four applications of Eq. (219) yields

$$\begin{aligned}
I(\gamma, \phi; \beta, \epsilon) &= s(\gamma) + s(\phi) - s(\beta) - s(\epsilon) \\
&= \{|\gamma| S(\bar{\gamma}) + |\phi| S(\bar{\phi}) - |\beta| S(\bar{\beta}) - |\epsilon| S(\bar{\epsilon})\} + \cdots \\
&\quad \cdots - \{|\gamma| \log |\gamma| + |\phi| \log |\phi| - |\beta| \log |\beta| - |\epsilon| \log |\epsilon|\}.
\end{aligned} \tag{226}$$

Under scaling $\lambda \mapsto k\lambda$ for any $k > 0$, observe that $\overline{k\lambda} = \bar{\lambda}$ and $|k\lambda| = k|\lambda|$. Therefore the above expression implies

$$I(k\gamma, k\phi; k\beta, k\epsilon) = kI(\gamma, \phi; \beta, \epsilon) + (|\gamma| + |\phi| - |\beta| - |\epsilon|)k \log k. \tag{227}$$

Since Eq. (223) is assumed to hold, the coefficient in front of the $k \log k$ term appearing in Eq. (227) vanishes and therefore $I(k\gamma, k\phi; k\beta, k\epsilon) = kI(\gamma, \phi; \beta, \epsilon)$ as claimed. $\square$

Theorem 58. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be the eigenvalues of Hermitian matrices (A, B, C, D, E, F) satisfying Eq. (3). Then necessarily $|\beta| + |\epsilon| = |\gamma| + |\phi|$ and furthermore*

$$I(\gamma, \phi; \beta, \epsilon) \geq 0. \tag{228}$$

Proof. The proof follows from asymptotics given by Lemma 57 in the limit as $n \rightarrow \infty$, the upper bound provided by Theorem 56, and the lower bound provided by Corollary 50. **TODO: Finish this** $\square$

5 The additive quantum entropy cone

The purpose of this section is to introduce the notion of an “additive” quantum entropy cone which may be viewed as an analogue to the ordinary quantum entropy cone considered by Pippenger [Pip03]. Formally, quantum states ρ_{ABC} and their reduced states (ρ_{AB} , etc.) are replaced by tuples of positive semidefinite operators (X, Y, Z) and their partial sums $(X + Y, \text{etc.})$. Moreover the von Neumann entropy of each reduced state is replaced by an unnormalized entropy of each partial sum. We prove that the set of all such “allocations” of unnormalized entropies of partial sums forms a convex cone, which we term the “additive quantum entropy cone”. Moreover, we prove that all balanced inequalities which are non-negative over the ordinary quantum entropy cone, such as subadditivity and strong subadditivity, are also non-negative over the additive quantum entropy cone.

In Section 5.1 we introduce our notion of an unnormalized entropy function of PSD matrices and delineate its properties. In Section 5.2, we state our main theorem (Theorem 68) which states that the unnormalized entropy function is strongly subadditive. Then, in Section 5.3 we consider the set of possible unnormalized entropies of partial sums and prove it forms a convex cone. In Section 5.4, we relate the additive quantum entropy cone to the ordinary quantum entropy cone and prove that positive and balanced functionals for the ordinary cone are also positive and balanced functionals for the additive cone, thus providing an alternative proof of Theorem 68. In Section 5.5, we explore how these inequalities related to the concavity of von Neumann entropy.

5.1 The shifted unnormalized entropy function

sec:unent

The primary quantity of interest will be a real-valued function of eigenvalues which resembles von Neumann entropy.

Definition 59. Let $\eta : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ be defined by

$$\eta(x) = \begin{cases} -x \log x & x > 0, \\ 0 & x = 0, \end{cases} \quad (229)$$

where $\log$ is always taken to be the natural logarithm.

Remark 60. For ease of notation, we will henceforth adopt the convention that $0 \log 0 := 0$ so that one can write $\eta(x) = -x \log x$ without consideration for the $x = 0$ case explicitly.

Remark 61. Note that η is a continuous function with maximum at $x = e^{-1}$ of value $\eta(e^{-1}) = e^{-1}$. Also, η is strictly positive on the interval $x \in (0, 1)$ and strictly negative for $x \in (1, \infty)$.

Lemma 62. The function $\eta : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ is subadditive meaning for all $x, y \geq 0$,

$$\eta(x + y) \leq \eta(x) + \eta(y), \quad (230)$$

and more generally strongly subadditive meaning for all $x, y, z \geq 0$,

$$\eta(x + y + z) + \eta(y) \leq \eta(x + y) + \eta(y + z). \quad (231)$$

Proof. This claim is a special case ($n = 1$) of a Theorem 29 which does not depend on this result. $\square$

Definition 63. Given $\beta \in \mathbb{R}$, let N_β be the real-valued function on PSD matrices given by

$$N_\beta(X) = \beta + \text{Tr}(\eta(X)) = \beta + \sum_{i=1}^n \eta(\lambda_i(X)), \quad \text{\texttt{\{eq:shifted_unnormalized_entropy\}}} \quad (232)$$

where $\lambda_i(X)$ denotes the i -th eigenvalue of X . The quantity $N_\beta(X)$ will be called the **(β -shifted) unnormalized entropy** of X .

Remark 64. It is worth remarking why N_β is termed the (shifted) unnormalized entropy. Suppose $(\lambda_1, \dots, \lambda_n)$ are the eigenvalues of X . When $\beta = 0$, we have the following

$$N_0(X) = - \sum_{i=1}^n \lambda_i \log \lambda_i. \quad (233)$$

We elect to call $N_0(X)$ the *unnormalized* entropy of X because it coincides with the von Neumann entropy when restricted to inputs X where $\text{Tr}(X) = 1$ (i.e., when X is a density matrix):

$$\text{Tr}(X) = 1 \implies N_0(X) = H(X). \quad (234)$$

In general, $N_0(X)$ is related to the von Neumann entropy of the *normalization* of X according to the formula

$$H\left(\frac{X}{\text{Tr}(X)}\right) = N_0\left(\frac{X}{\text{Tr}(X)}\right) = \frac{1}{\text{Tr}(X)} N_0(X) + \log \text{Tr}(X), \quad (235)$$

or equivalently,

$$N_0(X) = \text{Tr}(X) N_0\left(\frac{X}{\text{Tr}(X)}\right) + \eta(\text{Tr}(X)). \quad (236)$$

or equivalently,

$$H\left(\frac{X}{\text{Tr}(X)}\right) = \frac{N_{-\eta(\text{Tr}(X))}(X)}{\text{Tr}(X)}. \quad (237)$$

The term $N_\beta(X)$ is called the β -shifted unnormalized entropy because $N_\beta(X) = \beta + N_0(X)$.

The following proposition lists some basic properties of the function N_β .

Proposition 65. Let $\beta \in \mathbb{R}$, let N_β be as defined by Eq. (232) and let X be a PSD matrix.

- N_β is a spectral function: for all unitary matrices U ,

$$N_\beta(X) = N_\beta(UXU^\dagger). \quad (238)$$

- by subadditivity of η , we have

$$\eta(\text{Tr}(X)) \leq N_0(X) \quad (239)$$

- Let γI_n be the $n \times n$ scalar matrix with $\gamma \geq 0$ along the diagonal. Then

$$N_\beta(\gamma I_n) = \beta + d\eta(\gamma). \quad (240)$$

- Let $\beta' \in \mathbb{R}$ and $X' \geq 0$ a PSD matrix. Then,

$$N_{\beta+\beta'}(X \oplus X') = N_\beta(X) + N_{\beta'}(X'). \quad (241)$$

It is important to note that $N_\beta(X)$ is not a homogeneous function in X meaning in general

$$N_\beta(X) \neq N_\beta(\gamma X). \quad (242)$$

Nevertheless, there is a sense in which it is homogeneous up to a modification to the value of β which depends on the scaling parameter γ .

lem:deformation_homogeneous

Lemma 66. Let $\gamma > 0$ be non-zero and X PSD. Then N_β is deformation-homogeneous in the sense that

{eq:deformation_homogeneous}

$$N_\beta(\gamma X) = \beta + \gamma N_0(X) + \eta(\gamma) \text{Tr}(X) = \gamma N_{\beta - \text{Tr}(X) \log \gamma}(X). \quad (243)$$

As a special case, letting $\beta = \text{Tr}(X) \log \gamma$ produces

$$N_0(X) = \frac{1}{\gamma} N_{\text{Tr}(X) \log \gamma}(\gamma X), \quad (244)$$

and as a further special case, letting $\gamma = \text{Tr}(X)^{-1}$ yields

$$N_0(X) = \text{Tr}(X) N_{\eta(\text{Tr}(X))}\left(\frac{X}{\text{Tr}(X)}\right). \quad (245)$$

Proof. For any $\gamma > 0$ and $X \geq 0$ we have $\gamma X \geq 0$ and $\lambda_i(\gamma X) = \gamma \lambda_i(X)$. Therefore,

$$\begin{aligned}
N_0(\gamma X) &= \sum_{i=1}^n \eta(\lambda_i(\gamma X)) \\
&= \sum_{i=1}^n \eta(\gamma \lambda_i(X)) \\
&= - \sum_{i=1}^n \gamma \lambda_i(X) \log(\gamma \lambda_i(X)) \\
&= -\gamma \sum_{i=1}^n \lambda_i(X) (\log \gamma + \log \lambda_i(X)) \\
&= -(\gamma \log \gamma) \operatorname{Tr}(X) - \gamma \sum_{i=1}^n \lambda_i(X) \log \lambda_i(X) \\
&= \eta(\gamma) \operatorname{Tr}(X) + \gamma N_0(X).
\end{aligned} \tag{246}$$

□

It is also possible to conceptualize the shifts themselves as unnormalized entropies of scalar matrices.

Proposition 67. *Let $\beta \in \mathbb{R}$. Then there exists a sufficiently large integer $n \in \mathbb{N}_{>0}$ and a $\gamma \geq 0$ such that*

$$N_0(\gamma I_n) = \beta. \tag{247}$$

Proof. For any $\gamma \geq 0$ and integer $n \in \mathbb{N}_{>0}$,

$$N_0(\gamma I_n) = -n\gamma \log \gamma. \tag{248}$$

By varying $\gamma \in [0, \infty)$, the image under $\gamma \mapsto N_0(\gamma I_n)$ is $(-\infty, d/e]$. Therefore, if $\beta \leq 0$, then for all $n \in \mathbb{N}_{>0}$, there exists a γ such that $N_0(\gamma I_n) = \beta$. Otherwise, if $\beta > 0$, then let $n = \lceil e\beta \rceil$. This choice for n places β in the range of the map $\gamma \mapsto N_0(\gamma I_n)$. Then there exists a γ such that $\gamma^n = \exp(-\frac{\lceil e\beta \rceil}{e})$ which satisfies Eq. (247). □

5.2 Inequalities for the unnormalized entropy function

thm:unnormalized_entropyineq

Theorem 68. *Let $X, Y, Z \geq 0$ be PSD matrices. Then N_0 is both subadditive*

$$N_0(X + Y) \leq N_0(X) + N_0(Y), \tag{249}$$

{eq:subadditive_Hbeta}

and also strongly subadditive in the sense that

$$N_0(X + Y + Z) + N_0(Y) \leq N_0(X + Y) + N_0(Y + Z). \tag{250}$$

{eq:ssa_unnorm_ent}

Proof. We begin by noting that subadditivity (Eq. (249)) is a special case of strong subadditivity (Eq. (250)) under the substitution $(X, Y, Z) \mapsto (X, 0, Y)$ because $N_0(0) = 0$. Therefore it suffices to just prove strong subadditivity (Eq. (250)). The strong subadditivity of N_0 holds as an special case of Theorem 30 where the function f is taken to be the function

$$f(x) := -\eta(x) = x \log x. \tag{251}$$

The derivative of $-\eta$ is

$$(-\eta)'(x) = \log x + 1 \tag{252}$$

which is known to be operator monotone [Cha13] on $\mathbb{R}_{>0}$ as required. □

5.3 The additive unnormalized entropy cone

sec:unent_cone

We are now in a position to actually define allocations of unnormalized entropies of partial sums.

Definition 69. Let $\mathbf{X} := (X_1, \dots, X_n)$ be an n -tuple of PSD matrices, and let $\boldsymbol{\beta} := (\beta_1, \dots, \beta_n)$ be an n -tuple of non-negative real numbers in $\mathbb{R}_{\geq 0}^n$. Define the **allocation of unnormalized entropies** of $\mathbf{X}$ shifted by $\boldsymbol{\beta}$ be the element of $\mathbb{R}^{2^n-1}$ given by

$$\alpha_n(\mathbf{X}|\boldsymbol{\beta}) := (N_{\beta_S}(X_S))_{S \subseteq [n] \setminus \emptyset}, \quad (253)$$

where X_S and β_S respectively denote partial sums over the subset S :

$$X_S := \sum_{i \in S} X_i, \quad (254)$$

$$\beta_S := \sum_{i \in S} \beta_i. \quad (255)$$

Example 70. For example, when $n = 3$, the allocation of unnormalized entropies for $\mathbf{X} = (X, Y, Z)$ when $\boldsymbol{\beta} = (0, 0, 0)$ is

$$\alpha_3(X, Y, Z|0, 0, 0) = (N_0(X), N_0(Y), N_0(Z), N_0(Y + Z), N_0(X + Z), N_0(X + Y), N_0(X + Y + Z)). \quad (256)$$

Remark 71. The shifted unnormalized entropy can always be decomposed as $N_{\boldsymbol{\beta}}(X) = \boldsymbol{\beta} + N_0(X)$. Therefore, the allocation $\alpha_n(\mathbf{X}|\boldsymbol{\beta})$ can also be decomposed as

$$\alpha_n(\mathbf{X}|\boldsymbol{\beta}) = \alpha_n(\mathbf{0}|\boldsymbol{\beta}) + \alpha_n(\mathbf{X}|\mathbf{0}). \quad (257)$$

Note that if the eigenvalues of a PSD matrix X are larger than one, then it becomes possible for the unnormalized entropy $N_0(X)$ to become negative. In what follows, it will be desirable to consider allocations of unnormalized entropies which lie inside the positive cone $\mathbb{R}_{\geq 0}^{2^n-1} \subset \mathbb{R}^{2^n}$. To ensure positivity, we place explicit restrictions on the value of the permitted $\boldsymbol{\beta}$ -shifts.

Definition 72. /ori The collection of allocations of unnormalized entropies of n -tuples of PSD matrices is defined by

$$\mathcal{A}_n := \{\alpha_n(\mathbf{X}|\boldsymbol{\beta}) | \forall i \in [n] : X_i \geq 0, \beta_i \in \mathbb{R}, \forall S \subseteq [n], N_{\beta_S}(X_S) \geq 0\} \subseteq \mathbb{R}_{\geq 0}^{2^n}. \quad (258)$$

Our goal will be to prove that the set, $\mathcal{A}_n$, of the set of possible allocations of shifted unnormalized entropies forms a convex cone. To prove that $\mathcal{A}_n$ forms a convex cone, it suffices to show that it is both closed under scaling by positive constants $\gamma > 0$ and closed under addition.

lem:scalable

Lemma 73. Let $\gamma > 0$ and let $\alpha_n(\mathbf{X}|\boldsymbol{\beta}) \in \mathcal{A}_n$. Then,

$$\gamma \alpha_n(\mathbf{X}|\boldsymbol{\beta}) = \alpha_n(\gamma \mathbf{X} | \gamma \boldsymbol{\beta} + \log \gamma \boldsymbol{\tau}(\mathbf{X})) \in \mathcal{A}_n, \quad (259)$$

where $\boldsymbol{\tau}(\mathbf{X}) = (\text{Tr}(X_1), \dots, \text{Tr}(X_n))$. Moreover, as the origin belongs to $\mathcal{A}_n$, the set $\mathcal{A}_n$ is closed under scaling by non-negative $\gamma \geq 0$.

Proof. Let $\gamma \mathbf{X} = (\gamma X_1, \dots, \gamma X_n)$ be the n -tuple obtained from $\mathbf{X}$ by entry-wise multiplication by the same factor γ and likewise $\gamma \boldsymbol{\beta} = (\gamma \beta_1, \dots, \gamma \beta_n)$. If $\gamma = 0$, then the claim holds trivially because the origin belongs to $\mathcal{A}_n$. For $\gamma > 0$, the deformation-homogeneous property of shifted unnormalized entropies (Eq. (243)) implies for any $\boldsymbol{\beta}'$,

$$\alpha_n(\gamma \mathbf{X}|\boldsymbol{\beta}') = \gamma \alpha_n(\mathbf{X}|\boldsymbol{\beta}' - \log \gamma \boldsymbol{\tau}(\mathbf{X})). \quad (260)$$

Letting $\boldsymbol{\beta}' = \boldsymbol{\beta} + \log \gamma \boldsymbol{\tau}(\mathbf{X})$ yields

$$\gamma \alpha_n(\mathbf{X}|\boldsymbol{\beta}) = \alpha_n(\gamma \mathbf{X} | \boldsymbol{\beta} + \log \gamma \boldsymbol{\tau}(\mathbf{X})). \quad (261)$$

The additional claim that $\gamma \alpha_n(\gamma \mathbf{X}|\boldsymbol{\beta}) \in \mathcal{A}_n$ follows because $N_{\gamma \beta_S + \text{Tr}(X_S) \log \gamma}(\gamma X_S) = \gamma N_{\beta_S}(X_S) \geq 0$ where the last inequality follows from positivity of γ and the assumption that $\alpha_n(\mathbf{X}|\boldsymbol{\beta}) \in \mathcal{A}_n$. $\square$

Next we show that $\mathcal{A}_n$ is closed under addition.

lem:additive

Lemma 74. *Let $\mathbf{X} = (X_1, \dots, X_n)$ and $\mathbf{X}' = (X'_1, \dots, X'_n)$ be non-zero n -tuples of PSD matrices and let $\beta, \beta' \in \mathbb{R}^n$. Let $\mathbf{X} \oplus \mathbf{X}'$ be the n -tuple*

$$\mathbf{X} \oplus \mathbf{X}' := (X_1 \oplus X'_1, \dots, X_n \oplus X'_n). \quad (262)$$

Then

$$\alpha_n(\mathbf{X} \oplus \mathbf{X}', \beta + \beta') = \alpha_n(\mathbf{X} | \beta) + \alpha_n(\mathbf{X}' | \beta'). \quad (263)$$

eq:additive_with_shifts

Consequently, the set $\mathcal{A}_{\beta,n}$ is closed under addition.

Proof. First note that N_0 is additive across direct sums,

$$N_0(X \oplus X') = N_0(X) + N_0(X'), \quad (264)$$

and thus for any $\beta, \beta' \in \mathbb{R}$,

$$N_{\beta+\beta'}(X \oplus X') = N_{\beta}(X) + N_{\beta'}(X'). \quad (265)$$

Then, note that direct sums are distributive across sums so that

$$\sum_{i \in S} X_i \oplus X'_i = \left(\sum_{j \in S} X_j \right) \oplus \left(\sum_{k \in S} X'_k \right), \quad (266)$$

which directly implies Eq. (263). The proof that $\mathcal{A}_n$ is closed under addition follows by noting that the positivity condition is maintained under taking direct sums. $\square$

Theorem 75. *The set $\mathcal{A}_n \subseteq \mathbb{R}_{\geq 0}^{2^n - 1}$ of all possible allocations of (shifted) unnormalized entropies is a convex cone in $\mathbb{R}_{\geq 0}^{2^n - 1}$. We refer to $\mathcal{A}_n$ as the **additive quantum entropy cone**.*

Proof. This follows from Lemmas 73 and 74. $\square$

5.4 Connection to the ordinary quantum entropy cone

sec:qec

The purpose of this section is to compare the *additive* quantum entropy cone, $\mathcal{A}_n$, to the (*multiplicative*) quantum entropy cone previously studied by Pippenger [Pip03].

5.4.1 Embedding n PSD matrices into n -partite quantum states

Definition 76. Let n, n be positive integers and consider the complex vector space $(\mathbb{C}^{n+1})^{\otimes n}$. For each $p \in [n]$, let $\{|0\rangle_p, |1\rangle_p, \dots, |n\rangle_p\}$ be a basis for $\mathbb{C}^{n+1}$ (noting that $|i\rangle_p$ need not equal $|i\rangle_q$ for $p \neq q$). For each $k \in \{0, 1, \dots, n\}$, we define the **weight- k subspace**, $W_k \subset (\mathbb{C}^{n+1})^{\otimes n}$, to be the span of those vectors in $(\mathbb{C}^{n+1})^{\otimes n}$ of the form

$$|i_1\rangle_1 \otimes \dots \otimes |i_n\rangle_n \quad (267)$$

such that exactly k elements of the tuple $(i_1, \dots, i_p) \in [n+1]^n$ are non-zero.

Remark 77. For example, the weight zero subspace, W_0 , is one-dimension and is spanned by the vector

$$|\emptyset\rangle = |0\rangle_1 \otimes \dots \otimes |0\rangle_n. \quad (268)$$

eq:weight_zero

The weight one subspace, W_1 , is nn -dimensional and is spanned by the vectors $\|i_p\rangle$ where

$$\|i_p\rangle = |0\rangle_1 \otimes \dots \otimes |0\rangle_{p-1} \otimes |i\rangle \otimes |0\rangle_{p+1} \otimes \dots \otimes |0\rangle_n. \quad (269)$$

eq:weight_one

and where $i \in [d]$, $p \in [n]$. The weight two subspace, W_2 , is $n^2 \binom{n}{2}$ -dimensional and is spanned by vectors $\|i_p, j_q\rangle$ where

$$\|i_p, j_q\rangle = |0\rangle_1 \otimes \dots \otimes |0\rangle_{p-1} \otimes |i\rangle \otimes |0\rangle_{p+1} \otimes \dots \otimes |0\rangle_{q-1} \otimes |j\rangle_q \otimes |0\rangle_{q+1} \otimes \dots \otimes |0\rangle_n. \quad (270)$$

eq:weight_two

and where $1 \leq p < q \leq n$ and $i, j \in [d]$.

Remark 78. While the notion of a *weight- k* subspace is technically basis dependent, it is only partially basis dependent. In particular, it is invariant under independent local changes of basis which fix the zero-element, $|0\rangle_p$ of each party p . In other words, the weight k -subspace is invariant under the group $G = (U(1) \times U(n))^{\times n}$ where $U(1) \times U(n) \subset U(n+1)$ refers to the subgroup of the unitary group which preserves the subspace spanned by $|0\rangle_p$. In particular, a weight space decomposition of $(\mathbb{C}^{n+1})^{\otimes n}$ depends only on a fixed product vector $|\emptyset\rangle := |\phi\rangle_1 \otimes \cdots \otimes |\phi\rangle_n$.

[defn:spec_embed](#)

Definition 79. Let $\mathbf{X} = (X_1, \dots, X_n)$ be a n -tuple of $n \times n$ PSD operators and let $(\{|i\rangle_p\}_{i=0}^n)_{p \in [n]}$ be an n -tuple of bases for $\mathbb{C}^{n+1}$. Let $\Gamma(\mathbf{X}) = \Gamma(X_1, \dots, X_n)$ be the Hermitian operator on $(\mathbb{C}^{n+1})^{\otimes n}$ which is block-diagonal with respect to the weight decomposition of $(\mathbb{C}^{n+1})^{\otimes n}$ with non-zero entries given by

$$\langle\langle \emptyset | \Gamma(X_1, \dots, X_n) | \emptyset \rangle\rangle = 1 - \sum_{i=1}^n \text{Tr}(X_n), \quad (\text{eq:weight_zero_portion}) \quad (271)$$

and

$$\langle\langle i_p | \Gamma(X_1, \dots, X_n) | j_q \rangle\rangle = \langle i_p | X_p^{\frac{1}{2}} X_q^{\frac{1}{2}} | j \rangle_q. \quad (\text{eq:weight_one_portion}) \quad (272)$$

For example, the coefficients of $\Gamma_{[n]}$ in the weight-two subspace are zero,

$$\langle\langle i_p, i_{p'} | \Gamma(X_1, \dots, X_n) | j_q, j_{q'} \rangle\rangle = 0, \quad (273)$$

as well as all higher weight spaces.

Example 80. Given two PSD $n \times n$ matrices, (X_1, X_2) , the matrix $\Gamma_{[2]} := \Gamma(X_1, X_2)$ has the following coefficients for all $i, j, k, l \in \{1, \dots, d\}$.

$$\langle 00 | \Gamma | 00 \rangle = 1 - \text{Tr}(X_1) - \text{Tr}(X_2), \quad (274)$$

$$\langle i0 | \Gamma | j0 \rangle = \langle i | X_1 | j \rangle, \quad (275)$$

$$\langle 0i | \Gamma | 0j \rangle = \langle i | X_2 | j \rangle, \quad (276)$$

$$\langle i0 | \Gamma | 0j \rangle = \langle i | X_1^{\frac{1}{2}} X_2^{\frac{1}{2}} | j \rangle, \quad (277)$$

$$\langle 0i | \Gamma | j0 \rangle = \langle i | X_2^{\frac{1}{2}} X_1^{\frac{1}{2}} | j \rangle, \quad (278)$$

$$\langle ik | \Gamma | jl \rangle = 0. \quad (279)$$

Example 81. Given three $n \times n$ PSD matrices $X_1, X_2, X_3 \geq 0$, the operator $\Gamma_{[3]} := \Gamma(X_1, X_2, X_3)$ is the $(n+1)^3 \times (n+1)^3$ matrix with non-zero coefficients given for all $i, j \in \{1, \dots, d\}$ by the formulas:

$$\begin{aligned} \langle 000 | \Gamma | 000 \rangle &= 1 - \text{Tr}(X_1 + X_2 + X_3), \\ \langle i00 | \Gamma | j00 \rangle &= \langle i | X_1 | j \rangle, \\ \langle 0i0 | \Gamma | 0j0 \rangle &= \langle i | X_2 | j \rangle, \\ \langle 00i | \Gamma | 00j \rangle &= \langle i | X_3 | j \rangle, \\ \langle i00 | \Gamma | 0j0 \rangle &= \langle i | X_1^{\frac{1}{2}} X_2^{\frac{1}{2}} | j \rangle, \\ \langle 0i0 | \Gamma | 00j \rangle &= \langle i | X_2^{\frac{1}{2}} X_3^{\frac{1}{2}} | j \rangle, \\ \langle i00 | \Gamma | 00j \rangle &= \langle i | X_1^{\frac{1}{2}} X_3^{\frac{1}{2}} | j \rangle, \\ \langle 0i0 | \Gamma | j00 \rangle &= \langle i | X_2^{\frac{1}{2}} X_1^{\frac{1}{2}} | j \rangle, \\ \langle 00i | \Gamma | j00 \rangle &= \langle i | X_3^{\frac{1}{2}} X_1^{\frac{1}{2}} | j \rangle, \\ \langle 00i | \Gamma | 0j0 \rangle &= \langle i | X_3^{\frac{1}{2}} X_2^{\frac{1}{2}} | j \rangle. \end{aligned} \quad (280)$$

[lem:sum_spec_embed](#)

Lemma 82. Let $\Gamma_{[n]} := \Gamma(X_1, \dots, X_n)$ be defined as in Definition 79. Then the eigenvalues of $\Gamma_{[n]}$ are equal to the eigenvalues of the sum,

$$X_{[n]} := \sum_{i=1}^n X_i, \quad (281)$$

plus the additional value of $1 - \text{Tr}(X_{[n]})$:

$$\text{spec}(\Gamma_{[n]}) \simeq \text{spec}(X_{[n]}) \oplus \{1 - \text{Tr}(X_{[n]})\}. \quad (282)$$

Proof. By definition, $\Gamma_{[n]}$ is block diagonal with respect to the weight decomposition. As the weight-zero subspace is one-dimensional and spanned by $\|\emptyset\rangle = |0\rangle \cdots |0\rangle$, we see that $\|\emptyset\rangle$ is an eigenvector of $\Gamma_{[n]}$ with eigenvalue given by $\langle\langle\emptyset|\Gamma_{[n]}|\emptyset\rangle\rangle$ which is given by Eq. (272), i.e.,

$$\Gamma_{[n]} \|\emptyset\rangle = (1 - \text{Tr}(X_{[n]}) \|\emptyset\rangle. \quad (283)$$

For the remaining eigenvalues of $\Gamma_{[n]}$, let $\Phi_{[n]}$ be the operator $\Gamma_{[n]}$ with its weight-zero entry set to zero:

$$\Phi_{[n]} := \Gamma_{[n]} - (1 - \text{Tr}(X_{[n]}) \|\emptyset\rangle \langle\langle\emptyset|. \quad (284)$$

To prove the claim, it therefore suffices to show that the eigenvalues of $\Phi_{[n]}$ are identical to the eigenvalues of the sum $X_{[n]}$ (with multiplicity). Equivalently, it suffices to prove that the characteristic polynomials of $\Phi_{[n]}$ and $X_{[n]}$ have the same coefficients which is implied by proving that for any positive integer ℓ , $\text{Tr}(\Phi_{[n]}^\ell) = \text{Tr}(X_{[n]}^\ell)$. For $\ell = 1$, the proof follows from performing a single sum over the weight-one subspace

$$\text{Tr}(\Phi_{[n]}) = \sum_{p=1}^n \sum_{i=1}^n \langle\langle i_p | \Phi_{[n]} | i_p \rangle\rangle = \sum_{p=1}^n \sum_{i=1}^n \langle i_p | X_p | i \rangle_p = \sum_{p=1}^n \text{Tr}(X_p) = \text{Tr}(X_{[n]}). \quad (285)$$

For $\ell > 1$, the proof follows from an ℓ -fold sum over the weight-one subspace,

$$\text{Tr}(\Phi_{[n]}^\ell) = \sum_{p_1=1}^n \sum_{i_1=1}^n \cdots \sum_{p_\ell=1}^n \sum_{i_\ell=1}^n \langle\langle (i_1)_{p_1} | \Phi_{[n]} | (i_2)_{p_2} \rangle\rangle \cdots \langle\langle (i_\ell)_{p_\ell} | \Phi_{[n]} | (i_1)_{p_1} \rangle\rangle, \quad (286)$$

$$= \sum_{p_1=1}^n \sum_{i_1=1}^n \cdots \sum_{p_\ell=1}^n \sum_{i_\ell=1}^n \langle i_1 |_{p_1} X_{p_1}^{\frac{1}{2}} X_{p_2}^{\frac{1}{2}} | i_2 \rangle_{p_2} \cdots \langle i_\ell |_{p_\ell} X_{p_\ell}^{\frac{1}{2}} X_{p_1}^{\frac{1}{2}} | i_1 \rangle_{p_1}, \quad (287)$$

$$= \sum_{p_1=1}^n \cdots \sum_{p_\ell=1}^n \text{Tr}[X_{p_1}^{\frac{1}{2}} X_{p_2}^{\frac{1}{2}} \cdots X_{p_\ell}^{\frac{1}{2}} X_{p_1}^{\frac{1}{2}}], \quad (288)$$

$$= \sum_{p_1=1}^n \cdots \sum_{p_\ell=1}^n \text{Tr}[X_{p_1} X_{p_2} \cdots X_{p_\ell}], \quad (289)$$

$$= \text{Tr} \left[\left(\sum_{p_1=1}^n X_{p_1} \right) \cdots \left(\sum_{p_\ell=1}^n X_{p_\ell} \right) \right], \quad (290)$$

$$= \text{Tr}(X_{[n]}^\ell). \quad (291)$$

Therefore $X_{[n]}$ and $\Phi_{[n]}$ have the same eigenvalues with multiplicity which proves the claim. $\square$

cor:spec_of_embed

Corollary 83. *Let everything be as in Lemma 82. Then $\text{Tr}(\Gamma_{[n]}) = 1$. If the eigenvalues of the sum $X_{[n]}$ satisfy*

$$\text{Tr}(X_{[n]}) \leq 1, \quad (292)$$

then $\Gamma_{[n]}$ is positive semidefinite, and thus $\Gamma_{[n]}$ is a density matrix. In such cases, the von Neumann entropy of $\Gamma_{[n]}$ is related to unnormalized entropy of $X_{[n]}$ by

$$H(\Gamma_{[n]}) = (1 - \text{Tr}(X_{[n]})) \log(1 - \text{Tr}(X_{[n]})) + N_0(X_{[n]}), \quad (293)$$

$$= h_{\text{bin}}(\text{Tr}(X_{[n]})) + \text{Tr}(X_{[n]}) N_0 \left(\frac{X_{[n]}}{\text{Tr}(X_{[n]})} \right). \quad (294)$$

Moreover, if the eigenvalues of $X_{[n]}$ satisfy the stronger condition

$$\lambda_1(X_{[n]}) + \text{Tr}(X_{[n]}) \leq 1, \quad (295)$$

then the maximum eigenvalue of $\Gamma_{[n]}$ is $1 - \text{Tr}(X_{[n]})$.

Lemma 84. Let $\Gamma_{[n]} := \Gamma(X_1, \dots, X_n)$ be defined as in Definition 79. Let $S \subseteq \{1, \dots, n\}$ be any subset and let $\Gamma_S := \Gamma((X_i)_{i \in S})$ with $\Gamma_\emptyset := 1$. Then the partial trace of $\Gamma_{[n]}$ over $[n] \setminus S$ is equal to Γ_S :

$$\text{Tr}_{[n] \setminus S}(\Gamma_{[n]}) = \Gamma_S. \quad (296)$$

In particular, letting $S = \{p\}$ yields

$$\Gamma_p := \Gamma(X_p) = (1 - \text{Tr}(X_p)) \oplus X_p = \text{Tr}_{[n] \setminus \{p\}}(\Gamma_{[n]}). \quad (297)$$

Proof. To begin, notice that for any n -partite weight-block-diagonal operator X on $(\mathbb{C}^n)^{\otimes n}$, its partial trace over $S \subseteq [n]$, $\text{Tr}_{[n] \setminus S}(X)$, will also be weight-block-diagonal as an $|S|$ -partite operator because the weight-block-off-diagonal entries of $\text{Tr}_{[n] \setminus S}(X)$ are sums of weight-block-off-diagonal entries of X . Therefore, to prove the claim, it suffices to restrict focus to the weight-block-diagonal entries of $\text{Tr}_{[n] \setminus S}(\Gamma_{[n]})$.

Let $\omega : S \rightarrow \{1, \dots, |S|\}$ be the map which assigns to each $p \in S$ its position in S ordered relative to the ordering on $[n]$. Then the coefficients of $\Gamma_{[n]}$ in the weight-one subspace and the coefficients of Γ_S in the weight-one subspace are related for all $p, q \in S$ according to,

$$\langle\langle i_p \| \Gamma_{[n]} \| j_p \rangle\rangle = \langle i | X_p^{\frac{1}{2}} X_q^{\frac{1}{2}} | j \rangle = \langle\langle i_{\omega(p)} \| \Gamma_S \| j_{\omega(q)} \rangle\rangle \quad \{\text{eq:marginal_joint_correspondence}\} \quad (298)$$

Therefore, the coefficients of $\text{Tr}_{[n] \setminus S}[\Gamma_{[n]}]$ in the weight-one block for all $p, q \in S$ and $i, j \in [d]$ are given by

$$\langle\langle i_{\omega(p)} \| \text{Tr}_{[n] \setminus S}[\Gamma_{[n]}] \| j_{\omega(q)} \rangle\rangle = \langle\langle i_p \| \Gamma_{[n]} \| j_q \rangle\rangle = \langle\langle i_{\omega(p)} \| \Gamma_S \| j_{\omega(q)} \rangle\rangle. \quad (299)$$

Moreover, using Eq. (298) again shows that the coefficient of $\text{Tr}_{[n] \setminus S}[\Gamma_{[n]}]$ in the weight-zero block equals

$$\langle\langle \emptyset \| \text{Tr}_{[n] \setminus S}[\Gamma_{[n]}] \| \emptyset \rangle\rangle = \langle\langle \emptyset \| \Gamma_{[n]} \| \emptyset \rangle\rangle + \sum_{p \notin S} \sum_{i=1}^n \langle\langle i_p \| \Gamma_{[n]} \| i_p \rangle\rangle \quad (300)$$

$$= \langle\langle \emptyset \| \Gamma_{[n]} \| \emptyset \rangle\rangle + \sum_{p=1}^n \sum_{i=1}^n \langle\langle i_p \| \Gamma_{[n]} \| i_p \rangle\rangle - \sum_{p \in S} \sum_{i=1}^n \langle\langle i_p \| \Gamma_{[n]} \| i_p \rangle\rangle \quad (301)$$

$$= \text{Tr}(\Gamma_{[n]}) - \sum_{p \in S} \sum_{i=1}^n \langle\langle i_{\omega(p)} \| \Gamma_S \| i_{\omega(p)} \rangle\rangle \quad (302)$$

$$= \text{Tr}(\Gamma_{[n]}) - (\text{Tr}(\Gamma_S) - \langle\langle \emptyset \| \Gamma_S \| \emptyset \rangle\rangle) \quad (303)$$

$$= \langle\langle \emptyset \| \Gamma_S \| \emptyset \rangle\rangle \quad (304)$$

because $\text{Tr}(\Gamma_{[n]}) = \text{Tr}(\Gamma_S) = 1$. Therefore, $\Gamma_S = \text{Tr}_{[n] \setminus S}[\Gamma_{[n]}]$ which proves the claim. $\square$

It is worth emphasizing that the operators, $\Gamma_{[n]} = \Gamma(X_1, \dots, X_n)$, obtained from an n -tuple $(X_1, \dots, X_n)$ according to Definition 79 do not exhaust all n -partite operators. Nevertheless, there is an elegant characterization of those n -partite operators which can be obtained in this way. This characterization generalizes [CŞW18, Proposition 11].

Lemma 85. Let $\rho_{[n]}$ be an n -partite density matrix on $(\mathbb{C}^{n+1})^{\otimes n}$ with rank at most $n+1$ such that there exists a product eigenvector, denoted by $|\emptyset\rangle = |\phi\rangle_1 \otimes \dots \otimes |\phi\rangle_n$, such that $\rho_{[n]}$ is weight-block-diagonal with respect to $|\emptyset\rangle$ and only supported on the weight-zero and weight-one subspaces.

Then there exists an n -tuple of $n \times n$ PSD matrices $(X_1, \dots, X_n)$ such that up to a local unitary change of basis (which fixes $|\emptyset\rangle$), we have

$$\rho_{[n]} \simeq \Gamma(X_1, \dots, X_n). \quad (305)$$

Proof. Since $\rho_{[n]}$ is supported only on the weight-zero and weight-one subspaces and is block-diagonal with respect to these subspaces, the eigendecomposition of $\rho_{[n]}$ has the form:

$$\rho_{[n]} = \lambda_1 |\emptyset\rangle \langle\emptyset| + \sum_{k=1}^n \lambda_{k+1} |v_{k+1}\rangle \langle v_{k+1}| \quad (306)$$

where the eigenvector $|v_{k+1}\rangle$ belongs to the weight-one subspace and thus has the form

$$|v_{k+1}\rangle = \sum_{p=1}^n \sum_{i=1}^n V_{p,i,k} \|i_p\rangle \quad (307)$$

for some unknown $n \times n \times n$ tensor of coefficients denoted by $V_{p,i,k}$ and where $\|i_p\rangle$ is given by Eq. (269) taken with respect to some n -tuple of bases where $|0\rangle_p := |\phi\rangle_p$. Now, for each party p , let M_p be the $n \times n$ matrix defined by

$$(M_p)_{i,k} := \lambda_{k+1}^{\frac{1}{2}} V_{p,i,k}. \quad (308)$$

With this definition, one obtains

$$\langle\langle i_p \| \rho_{[n]} \| j_q \rangle\rangle = \sum_{k=1}^n \lambda_{k+1} V_{p,i,k} V_{q,j,k}^* = \sum_{k=1}^n (M_p)_{i,k} (M_q)_{j,k}^* = \langle i |_p M_p M_q^\dagger | j \rangle_q \quad (309)$$

Now by a polar decomposition, for each party p , there exists a PSD $n \times n$ matrix X_p and an $n \times n$ unitary U_p such that

$$M_p = U_p X_p^{\frac{1}{2}}, \quad (310)$$

and therefore,

$$\langle\langle i_p \| \rho_{[n]} \| j_q \rangle\rangle = \langle i |_p U_p X_p^{\frac{1}{2}} X_q^{\frac{1}{2}} U_q^\dagger | j \rangle_q. \quad (311)$$

Letting $|i\rangle'_p = U_p^\dagger |i\rangle_p$ be a modified basis for party p and letting

$$\mathcal{U}_{[n]} := (1 \oplus U_1) \otimes (1 \oplus U_2) \otimes \cdots \otimes (1 \oplus U_n), \quad (312)$$

we observe

$$\langle\langle i_p \|' \mathcal{U}_{[n]}^\dagger \rho_{[n]} \mathcal{U}_{[n]} \| j_q \rangle\rangle' = \langle\langle i_p \| \rho_{[n]} \| j_q \rangle\rangle = \langle i |_p U_p X_p^{\frac{1}{2}} X_q^{\frac{1}{2}} U_q^\dagger | j \rangle_q = \langle\langle i_p \|' X_p^{\frac{1}{2}} X_q^{\frac{1}{2}} \| j_q \rangle\rangle', \quad (313)$$

which implies $\mathcal{U}_{[n]}^\dagger \rho_{[n]} \mathcal{U}_{[n]} = \Gamma(X_1, \dots, X_p)$ as claimed. $\square$

5.4.2 Comparing allocations of entropy

As a direct consequence of Corollary 83 and Lemma 84 we can compare allocations of unnormalized entropies against allocations of von Neumann entropies.

Corollary 86. *Let $\mathbf{X} = (X_1, \dots, X_n)$ be an n -tuple of PSD matrices such that $\text{Tr}(X_{[n]}) \leq 1$ and let $\Gamma(\mathbf{X})$ denote the corresponding n -partite quantum state. Then the allocation $\alpha_n(\mathbf{X}|\mathbf{0})$ of unnormalized entropies of $\mathbf{X}$ is related to the allocation $\mu_n(\Gamma(\mathbf{X}))$ of von Neumann entropies of $\Gamma(\mathbf{X})$ according to the formula*

$$\mu_n(\Gamma(\mathbf{X}))_S = \alpha_n(\mathbf{X}|\mathbf{0})_S - (1 - \text{Tr}(X_S)) \log(1 - \text{Tr}(X_S)) \quad (314)$$

where $S \subseteq [n]$ and where

$$\mu_n(\Gamma(\mathbf{X}))_S = H(\text{Tr}_{[n] \setminus S}(\Gamma(\mathbf{X}))), \quad (315)$$

$$\alpha_n(\mathbf{X}|\mathbf{0})_S = N_0(X_S). \quad (316)$$

thm:almost_subcone

Theorem 87. *Let $\mathbf{X} = (X_1, \dots, X_n)$ be an n -tuple of PSD matrices and let $\Gamma(\mathbf{X})$ denote the n -partite density matrix given by Definition 79. Then for all sufficiently small $\gamma > 0$,*

$$\|\alpha_n(\mathbf{X}|\tau(\mathbf{X})(1 - \log \gamma)) - \gamma^{-1} \mu_n(\Gamma(\gamma \mathbf{X}))\|_\infty \leq \gamma \text{Tr}(X_{[n]})^2 \quad (317)$$

Proof. By Lemma 84, it is known that the von Neumann entropy of the reduced state of $\Gamma(\mathbf{X})$ (whenever $\text{Tr}(X_{[n]}) \leq 1$ so that $\Gamma(\mathbf{X})$ is a density matrix) over $S \subseteq [n]$ is given by

$$H(\Gamma(\mathbf{X})_S) = N_0(X_S) - (1 - \text{Tr}(X_S)) \log(1 - \text{Tr}(X_S)). \quad (318)$$

Then, mapping $\mathbf{X}$ to $\gamma\mathbf{X}$ and multiplying by γ^{-1} yields

$$\gamma^{-1}H(\Gamma(\gamma\mathbf{X})_S) = \gamma^{-1}N_0(\gamma X_S) - \gamma^{-1}(1 - \gamma \text{Tr}(X_S)) \log(1 - \gamma \text{Tr}(X_S)). \quad (319)$$

Define $f(x)$ over $x \in [0, 1]$ as

$$f(x) := 1 + x^{-1}(1 - x) \log(1 - x). \quad (320)$$

with $f(0) = 0$ and $f(1) = 1$ noting that $f(x) \leq x$ holds for all $x \in [0, 1]$. Now take $\gamma \geq 0$ sufficiently small such that $\gamma \text{Tr}(X_S) \in [0, 1]$ holds for all $S \subseteq [n]$ and thus $\Gamma(\gamma\mathbf{X})$ is a well-defined density matrix, Then,

$$\gamma^{-1}H(\Gamma(\gamma\mathbf{X})_S) = \gamma^{-1}N_0(\gamma X_S) + \text{Tr}(X_S) - \text{Tr}(X_S)f(\gamma \text{Tr}(X_S)), \quad (321)$$

Since $\gamma^{-1}N_0(\gamma X_S) + \text{Tr}(X_S) = N_{\text{Tr}(X_S)(1-\log \gamma)}(X_S)$, we conclude that

$$N_{\text{Tr}(X_S)(1-\log \gamma)}(X_S) - \gamma^{-1}H(\Gamma(\gamma\mathbf{X})_S) = \text{Tr}(X_S)f(\gamma \text{Tr}(X_S)) \leq \gamma \text{Tr}(X_S)^2 \quad (322)$$

□
cor:balanced_ineqs

Corollary 88. Let $I : \mathbb{R}_{\geq 0}^{2^n-1} \rightarrow \mathbb{R}$ be a balanced linear functional which is non-negative on the ordinary quantum entropy cone. Then I is also non-negative on the additive quantum entropy cone.

Proof. Let $\mathbf{X} = (X_1, \dots, X_n)$ be an arbitrary n -tuple of PSD matrices. That I is balanced is equivalent to the claim that $I(\alpha_n(\mathbf{X}|\boldsymbol{\beta}))$ is independent of shifting by $\boldsymbol{\beta}$, meaning for any $\boldsymbol{\beta} \in \mathbb{R}$ and thus any γ taken from Theorem 87,

$$I(\alpha_n(\mathbf{X}|\tau(\mathbf{X})(1 - \log \gamma))) = I(\alpha_n(\mathbf{X}|\boldsymbol{\beta})) = I(\alpha_n(\mathbf{X}|\mathbf{0})). \quad (323)$$

That I is linear and positive on the ordinary quantum entropy cone implies

$$I(\gamma^{-1}\mu_n(\Gamma(\gamma\mathbf{X}))) = \gamma^{-1}I(\mu_n(\Gamma(\gamma\mathbf{X}))) \geq 0. \quad (324)$$

Furthermore, by Theorem 87, the value of $I(\alpha_n(\mathbf{X}|\boldsymbol{\beta}))$ is also positive because

$$I(\alpha_n(\mathbf{X}|\boldsymbol{\beta})) = I(\alpha_n(\mathbf{X}|\tau(\mathbf{X})(1 - \log \gamma))) \quad (325)$$

$$= I(\omega(\gamma, \mathbf{X})) + I(\gamma^{-1}\mu_n(\Gamma(\gamma\mathbf{X}))) \quad (326)$$

$$\geq I(\omega(\gamma, \mathbf{X})), \quad (327)$$

where $\omega(\gamma, \mathbf{X}) \in \mathbb{R}^{2^n-1}$ is the vector given by:

$$\omega := \alpha_n(\mathbf{X}|\tau(\mathbf{X})(1 - \log \gamma)) - \gamma^{-1}\mu_n(\Gamma(\gamma\mathbf{X})). \quad (328)$$

However, $\alpha_n(\mathbf{X}|\boldsymbol{\beta})$ is independent of γ so we also have

$$I(\alpha_n(\mathbf{X}|\boldsymbol{\beta})) \geq \lim_{\gamma \rightarrow 0} I(\omega(\gamma, \mathbf{X})) = I(\lim_{\gamma \rightarrow 0} \omega(\gamma, \mathbf{X})). \quad (329)$$

By the proof of Theorem 87, we know the coefficients of $\omega(\gamma, \mathbf{X})$ are non-negative and bounded above by $\gamma \text{Tr}(X_{[n]})^2$. Therefore, $\lim_{\gamma \rightarrow 0} \omega(\gamma, \mathbf{X}) = 0$ and therefore, $I(\alpha_n(\mathbf{X}|\boldsymbol{\beta})) \geq I(0) = 0$ as claimed. □

Remark 89. Corollary 88 serves as an alternative proof of that unnormalized entropies satisfy strong subadditivity Eq. (250). Specifically, let $S, T \subseteq [n]$ be subsets and consider the functional $I_{S,T} : \mathbb{R}^{2^n-1} \rightarrow \mathbb{R}$ defined for $v \in \mathbb{R}^{2^n-1}$ by

$$I_{S,T}(v) := v_S + v_T - v_{S \cup T} - v_{S \cap T} \quad (330)$$

where $v_\emptyset := 0$. This functional is balanced and linear. When evaluated on allocations of von Neumann entropies of n -partite quantum states, it corresponds to the conditional mutual information of between subsystems $A = S \setminus T$ and $B = T \setminus S$ given $C = T \cap S$. Therefore $I_{S,T}$ is non-negative on the ordinary quantum entropy cone because of strong subadditivity of von Neumann entropy. Therefore, by Corollary 88, we conclude that $I_{S,T}$ it is also non-negative on the additive quantum entropy cone which is equivalent to Eq. (250).

Remark 90. Corollary 88 demonstrates that all balanced inequalities which are valid for the ordinary quantum entropy cone are valid inequalities for the additive quantum entropy cone. It is currently a major open problem in quantum information theory whether or not there exists (unconstrained) inequalities for the quantum entropy cone which are not implied by strong subadditivity [LW05]. If there exists balanced inequalities for the quantum entropy cone which go beyond strong subadditivity, then Corollary 88 proves such inequalities also hold for the additive quantum entropy cone. For instance, the following inequality, due to Zhang and Yeung [ZY98], holds for the Shannon entropies of classical probability distributions and is balanced:

$$2I_{C,D} \leq I_{A,B} + I_{A,CD} + 3I_{AC,AD} + I_{BC,BD}. \quad \text{\texttt{eq:zhang-yeung}} \quad (331)$$

It is not known whether this inequality holds for the quantum entropy cone also [Wal14]. If the inequality in Eq. (331) does hold for the von Neumann entropies of quantum states, then Corollary 88 would imply the following inequality constraining the unnormalized entropies of a four-tuple $\mathbf{X} = (A, B, C, D)$ of PSD matrices is valid:

$$\begin{aligned} N_0(A) + 2N_0(C) + 2N_0(D) + N_0(A+B) + N_0(B+C+D) + 4N_0(A+B+C+D) \\ \leq 3N_0(A+C) + 3N_0(A+D) + 3N_0(C+D) + N_0(B+C) + N_0(B+D). \end{aligned} \quad \text{\texttt{eq:zhang-yeung-conjecture}} \quad (332)$$

At the moment, it is unknown whether or not Eq. (332) holds for all PSD matrices (A, B, C, D) .

On a related note, it has been shown by Gross and Walter [GW13] (and also reported by Linden et al. [Lin+13]) that the von Neumann entropies of stabilizer quantum states automatically satisfy every balanced information inequality of Shannon entropies (such as the Zhang-Yeung inequality, Eq. (331), above). In [CLW12], Cadney, Linden, and Winter conjecture that *every* balanced inequality which holds for Shannon entropies of joint probability distributions, also holds for the von Neumann entropies of *every* multipartite quantum states. In light of Corollary 88, this extends to the conjecture that every balanced inequality which holds for Shannon entropies of joint probability distributions also holds for the unnormalized entropies of sums of PSD matrices.

5.5 Comparison with concavity of von Neumann entropy

\texttt{sec:unent_concavity}

At this stage it is worth comparing how the inequalities in Theorem 68 compare to known inequalities concerning von Neumann entropy. Suppose ρ_X and ρ_Y are two density matrices and $\lambda \in [0, 1]$ is a convex parameter. Then it is known that the von Neumann entropy satisfies *concavity*:

$$\lambda H(\rho_X) + (1 - \lambda)H(\rho_Y) \leq H(\lambda\rho_X + (1 - \lambda)\rho_Y), \quad (333)$$

but also an associated upper bound:

$$H(\lambda\rho_X + (1 - \lambda)\rho_Y) \leq \lambda H(\rho_X) + (1 - \lambda)H(\rho_Y) + h_{\text{bin}}(\lambda) \quad (334)$$

where $h_{\text{bin}}(\lambda)$ is the binary entropy function:

$$h_{\text{bin}}(\lambda) = -\lambda \log \lambda - (1 - \lambda) \log \lambda. \quad (335)$$

5.5.1 Concavity upper bound

We first show that this upper bound is related to subadditivity of N_0 . If we let $p_X, p_Y \geq 0$ be weights such that

$$\lambda = \frac{p_X}{p_X + p_Y}, \quad 1 - \lambda = \frac{p_Y}{p_X + p_Y}, \quad \text{\texttt{eq:binary_weights}} \quad (336)$$

then the concavity upper bound above can be equivalently written in terms of the unnormalized entropy N_0 as

$$N_0(\sigma_X + \sigma_Y) \leq N_0(\sigma_X) + N_0(\sigma_Y), \quad (337)$$

where

$$\sigma_X := p_X \rho_X, \quad \sigma_Y = p_Y \rho_Y. \quad \text{\texttt{eq:normalization_scheme}} \quad (338)$$

We note that this expression is equivalent to the subadditivity of N_0 (Eq. (249) from Theorem 68). This is because every non-zero PSD matrix σ_X can be equivalently expressed in terms of a positive weight p_X and a density matrix ρ_X according to the formula:

$$(p_X, \rho_X) := \left(\text{Tr}(\sigma_X), \frac{\sigma_X}{\text{Tr}(\sigma_X)} \right). \quad (339)$$

5.5.2 Concavity lower bound

It is also possible to express the concavity of von Neumann entropy as a kind of superadditivity of a different function, namely the function

$$g(\sigma) = \text{Tr}(\sigma) H\left(\frac{\sigma}{\text{Tr}(\sigma)}\right) = N_0(\sigma) - \eta(\text{Tr}(\sigma)), \quad (340)$$

for PSD σ . The non-negativity of $g(\sigma)$ follows because von Neumann entropy is non-negative, or equivalently the non-negativity of $g(\sigma)$ follows from the subadditivity of η : ($N(\sigma) \geq \eta(\text{Tr}(\sigma))$ and thus $g(\sigma) \geq 0$). In this notation, the concavity of von Neumann entropy is equivalent to

$$g(\sigma_X + \sigma_Y) \geq g(\sigma_X) + g(\sigma_Y), \quad (341)$$

because Eq. (338) and Eq. (336) implies

$$\frac{g(\sigma_X + \sigma_Y)}{\text{Tr}(\sigma_X + \sigma_Y)} = H\left(\frac{\sigma_X + \sigma_Y}{\text{Tr}(\sigma_X + \sigma_Y)}\right) = H(\lambda\rho_X + (1-\lambda)\rho_Y), \quad (342)$$

$$\frac{g(\sigma_X)}{\text{Tr}(\sigma_X + \sigma_Y)} = \frac{\text{Tr}(\sigma_X)}{\text{Tr}(\sigma_X + \sigma_Y)} H\left(\frac{\sigma_X}{\text{Tr}(\sigma_X)}\right) = \lambda H(\rho_X), \quad (343)$$

$$\frac{g(\sigma_Y)}{\text{Tr}(\sigma_X + \sigma_Y)} = \frac{\text{Tr}(\sigma_Y)}{\text{Tr}(\sigma_X + \sigma_Y)} H\left(\frac{\sigma_Y}{\text{Tr}(\sigma_Y)}\right) = (1-\lambda)H(\rho_Y). \quad (344)$$

TC: At the moment it is unclear if the function g satisfies a kind of strong superadditivity, i.e.,

$$g(\sigma_X + \sigma_Y) + g(\sigma_Y + \sigma_Z) \stackrel{?}{\leq} g(\sigma_Y) + g(\sigma_X + \sigma_Y + \sigma_Z) \quad (345)$$

or not, but it is probably worth finding out. The strong superadditivity of g is equivalent to the following upper bound:

$$0 \leq N_0(X + Y) + N_0(Y + Z) - N_0(X + Y + Z) - N_0(Y), \quad (346)$$

$$\stackrel{?}{\leq} \eta(\text{Tr}(X + Y)) + \eta(\text{Tr}(Y + Z)) - \eta(\text{Tr}(X + Y + Z)) - \eta(\text{Tr}(Y)). \quad (347)$$

This inequality (or something very close) may or may not hold.

Below are some more calculations which may prove useful. Let $\sigma_{AY_1Y_2Y_3}$ be an 4-party state defined by PSD X_1, X_2, X_3 as follows:

$$\sigma_{AY_1Y_2Y_3} = \frac{1}{\text{Tr}(X_1 + X_2 + X_3)} \left(X_1 \otimes |1\rangle\langle 1| \otimes |0\rangle\langle 0| \otimes |0\rangle\langle 0| + \right. \\ \left. X_2 \otimes |0\rangle\langle 0| \otimes |1\rangle\langle 1| \otimes |0\rangle\langle 0| + \right. \\ \left. X_3 \otimes |0\rangle\langle 0| \otimes |0\rangle\langle 0| \otimes |1\rangle\langle 1| \right). \quad (348)$$

For each subset $S \subseteq [n]$, let $X_S = \sum_{i \in S} X_i$ denote the partial sum. Then

$$\text{Tr}(X_{123})H(\sigma_A) = g(X_{123}) \quad (349)$$

$$\text{Tr}(X_{123})H(\sigma_{Y_1}) = \eta(\text{Tr}(X_1)) + \eta(\text{Tr}(X_{23})) - \eta(\text{Tr}(X_{123})) \quad (350)$$

$$\text{Tr}(X_{123})H(\sigma_{AY_1}) = g(X_1) + g(X_{23}) + \eta(\text{Tr}(X_1)) + \eta(\text{Tr}(X_{23})) - \eta(\text{Tr}(X_{123})) \quad (351)$$

$$\text{Tr}(X_{123})H(\sigma_{AY_1Y_2}) = g(X_1) + g(X_2) + g(X_3) + \eta(\text{Tr}(X_1)) + \eta(\text{Tr}(X_2)) + \eta(\text{Tr}(X_3)) - \eta(\text{Tr}(X_{123})) \quad (352)$$

$$\text{Tr}(X_{123})H(\sigma_{AY_1Y_2Y_3}) = \text{Tr}(X_{123})H(\sigma_{AY_1Y_2}). \quad (353)$$

(i) Subadditivity between A and Y_1 yields

$$H(\sigma_{AY_1}) \leq H(\sigma_A) + H(\sigma_{Y_1}). \quad (354)$$

In terms of X_1, X_2, X_3 , this is equivalent to concavity:

$$g(X_1) + g(X_{23}) \leq g(X_{123}). \quad (355)$$

(ii) Strong subadditivity implies

$$I_{Y_1, Y_3|A} = H(\sigma_{AY_1}) + H(\sigma_{AY_3}) - H(\sigma_A) - H(\sigma_{AY_1Y_3}) \geq 0. \quad (356)$$

In terms of X_1, X_2, X_3 , this is equivalent to

$$\begin{aligned} & [g(X_1) + g(X_{23}) + \eta(\text{Tr}(X_1)) + \eta(\text{Tr}(X_{23})) - \eta(\text{Tr}(X_{123}))] \\ & + [g(X_{12}) + g(X_3) + \eta(\text{Tr}(X_{12})) + \eta(\text{Tr}(X_3)) - \eta(\text{Tr}(X_{123}))] \\ & \geq [g(X_{123})] + [g(X_1) + g(X_2) + g(X_3) + \eta(\text{Tr}(X_1)) + \eta(\text{Tr}(X_2)) + \eta(\text{Tr}(X_3)) - \eta(\text{Tr}(X_{123}))] \end{aligned} \quad (357)$$

which is equivalent to

$$N_0(X_{12}) + N_0(X_{23}) \geq N_0(X_2) + N_0(X_{123}). \quad (358)$$

(iii) Strong subadditivity implies

$$I_{Y_1, A|Y_3} = H(\sigma_{Y_1Y_3}) + H(\sigma_{AY_3}) - H(\sigma_{Y_3}) - H(\sigma_{AY_1Y_3}) \geq 0. \quad (359)$$

In terms of X_1, X_2, X_3 , this is equivalent to

$$\begin{aligned} & [\eta(\text{Tr}(X_1)) + \eta(\text{Tr}(X_2)) + \eta(\text{Tr}(X_3)) - \eta(\text{Tr}(X_{123}))] \\ & + [g(X_{12}) + g(X_3) + \eta(\text{Tr}(X_{12})) + \eta(\text{Tr}(X_3)) - \eta(\text{Tr}(X_{123}))] \\ & \geq [\eta(\text{Tr}(X_{12})) + \eta(\text{Tr}(X_3)) - \eta(\text{Tr}(X_{123}))] \\ & + [g(X_1) + g(X_2) + g(X_3) + \eta(\text{Tr}(X_1)) + \eta(\text{Tr}(X_2)) + \eta(\text{Tr}(X_3)) - \eta(\text{Tr}(X_{123}))] \end{aligned} \quad (360)$$

which is equivalent to concavity again

$$g(X_{12}) \geq g(X_1) + g(X_2) \quad (361)$$

5.6 Entropies of ensembles & coarse-grained ensembles

Let $(X_1, \dots, X_n)$ be an n -tuple of PSD matrices or equivalently, let

$$\mathcal{E} = \{(p_1, \rho_1), \dots, (p_n, \rho_n)\} \quad \text{\texttt{\{eq:n_ensemble_states\}}} \quad (362)$$

be an ensemble of density matrices (where $\rho_i \geq 0$, $p_i \geq 0$, $\text{Tr}(\rho_i) = 1$ and $\sum_i p_i = 1$).

The *Holevo information*, denoted by $\chi(\mathcal{E})$, of such an ensemble is the quantity

$$\chi(\mathcal{E}) := H\left(\sum_{i=1}^n p_i \rho_i\right) - \sum_{i=1}^n p_i H(\rho_i). \quad (363)$$

A straight-forward calculation reveals that $\chi(\mathcal{E})$ is equal to the mutual information of a classical-quantum (CQ) state of the form

$$\Gamma_{CQ} = \sum_{i=1}^n p_i |i\rangle\langle i| \otimes \rho_i \cong \bigoplus_{i=1}^n p_i \rho_i \quad (364)$$

The mutual information between the classical register C supporting the disjoint states $\{|1\rangle\langle 1|, \dots, |n\rangle\langle n|\}$ and the quantum register Q supporting the states $\{\rho_1, \dots, \rho_n\}$ for the state Γ_{CQ} can be shown to be equal to

$$I_{C,Q}(\Gamma_{CQ}) := H(\text{Tr}_Q(\Gamma_{CQ})) + H(\text{Tr}_C(\Gamma_{CQ})) - H(\Gamma_{CQ}) \quad (365)$$

$$= H\left(\sum_{i=1}^n p_i |i\rangle\langle i|\right) + H\left(\sum_{i=1}^n p_i \rho_i\right) - H\left(\sum_{i=1}^n p_i |i\rangle\langle i| \otimes \rho_i\right) \quad (366)$$

$$= \chi(\mathcal{E}). \quad (367)$$

because the von Neumann entropy of a CQ state associated to the ensemble $\mathcal{E}$ satisfies

$$\mu(\mathcal{E}) := H(\Gamma_{CQ}) = H\left(\sum_{i=1}^n p_i |i\rangle\langle i| \otimes \rho_i\right) = H\left(\sum_{i=1}^n p_i |i\rangle\langle i|\right) + \sum_{i=1}^n p_i H(\rho_i). \quad (368)$$

In what follows, we will be interested in how such information can be *distributed* across multiple classical registers. In particular, for an ensemble $\mathcal{E}$ of n states as in Eq. (362), we consider n classical registers $C_1, \dots, C_n$ along with the CQ state

$$\Gamma_{C_1 \dots C_n Q} = \sum_{i=1}^n p_i \Delta_i \otimes \rho_i \quad (369)$$

where Δ_i is an indicator state of the form

$$\Delta_i = |0\rangle\langle 0|^{\otimes i-1} \otimes |1\rangle\langle 1| \otimes |0\rangle\langle 0|^{\otimes n-i}. \quad (370)$$

Now, given a subset $S \subseteq \{1, \dots, n\}$ and subset of classical registers $C_S := \{C_j \mid j \in S\}$, the reduced state on $C_S \cup Q$ is given by

$$\Gamma_{C_S Q} = \text{Tr}_{C_{-S}}(\Gamma_{C_1 \dots C_n Q}) \quad (371)$$

$$= \sum_{i=1}^n p_i \text{Tr}_{C_{-S}}(\Delta_i) \otimes \rho_i \quad (372)$$

$$= \sum_{i \in S} p_i \text{Tr}_{C_{-S}}(\Delta_i) \otimes \rho_i + \sum_{i \notin S} p_i \text{Tr}_{C_{-S}}(\Delta_i) \otimes \rho_i \quad (373)$$

$$= \sum_{i \in S} p_i \Delta'_i \otimes \rho_i + |0\rangle\langle 0|^{\otimes |S|} \otimes \sum_{i \notin S} p_i \rho_i. \quad (374)$$

Note that $\Gamma_{C_S Q}$ is therefore a CQ state for the coarse-grained ensemble $\mathcal{E}_S$ where

$$\mathcal{E}_S := \{(p_j, \rho_j)\}_{j \in S} \cup \{(p_{-S}, \rho_{-S})\}, \quad \text{where} \quad p_{-S} := \sum_{j \notin S} p_j, \quad \rho_{-S} := \frac{1}{p_{-S}} \sum_{j \notin S} p_j \rho_j. \quad (375)$$

Therefore, the von Neumann entropy of the reduced state $\Gamma_{C_S Q}$ is given by

$$\mu(\mathcal{E}_S) = H(\Gamma_{C_S Q}) \quad (376)$$

$$= H\left(\sum_{i \in S} p_i \Delta'_i + p_{-S} |0\rangle\langle 0|^{\otimes |S|}\right) + \sum_{i \in S} p_i H(\rho_i) + p_{-S} H(\rho_{-S}), \quad (377)$$

$$= \sum_{i \in S} \eta(p_i) + \eta(p_{-S}) + \sum_{i \in S} p_i H(\rho_i) + p_{-S} H(\rho_{-S}), \quad (378)$$

$$= \sum_{i \in S} N_{\mathcal{E}}(\{i\}) + N_{\mathcal{E}}(\neg S), \quad (379)$$

where $N_{\mathcal{E}}(S) := \eta(p_S) + p_S H(\rho_S)$. Moreover, the mutual information between C_S and Q is equal to

$$I_{C_S, Q}(\Gamma_{C_S Q}) = H(\text{Tr}_Q(\Gamma_{C_S Q})) + H(\text{Tr}_{C_S}(\Gamma_{C_S Q})) - H(\Gamma_{C_S Q}), \quad (380)$$

$$= \sum_{i \in S} \eta(p_i) + \eta(p_{\neg S}) + H\left(\sum_{i=1}^n p_i \rho_i\right) - \sum_{i \in S} \eta(p_i) - \eta(p_{\neg S}) - \sum_{i \in S} p_i H(\rho_i) - p_{\neg S} H(\rho_{\neg S}), \quad (381)$$

$$= H\left(\sum_{i=1}^n p_i \rho_i\right) - \sum_{i \in S} p_i H(\rho_i) - p_{\neg S} H(\rho_{\neg S}), \quad (382)$$

$$= \chi(\mathcal{E}_S). \quad (383)$$

Theorem 91. Let $\mathcal{E} = \{(p_1, \rho_1), \dots, (p_n, \rho_n)\}$ be an n -ensemble and for each subset $S \subseteq \{1, \dots, n\}$ let $\mathcal{E}_S$ be the coarse-grained ensemble. Then for every pair of subsets $S, T \subseteq \{1, \dots, n\}$,

$$\mu(\mathcal{E}_S) + \mu(\mathcal{E}_T) - \mu(\mathcal{E}_{S \cup T}) - \mu(\mathcal{E}_{S \cap T}) \geq 0. \quad (384) \quad \{\text{eq:}\mu_{\text{SSA}}\}$$

Moreover, one has

$$N_{\mathcal{E}}(S) + N_{\mathcal{E}}(T) - N_{\mathcal{E}}(S \cup T) - N_{\mathcal{E}}(S \cap T) \geq 0. \quad (385) \quad \{\text{eq:}N_{\text{SSA}}\}$$

where $N_{\mathcal{E}}(S) := \eta(p_S) + p_S H(\rho_S)$.

Proof. By strong subadditivity of von Neumann entropy applied to $\Gamma_{C_1 \dots C_n Q}$ we have for every pair of subsets $S, T \subseteq \{1, \dots, n\}$,

$$I_{C_S, C_T | Q}(\Gamma_{C_1 \dots C_n Q}) := H(\Gamma_{C_S Q}) + H(\Gamma_{C_T Q}) - H(\Gamma_{C_{S \cup T} Q}) - H(\Gamma_{C_{S \cap T} Q}) \geq 0, \quad (386)$$

which is equivalent to Eq. (384) because $\mu(\mathcal{E}_S) = H(\Gamma_{C_S Q})$. Now to prove Eq. (385), note that $\mu(\mathcal{E}_S)$ can also be written as

$$\mu(\mathcal{E}_S) = \sum_{i \in S} (\eta(p_i) + p_i H(\rho_i)) + \left(\eta(p_{\neg S}) + \sum_{p \in \neg S} H(\rho_{\neg S}) \right), \quad (387)$$

$$= \sum_{i \in S} N_{\mathcal{E}}(\{i\}) + N_{\mathcal{E}}(\neg S). \quad (388)$$

Which implies for any subsets $S, T \subseteq \{1, \dots, n\}$

$$0 \leq \mu(\mathcal{E}_S) + \mu(\mathcal{E}_T) - \mu(\mathcal{E}_{S \cup T}) - \mu(\mathcal{E}_{S \cap T}) \quad (389)$$

$$= \left(\sum_{i \in S} N_{\mathcal{E}}(\{i\}) + \sum_{i \in T} N_{\mathcal{E}}(\{i\}) - \sum_{i \in S \cup T} N_{\mathcal{E}}(\{i\}) - \sum_{i \in S \cap T} N_{\mathcal{E}}(\{i\}) \right) + \dots \quad (390)$$

$$\dots + (N_{\mathcal{E}}(\neg S) + N_{\mathcal{E}}(\neg T) - N_{\mathcal{E}}(\neg(S \cup T)) - N_{\mathcal{E}}(\neg(S \cap T))), \quad (391)$$

$$= N_{\mathcal{E}}(\neg S) + N_{\mathcal{E}}(\neg T) - N_{\mathcal{E}}(\neg(S \cup T)) - N_{\mathcal{E}}(\neg(S \cap T)), \quad (392)$$

Since the above inequality holds for any subsets S, T , it also holds for $S' = \neg S$ and $T' = \neg T$. Therefore, we conclude Eq. (385) holds also. $\square$

TC : We have, as a corollary of the previous theorem, Corollary 88 itself without needing to employ some sort of limiting argument. Namely, any balanced inequality of von Neumann entropies of subsystems of multipartite quantum states is also a valid inequality of both μ and $N_{\mathcal{E}}$ quantities of subsets of ensembles of PSD matrices. This is already a large improvement on the previous proof of Corollary 88. What remains is to find a similarly elegant treatment of the notion of the additive entropy cone. I suspect it will involve some sort of downscaling of ensemble probabilities followed by an extension by a “marked” element so as to preserve normalization.

Given an n -ensemble $\mathcal{E} = \{(p_1, \rho_1), \dots, (p_n, \rho_n)\}$ of $n \times n$ density matrices, and $\epsilon \in (0, 1)$, we define an ϵ -extended n -ensemble of $n + 1 \times n + 1$ density matrices by

$$\mathcal{E}^{\epsilon} := \{(p_1, (1 - \epsilon) \oplus \epsilon \rho_1), \dots, (p_n, (1 - \epsilon) \oplus \epsilon \rho_n)\}. \quad (393)$$

Then, for any subset $S \subseteq \{1, \dots, n\}$, if ρ_S is the normalized mixture over S for the original ensemble,

$$\rho_S = \frac{\sum_{i \in S} p_i \rho_i}{\sum_{i \in S} p_i}, \quad (394)$$

then the normalized mixture over S for the ϵ -extended ensemble satisfies

$$\rho_S^\epsilon = \frac{\sum_{i \in S} p_i [(1 - \epsilon) \oplus \epsilon \rho_i]}{\sum_{i \in S} p_i} = (1 - \epsilon) \oplus \epsilon \rho_S. \quad (395)$$

The von Neumann entropy of ρ_S is related to the von Neumann entropy of ρ_S by

$$H(\rho_S^\epsilon) = h(\epsilon) + \epsilon H(\rho_S). \quad (396)$$

Moreover, as the weights in the ϵ -extended ensemble are the same as the weights in the original ensemble, we also have

$$N_{\mathcal{E}^\epsilon}(S) = \eta(p_S) + p_S H(\rho_S^\epsilon) = \eta(p_S) + p_S h(\epsilon) + p_S \epsilon H(\rho_S) \quad (397)$$

$$\mu(\mathcal{E}_S) = \sum_{i \in S} N_{\mathcal{E}}(\{i\}) + N_{\mathcal{E}}(\neg S) \quad (398)$$

$$\mu(\mathcal{E}_S^\epsilon) = \sum_{i \in S} N_{\mathcal{E}^\epsilon}(\{i\}) + N_{\mathcal{E}^\epsilon}(\neg S) \quad (399)$$

$$\Gamma_{C_1 \dots C_n Q}^\epsilon = \sum_{i=1}^n p_i \Delta_i^\epsilon \otimes \rho_i \quad (400)$$

$$\Delta_i^\epsilon = (1 - \epsilon) |\omega\rangle\langle\omega|^{\otimes n} + \epsilon \Delta_i. \quad (401)$$

6 Conclusion

A Multinomial distribution inequalities

In this section, we recall a handful of identities and inequalities concerning multinomial coefficients and the multinomial distribution in the context of Sanov's theorem [DZ10, Chapter 2]. Let $q = (q_1, \dots, q_n)$ be a probability distribution such that for all $i \in \{1, \dots, n\}$, kq_i is an integer. Let $M_k(q)$ denote the following multinomial coefficient:

$$M_k(q) := \binom{k}{kq_1, \dots, kq_n} = \frac{k!}{(kq_1)! \dots (kq_n)!}, \quad \text{\texttt{eq:multinomial_coefficient}} \quad (402)$$

which can be bounded (see [DZ10, Lemma 2.1.8]) in terms of $H(q)$ by ,

$$(k+1)^{-n} e^{kH(q)} \leq M_k(q) \leq e^{kH(q)}, \quad \text{\texttt{eq:multinomial_coefficient_bound}} \quad (403)$$

where $H(q) = -\sum_{i=1}^n q_i \log q_i$ is the Shannon entropy of q . Additionally, we have for any probability distribution $p = (p_1, \dots, p_n)$,

$$p_1^{kq_1} \dots p_n^{kq_n} = e^{-k(H(q) + H(q|p))} \quad \text{\texttt{eq:multinomial_probability}} \quad (404)$$

where $H(q|p) = \sum_{i=1}^n q_i \log q_i/p_i$ denotes the Kullback-Leibler divergence or relative entropy of q from p with the convention that $H(q|p) = \infty$ if $q_i > 0$ whenever $p_i = 0$. Note that $H(q|p) = 0$ if and only if $p = q$. Combining Eq. (403) with Eq. (404) places bounds on the multinomial distribution

$$(k+1)^{-n} e^{-kH(q|p)} \leq \binom{k}{kq_1, \dots, kq_n} p_1^{kq_1} \dots p_n^{kq_n} \leq e^{-kH(q|p)} \quad (405)$$

If needed, it is possible to relax these bounds further using Pinsker's inequality for $H(q|p)$ in conjunction with its reverse (which only holds for $H(q|p) < \infty$) [GSS19]:

$$2d_{\text{TV}}^2(p, q) \leq H(q|p) \leq \frac{4}{p_{\min}} d_{\text{TV}}^2(p, q) \quad (406)$$

where

$$d_{\text{TV}}(p, q) = \max_{1 \leq j \leq n} |p_j - q_j| = \frac{1}{2} \sum_{j=1}^n |p_j - q_j| \quad \text{\texttt{eq:pinskera}} \quad (407)$$

is known as the *total variation distance* between p and q . Now for any distribution $p = (p_1, \dots, p_n)$ and any positive integer k , it is possible to find another distribution $p' = (p'_1, \dots, p'_n)$ such that (i) kp'_i is an integer for all $i \in \{1, \dots, n\}$, (ii) $H(p'|p) < \infty$ is finite, and (iii) p' approximates p in the sense that for some constant $C \geq 0$,

$$d_{\text{TV}}(p, p') \leq \frac{C}{k}. \quad \text{\texttt{eq:tv_approx}} \quad (408)$$

Concretely, one can choose p' such that

$$kp'_j := \text{nint} \left(k \sum_{i=1}^j p_i \right) - \text{nint} \left(k \sum_{i=1}^{j-1} p_i \right) \quad \text{\texttt{eq:nint_approx}} \quad (409)$$

where $\text{nint} : \mathbb{R} \rightarrow \mathbb{Z}$ is the function which rounds each real number to the nearest integer (with half integers always rounded up). Using this approximation, we have $|kp_j - kp'_j| \leq 1$ because each rounding operation in Eq. (409) introduces an error of at most $1/2$ and thus Eq. (408) holds with $C = 1$. In any case, when Eq. (408) holds for some constant $C \geq 0$, Eq. (407) reveals $H(p'|p) \leq \frac{4C}{p_{\min} k^2}$ which yields a useful, albeit crude, lower bound for the multinomial distribution:

$$\binom{k}{kp'_1, \dots, kp'_n} p_1^{kp'_1} \dots p_n^{kp'_n} \geq \exp \left(-\frac{4C^2}{p_{\min} k} - n \log(k+1) \right). \quad (410)$$

\texttt{lem:multinomial_dist_lower_bound}

Lemma 92. Let $n \geq 2$ and let $(w_1, \dots, w_n) \in \mathbb{R}_{>0}^n$ have minimum $w_{\min} := \min\{w_1, \dots, w_n\} > 0$ and let k be a positive integer satisfying $k \geq w_{\min}^{-1}$ and define positive integers $(z_1, \dots, z_n) \in \mathbb{N}_{>0}^n$ by the equation

$$z_i := \lceil kw_i \rceil. \quad (411)$$

Let $Z = z_1 + \dots + z_n$ and $W = w_1 + \dots + w_n$ and defined probability distributions $\bar{z}$ and $\bar{w}$ by

$$\bar{z}_i = \frac{z_i}{Z}, \quad \text{and} \quad \bar{w}_i = \frac{w_i}{W}, \quad (412)$$

for all $i \in \{1, \dots, n\}$. Then

$$\frac{Z!}{z_1! \dots z_n!} \frac{w_1^{z_1} \dots w_n^{z_n}}{W^Z} \geq e^{-4(n-1)^2} (kW + n + 1)^{-n} \quad \text{\textcolor{blue}{\code{eq:multinomial_dist_lower_bound}}} \quad (413)$$

Proof. Let $\delta_j := z_j - kw_j$ denote the rounding surplus such that $0 \leq \delta_j < 1$. Note that for $k \gg n$, we have $Z \approx kW$ because

$$kW \leq Z < kW + n. \quad (414)$$

In this regime, we wish to show that $\bar{z}_i \approx \bar{w}_i$.

$$|\bar{z}_i - \bar{w}_i| = \left| \frac{z_i}{Z} - \frac{w_i}{W} \right| = \frac{|Wz_i - Zw_i|}{ZW}. \quad (415)$$

The numerator above can be bounded independently of k by,

$$|Wz_i - Zw_i| \quad (416)$$

$$= |(kw_i + \delta_i)(w_1 + \dots + w_n) - w_i(\delta_1 + kw_1 + \dots + \delta_n + kw_n)|, \quad (417)$$

$$= |\delta_i(w_1 + \dots + w_n) - w_i(\delta_1 + \dots + \delta_n)|, \quad (418)$$

$$\leq |\delta_i(w_1 + \dots + w_n)| + |w_i(\delta_1 + \dots + \delta_n)|, \quad (419)$$

$$\leq (w_1 + \dots + w_n) + (n-1)(w_i), \quad (420)$$

$$\leq (n-1)(w_1 + \dots + w_n), \quad (421)$$

$$= (n-1)W. \quad (422)$$

Consequently,

$$d_{\text{TV}}(\bar{z}, \bar{w}) = \max_{1 \leq i \leq n} |\bar{z}_i - \bar{w}_i| \leq \frac{n-1}{Z} \quad (423)$$

and therefore

$$\frac{Z!}{z_1! \dots z_n!} \frac{w_1^{z_1} \dots w_n^{z_n}}{W^Z} \geq (Z+1)^{-n} \exp(-ZH(\bar{z}|\bar{w})), \quad (424)$$

$$\geq (Z+1)^{-n} \exp \left[-\frac{4ZW}{w_{\min}} \left(\frac{n-1}{Z} \right)^2 \right], \quad (425)$$

$$\geq (kW + n + 1)^{-n} \exp \left[-\frac{4(n-1)^2}{kw_{\min}} \right], \quad (426)$$

$$\geq (kW + n + 1)^{-n} \exp(-4(n-1)^2). \quad (427)$$

□

The following is an alternative lower bound on the multinomial distribution based on the Stirling approximation which would work just as well in the main text. TODO: Figure out the best lower bound on the multinomial distribution to use here.

\code{lem:multinomial_dist_lower_bound_alt}

Lemma 93. *Let everything be as in Lemma 92. Then*

$$\frac{Z!}{z_1! \dots z_n!} \frac{w_1^{z_1} \dots w_n^{z_n}}{W^Z} \geq \left(\frac{1}{20} \right)^n \left(\frac{W}{w_1 \dots w_n} \right)^{\frac{1}{2}} \frac{\text{\textcolor{blue}{\code{eq:multinomial_dist_lower_bound_alt}}}}{k^{\frac{3n-1}{2}}} \quad (428)$$

Proof. The proof follows from a version of Stirling's approximation due to Robbins [Rob55] which states

$$k! = \sqrt{2\pi} k^{k+\frac{1}{2}} e^{-k} e^{r_k}, \quad \text{where} \quad \frac{1}{12k+1} \leq r_k \leq \frac{1}{12k}. \quad (429)$$

Therefore,

$$\frac{Z!}{z_1! \cdots z_n!} \frac{w_1^{z_1} \cdots w_n^{z_n}}{W^Z} = \frac{Z!}{W^Z} \frac{w_1^{z_1}}{z_1!} \cdots \frac{w_n^{z_n}}{z_n!} \quad (430)$$

$$= \left(\frac{1}{2\pi}\right)^{\frac{n-1}{2}} \left(\frac{W}{w_1 \cdots w_n}\right)^{\frac{1}{2}} \left(\frac{Z}{W}\right)^{Z+\frac{1}{2}} \left(\frac{w_1}{z_1}\right)^{z_1+\frac{1}{2}} \cdots \left(\frac{w_n}{z_n}\right)^{z_n+\frac{1}{2}} \exp\left(r_Z - \sum_i r_{z_i}\right). \quad (431)$$

In this setting, we have assumed $kw_i \geq 1$ which implies

$$kw_i \leq z_i < kw_i + 1 \leq 2kw_i, \quad \text{and} \quad kW \leq Z < kW + n < 2kW. \quad (432)$$

Consequently⁸,

$$\left(\frac{Z}{W}\right)^{Z+\frac{1}{2}} \geq k^{kW+\frac{1}{2}}, \quad \left(\frac{w_i}{z_i}\right)^{z_i+\frac{1}{2}} \geq \left(\frac{w_i}{kw_i+1}\right)^{kw_i+\frac{3}{2}} \geq \frac{2^{-\frac{5}{2}}}{k^{kw_i+\frac{3}{2}}}. \quad (433)$$

and the remainder term satisfies

$$r_Z - \sum_i r_{z_i} \geq \frac{1}{12Z+1} - \sum_i \frac{1}{12z_i} \geq -\frac{n}{12}. \quad (434)$$

Altogether, we have

$$\frac{Z!}{z_1! \cdots z_n!} \frac{w_1^{z_1} \cdots w_n^{z_n}}{W^Z} = \left(\frac{1}{2\pi}\right)^{\frac{n-1}{2}} \left(\frac{W}{w_1 \cdots w_n}\right)^{\frac{1}{2}} \frac{2^{-\frac{5n}{2}}}{k^{\frac{3n-1}{2}}} \exp\left(-\frac{n}{12}\right). \quad (435)$$

Moreover,

$$\left(\frac{1}{2\pi}\right)^{\frac{n-1}{2}} 2^{-\frac{5n}{2}} \exp\left(-\frac{n}{12}\right) \geq \left(\frac{1}{20}\right)^n \quad (436)$$

□

⁸Here we use the identity that $x \geq 1$ implies $\left(\frac{x}{x+1}\right)^{x+\frac{3}{2}} \geq 2^{-5/2}$

B Thompson's conjecture for 2×2 matrices

[sec:2by2case_intuitively](#)

The purpose of this section is to sketch how to prove Thompson's conjecture for the case of 2×2 matrices. Given $\lambda \geq 0$, let $\mathcal{O}_\lambda$ denotes the set of 2×2 traceless Hermitian operators with eigenvalues $\pm\lambda$ and $\mathcal{C}_{\exp(\lambda)}$ denote the set of 2×2 unit determinant Hermitian operators with singular-values $\exp(\pm\lambda)$. Note that the matrix exponential $\exp : \mathcal{O}_\lambda \rightarrow \mathcal{C}_{\exp(\lambda)}$ here is a bijective map with inverse $\log : \mathcal{C}_{\exp(\lambda)} \rightarrow \mathcal{O}_\lambda$.

The central idea is to show that exists a diffeomorphism on $\mathcal{O}_\alpha \times \mathcal{O}_\beta$, sending each pair of matrices (A, B) with eigenvalues $(\pm\alpha, \pm\beta)$ to another pair of matrices

$$(A, B) \mapsto (\tilde{A}, \tilde{B}) \in \mathcal{O}_\alpha \times \mathcal{O}_\beta \quad (437)$$

with the same eigenvalues as (A, B) but with the additional property:

$$A + B \in \mathcal{O}_\gamma \iff \exp(\tilde{A}) \exp(\tilde{B}) \in \mathcal{C}_{\exp(\gamma)}. \quad (438)$$

From the existence of this map one can therefore prove Thompson's conjecture: there exists $(A, B) \in \mathcal{O}_\alpha \times \mathcal{O}_\beta$ satisfying $A + B \in \mathcal{O}_\gamma$ if and only if there exists $(X, Y) \in \mathcal{C}_{\exp(\alpha)} \times \mathcal{C}_{\exp(\beta)}$ satisfying $XY \in \mathcal{C}_{\exp(\gamma)}$. For the only if portion of the proof, simply take (X, Y) to be $(\exp(\tilde{A}), \exp(\tilde{B}))$. For the if portion of the proof, define $(\tilde{A}, \tilde{B})$ to equal $(\log X, \log Y)$ and then apply the inverse of the above diffeomorphism to obtain the desired (A, B) from $(\tilde{A}, \tilde{B})$.

B.1 Euclidean triangles and eigenvalues

Letting σ_1, σ_2 and σ_3 be the 2×2 Pauli matrices,

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (439)$$

it becomes possible to identify the space of traceless 2×2 Hermitian matrices, $\mathfrak{su}(2)$, with $\mathbb{R}^3$ via the standard parameterization associating $(x_1, x_2, x_3) \in \mathbb{R}^3$ to

$$X = x \cdot \sigma = x_1\sigma_1 + x_2\sigma_2 + x_3\sigma_3 = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}. \quad (440)$$

The spectrum of X obtained in this way is simply $(+\chi, -\chi)$ where $\chi = \|x\|$ and $\|x\|$ the Euclidean norm of the vector $x \in \mathbb{R}^3$

$$\|x\| = \sqrt{x_1^2 + x_2^2 + x_3^2}. \quad (441)$$

From this point of view, it is possible to characterize the set of possible spectra, $(\alpha, \beta, \gamma) \in \mathbb{R}^3$, for the set of two Hermitian operators,

$$A = a_1\sigma_1 + a_2\sigma_2 + a_3\sigma_3, \quad B = b_1\sigma_1 + b_2\sigma_2 + b_3\sigma_3 \quad (442)$$

with spectra $(\alpha, \beta) = (\|a\|, \|b\|)$ and there sum $A + B$

$$A + B = (a_1 + b_1)\sigma_1 + (a_2 + b_2)\sigma_2 + (a_3 + b_3)\sigma_3, \quad (443)$$

with spectrum $\gamma = \|a + b\|$ where

$$\|a + b\|^2 = \|a\|^2 + \|b\|^2 + 2\|a\|\|b\|\cos\theta, \quad \cos\theta = \frac{a \cdot b}{\|a\|\|b\|}. \quad (444)$$

As the only parameter not fixed by spectra is the angle θ , it is easy to see that $(\alpha, \beta, \gamma) \in \mathbb{R}^3$ is precisely determined by the triangle inequalities for the Euclidean norm in $\mathbb{R}^3$:

$$|||a\| - \|b\|| \leq \|a + b\| \leq \|a\| + \|b\|, \quad (445)$$

$$|\alpha - \beta| \leq \gamma \leq \alpha + \beta, \quad (446)$$

where the extreme cases are characterized by anti-alignment $\cos \theta = -1$ and alignment $\cos \theta = 1$ respectively.

$$(447)$$

In summary, given a pair $(A, B) = (a \cdot \sigma, b \cdot \sigma)$ of traceless 2×2 Hermitian operators with spectra $(\pm\alpha, \pm\beta)$ we saw that the spectrum $\pm\gamma$ of their sum, $A + B$, is given by the Euclidean cosine law with $\cos \theta = \frac{a \cdot b}{\alpha\beta}$ ⁹

$$\gamma^2 = \alpha^2 + \beta^2 + 2\alpha\beta \cos \theta. \quad (448)$$

B.2 Hyperbolic triangles and singular values

An analogous calculation can be performed to character the spectra of certain invertible matrices in $\text{SL}(2, \mathbb{C})$ (obtained by exponentiation of the traceless Hermitian matrices previously considered) and their products. Given the traceless Hermitian matrix $X = x \cdot \sigma$, the identity $X^2 = \|x\|^2 I$ yields a well-known pair of results (see [FT14]):

$$\exp(iX) = \cos \|x\| I + i \frac{\sin \|x\|}{\|x\|} X, \quad \exp(X) = \cosh \|x\| I + \frac{\sinh \|x\|}{\|x\|} X. \quad (449)$$

As before, consider traceless Hermitian matrices, $A = a \cdot \sigma$ and $B = b \cdot \sigma$ with $\alpha = \|a\|$ and $\beta = \|b\|$. Now consider their exponentials $\exp(A)$ and $\exp(B)$ and their product $\exp(A)\exp(B)$:

$$\exp(A)\exp(B) = \left(\cosh \alpha I + \frac{\sinh \alpha}{\alpha} A \right) \left(\cosh \beta I + \frac{\sinh \beta}{\beta} B \right) \quad (450)$$

Notice the singular values of $\exp(A)$ and $\exp(B)$ are given by $\exp(\pm\alpha)$ and $\exp(\pm\beta)$. The singular values of their product, $\exp(A)\exp(B)$, are however equal to the positive square-root of the eigenvalues of

$$\exp(A)\exp(B)(\exp(A)\exp(B))^\dagger = \exp(A)\exp(B)^2\exp(A). \quad (451)$$

Using the identity $\{A, B\} = AB + BA = 2(a \cdot b)I$ yields

$$\exp(A)\exp(B)^2\exp(A) \quad (452)$$

$$= \cosh(2\alpha) \cosh(2\beta) I + \frac{\sinh(2\alpha) \cosh(2\beta)}{\alpha} A + \frac{\sinh(2\beta)}{\beta} B \dots \quad (453)$$

$$\dots + \sinh(2\alpha) \sinh(2\beta) \cos \theta I + \frac{2 \sinh^2 \alpha \sinh(2\beta) \cos \theta}{\alpha} A \quad (454)$$

While there is an asymmetry in $\exp(A)\exp(B)^2\exp(A)$ with respect to $A \leftrightarrow B$, this asymmetry disappears when calculating the eigenvalues; the eigenvalues are given by $\eta \pm \delta$ where $\eta^2 - \delta^2 = 1$ and

$$\eta = \cosh(2\alpha) \cosh(2\beta) + \sinh(2\alpha) \sinh(2\beta) \cos \theta, \quad (455)$$

$$\delta = \frac{1}{4} \left(8 \sinh^2(2\alpha) \sinh^2(2\beta) \cos(2\theta) + 8 \sinh(4\alpha) \sinh(4\beta) \cos(\theta) + \dots \right. \quad (456)$$

$$\left. \dots + 3 \cosh(4(\alpha - \beta)) + 3 \cosh(4(\alpha + \beta)) + 2 \cosh(4\alpha) + 2 \cosh(4\beta) - 10 \right)^{\frac{1}{2}}. \quad (457)$$

Alternatively, defining $\gamma' \geq 0$ such that

$$\cosh(2\gamma') = \eta, \quad \sinh(2\gamma') = \delta, \quad (458)$$

⁹Note that the angle θ used here is not the interior angle typically used in the statement of cosine law, hence the difference in minus sign.

One has that the singular values of $\exp(A)\exp(B)$ are equal to $\exp(\pm\gamma')$. Once again, there exists only one parameter that is not fixed by specifying the singular values ($\exp(\pm\alpha), \exp(\pm\beta)$) of $\exp(A)$ and $\exp(B)$; specifically the angle $\theta \in [0, 2\pi)$ fixes the singular values of $\exp(A)\exp(B)$ by

$$\cosh(2\gamma') = \cosh(2\alpha) \cosh(2\beta) + \sinh(2\alpha) \sinh(2\beta) \cos \theta, \quad (459)$$

which is the *hyperbolic* cosine law with Gaussian curvature -4 (again note that the angle θ used is not interior to the hyperbolic triangle, hence the difference in minus sign).

Also note the singular values $\exp(\pm\gamma')$ of $\exp(A)\exp(B)$ at the values of $\theta \in \{0, \pi\}$ are

$$\theta = 0 \implies \exp(\pm(\alpha + \beta)), \quad \theta = \pi \implies \exp(\pm(\alpha - \beta)). \quad (460)$$

B.3 Thompson's conjecture via angle modifications

Thompson's conjecture in the context of 2×2 matrices corresponds to the following conjecture: there exists a pair (A, B) of traceless 2×2 Hermitian matrices with spectra $(\pm\alpha, \pm\beta)$ such that $A + B$ has spectra $\pm\gamma$ if and only if there exists a pair (X, Y) of unit determinant positive-definite Hermitian matrices with singular values $(\exp(\pm\alpha), \exp(\pm\beta))$ such that XY has singular values $\exp(\pm\gamma)$.

By the above analysis, letting $(X, Y) = (\exp A, \exp B)$ does *not* prove the conjecture; while the singular values of $(\exp A, \exp B)$ are $(\exp(\pm\alpha), \exp(\pm\beta))$ as desired, the singular values of their product $\exp A \exp B$ are equal to $\exp(\pm\gamma')$ which, in general, are not equal to $\exp(\pm\gamma)$ (the singular values of $\exp(A + B)$). This difference is to be expected because in general, one has

$$\exp(A)\exp(B) = \exp(A + B + \frac{1}{2}[A, B] + \dots) \neq \exp(A + B), \quad (461)$$

or alternatively

$$\exp(A)\exp(B)^2\exp(A) \neq \exp(2(A + B)). \quad (462)$$

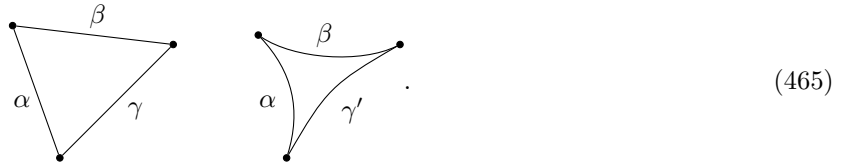
Nevertheless, there is a relationship between γ and γ' provided by the Golden-Thompson inequality $\text{Tr}(\exp(A' + B')) \leq \text{Tr}(\exp(A')\exp(B'))$. Letting $A' = 2A$ and $B' = 2B$ implies

$$\text{Tr}(\exp(2(A + B))) \leq \text{Tr}(\exp(2A)\exp(2B)) = \text{Tr}(\exp(A)\exp(B)^2\exp(A)). \quad (463)$$

Or by dividing by 2, one obtains

$$\cosh(2\gamma) \leq \cosh(2\gamma') \implies \gamma \leq \gamma', \quad (464)$$

which essentially states that the side-lengths, γ' , of hyperbolic triangles are generally longer than their Euclidean counter-parts, γ :



Therefore, in order to prove Thompson's conjecture in the context of 2×2 matrices, one needs to pick the pair (X, Y) to be something other than $(\exp(A), \exp(B))$ but with the *same* singular values. The solution comes from choosing a different pair $(\tilde{A}, \tilde{B}) = (\tilde{a} \cdot \sigma, \tilde{b} \cdot b)$ of Hermitian matrices with the *same* spectra $(\pm\alpha, \pm\beta)$ as (A, B) such that $\exp(\tilde{A})\exp(\tilde{B})$ has as singular values $\exp(\pm\gamma)$ where $(\pm\gamma)$ are the eigenvalues of $A + B$.

As we have shown, the only freedom in this choice for $(\tilde{A}, \tilde{B})$ is the angle $\tilde{\theta}$ between $\tilde{a}$ and $\tilde{b}$:

$$\cos \tilde{\theta} = \frac{\tilde{a} \cdot \tilde{b}}{\alpha\beta}. \quad (466)$$

The value $\tilde{\theta}$ determines the value $\tilde{\gamma}'$, and thus the singular values of $\exp(\tilde{A})\exp(\tilde{B})$ by the previously derived formula:

$$\cosh(2\tilde{\gamma}') = \cosh(2\alpha) \cosh(2\beta) + \sinh(2\alpha) \sinh(2\beta) \cos \tilde{\theta}. \quad (467)$$

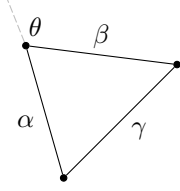
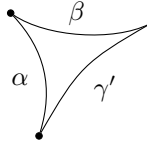
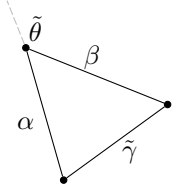
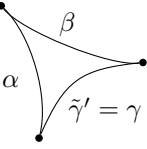
Setting $\tilde{\gamma}' = \gamma$ and solving for $\tilde{\theta}$ yields

$$\cos \tilde{\theta} = \cosh(2\gamma) \operatorname{csch}(2\alpha) \operatorname{csch}(2\beta) - \tanh(2\alpha) \tanh(2\beta). \quad (468)$$

Recalling that γ itself is fixed by the angle θ between the original (A, B) , one obtains a formula for $\tilde{\theta}$ in terms of (θ, α, β) :

$$\cos \tilde{\theta} = \cosh(2\sqrt{\alpha^2 + \beta^2 + 2\alpha\beta \cos \theta}) \operatorname{csch}(2\alpha) \operatorname{csch}(2\beta) - \tanh(2\alpha) \tanh(2\beta). \quad (469)$$

In summary, when the spectra α, β , and γ are fixed and there exists a pair of Hermitian matrices (A, B) with spectra $(\pm\alpha, \pm\beta)$ whose sum $A + B$ has spectra $\pm\gamma$ (thus there exists a value for θ), the above formula describes precisely how to reduce the angle θ between A and B to the value $\tilde{\theta}$ obtaining another pair of Hermitian matrices $(\tilde{A}, \tilde{B})$ such that $(\exp(\tilde{A}), \exp(\tilde{B}))$ have singular values $(\exp(\pm\alpha), \exp(\pm\beta))$ and their product $\exp(\tilde{A})\exp(\tilde{B})$ has singular values $\exp(\pm\gamma)$.

 $\lambda(A) = \pm\alpha$ $\lambda(B) = \pm\beta$ $\lambda(A + B) = \pm\gamma$	 $\sigma(\exp(A)) = \exp(\pm\alpha)$ $\sigma(\exp(B)) = \exp(\pm\beta)$ $\sigma(\exp(A)\exp(B)) = \exp(\pm\gamma')$	$\gamma \leq \gamma'$ $\tilde{\gamma} \leq \tilde{\gamma}'$
(470)		
 $\lambda(\tilde{A}) = \pm\alpha$ $\lambda(\tilde{B}) = \pm\beta$ $\lambda(\tilde{A} + \tilde{B}) = \pm\tilde{\gamma}$	 $\sigma(\exp(\tilde{A})) = \exp(\pm\alpha)$ $\sigma(\exp(\tilde{B})) = \exp(\pm\beta)$ $\sigma(\exp(\tilde{A})\exp(\tilde{B})) = \exp(\pm\gamma)$	$\tilde{\gamma} \leq \gamma$ $\tilde{\gamma}' = \gamma$

Of course, this only proves one half of Thompson's conjecture in this setting, one also needs to prove the reverse: given positive definite (X, Y, XY) with singular values $\exp(\pm\alpha), \exp(\pm\beta)$ and $\exp(\pm\gamma)$, find the corresponding Hermitian matrices (A, B) . This can be done simply by defining $(\tilde{A}, \tilde{B}) = (\log X, \log Y)$ with separated by angle $\tilde{\theta}$ and then modifying this angle to obtain (A, B) .

B.4 Scaling hyperbolic curvature

In the previous section, it was noted that in general $\exp(A)\exp(B) \neq \exp(A + B)$ because of the Baker-Campbell-Hausdorff term:

$$\exp(A)\exp(B) = \exp(A + B + \frac{1}{2}[A, B] + \dots) \neq \exp(A + B). \quad (471)$$

If $(A, B) \in \mathcal{O}_\alpha \times \mathcal{O}_\beta$ is replaced with $(sA, sB) \in \mathcal{O}_{s\alpha} \times \mathcal{O}_{s\beta}$ above for some $s \geq 0$, then one obtains

$$\exp(sA)\exp(sB) = \exp(sA + sB + \frac{s^2}{2}[A, B] + \dots) \neq \exp(s(A + B)). \quad (472)$$

Consequently, one still should expect the singular values of $\exp(sA)\exp(sB)$ to be different from the singular values, $\exp(\pm s\gamma)$, of $\exp(s(A + B))$. Nevertheless, for small $s \approx 0$, one might expect an approximation of the form

$$\exp(sA)\exp(sB) \approx \exp(s(A + B)). \quad (473)$$

and thus the singular values of $\exp(sA)\exp(sB)$ might approximate $\exp(\pm s\gamma)$ in a suitable sense. This intuition is also encoded in the hyperbolic geometry; rescaling $(\alpha, \beta) \mapsto (s\alpha, s\beta)$ amounts to changing the singular values $\exp(\pm\gamma')$ of $\exp(A)\exp(B)$ to $\exp(\pm s\gamma'_s)$ where γ'_s is given by the formula

$$\cosh(2s\gamma'_s) = \cosh(2s\alpha) \cosh(2s\beta) + \sinh(2s\alpha) \sinh(2s\beta) \cos \theta. \quad (474)$$

This formula is the hyperbolic cosine law for triangles in hyperbolic space whose Gaussian curvature is $-4s^2$, meaning γ'_s is the side length of a hyperbolic triangle whose opposite sides are α, β with opposite angle $\pi - \theta$ in a space with Gaussian curvature $-4s^2$. For small $s \approx 0$, space is approximately Euclidean: $\cosh(2sx) = 1 + 2s^2x^2 + O(s^4)$ and $\sinh(2sx) = 2sx + O(s^3)$ together yield

$$1 + 2s^2(\gamma'_s)^2 = 1 + 2s^2\alpha^2 + 2s^2\beta^2 + 4s^2\alpha\beta \cos \theta + O(s^3). \quad (475)$$

Simplifying yields the expected result that $\gamma'_s \approx \gamma$ when $s \approx 0$:

$$(\gamma'_s)^2 = \alpha^2 + \beta^2 + 2\alpha\beta \cos \theta + O(s) = \gamma^2 + O(s). \quad (476)$$

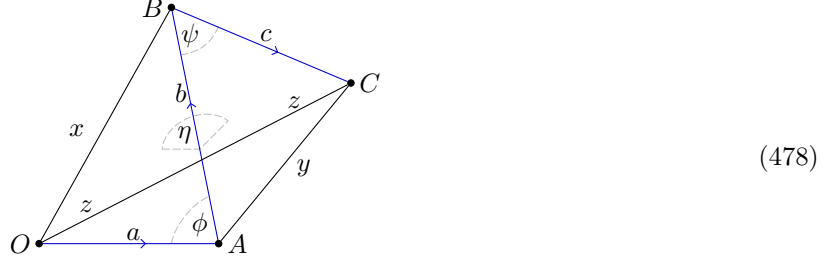
B.5 Admissibility of Hyperbolic tetrahedron

TODO, TC: Type up the proof from my tablet notes “Positive definite products of 2 by 2 matrices” The space of admissible (a, b, c, x, y, z) is given by triangle inequalities on x, y in terms of a, b, c as well as the following constraint on z in terms of x, y, a, b, c :

$$\begin{aligned} & + 2 \cosh(2s(a - b - c - z)) + 2 \cosh(2s(a + b - c - z)) + 2 \cosh(2s(a - b + c - z)) \\ & + 2 \cosh(2s(a + b + c - z)) + 2 \cosh(2s(a - b - c + z)) + 2 \cosh(2s(a + b - c + z)) \\ & + 2 \cosh(2s(a - b + c + z)) + 2 \cosh(2s(a + b + c + z)) + 2 \cosh(2s(a - c - x - y)) \\ & + 2 \cosh(2s(a + c - x - y)) + 2 \cosh(2s(a - c + x - y)) + 2 \cosh(2s(a + c + x - y)) \\ & + 2 \cosh(2s(a - c - x + y)) + 2 \cosh(2s(a + c - x + y)) + 2 \cosh(2s(a - c + x + y)) \\ & + 2 \cosh(2s(a + c + x + y)) + 2 \cosh(2s(b - x - y - z)) + 2 \cosh(2s(b + x - y - z)) \\ & + 2 \cosh(2s(b - x + y - z)) + 2 \cosh(2s(b + x + y - z)) + 2 \cosh(2s(b - x - y + z)) \\ & + 2 \cosh(2s(b + x - y + z)) + 2 \cosh(2s(b - x + y + z)) + 2 \cosh(2s(b + x + y + z)) \\ & + 2 \cosh(4as) + 2 \cosh(4bs) + 2 \cosh(4cs) + 2 \cosh(4sx) + 2 \cosh(4sy) + 2 \cosh(4sz) + 10 \\ & \geq \\ & + 4 \cosh(2s(a - b - x)) + 4 \cosh(2s(a + b - x)) + 4 \cosh(2s(a - b + x)) \\ & + 4 \cosh(2s(a + b + x)) + 4 \cosh(2s(a - y - z)) + 4 \cosh(2s(a + y - z)) \\ & + 4 \cosh(2s(a - y + z)) + 4 \cosh(2s(a + y + z)) + 4 \cosh(2s(b - c - y)) \\ & + 4 \cosh(2s(b + c - y)) + 4 \cosh(2s(b - c + y)) + 4 \cosh(2s(b + c + y)) \\ & + 4 \cosh(2s(c - x - z)) + 4 \cosh(2s(c + x - z)) + 4 \cosh(2s(c - x + z)) \\ & + 4 \cosh(2s(c + x + z)) \\ & + \cosh(4s(x - y)) + \cosh(4s(x + y)) + \cosh(4s(a - c)) \\ & + \cosh(4s(a + c)) + \cosh(4s(b - z)) + \cosh(4s(b + z)) \end{aligned} \quad (477)$$

B.6 Joint statistics for partial sums of traceless 2×2 matrices

In this section we consider a three-step random walk in the symmetric space $\mathfrak{su}(2) \simeq \mathbb{E}^3$ starting at the origin $O = (0,0,0)$ and terminating at C with intermediate steps A, B .



Supposing the step sizes $(a, b, c) \in \mathbb{R}_{\geq 0}^3$ are fixed, we are interested in the induced correlations between the interval sizes $(x, y, z) \in \mathbb{R}_{\geq 0}^3$ when the angles planar angles $(\phi, \psi) \in [0, \pi]$ and dihedral angle $\eta \in [0, \pi]$ are sampled independently according to the measures $\frac{1}{2} \sin \phi d\phi$, $\frac{1}{2} \sin \psi d\psi$ and $\pi^{-1} d\eta$ respectively (note it is a non-trivial observation that sampling the vectors A, B , and C uniformly from spheres of radii a, b , and c induces the same correlations between x, y , and z as the correlations induced by the random angles ϕ, ψ and η).

Begin by noting that x and y are fixed by ϕ and ψ respectively by the law of cosines:

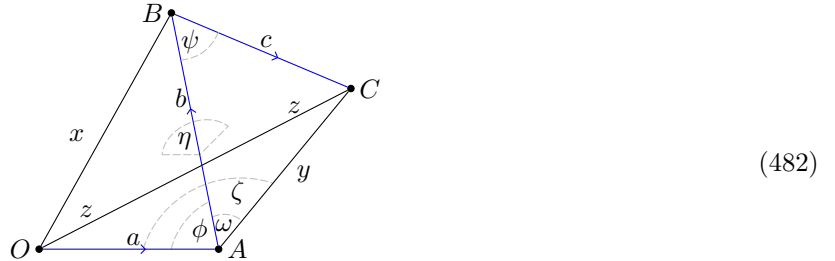
$$x^2 = a^2 + b^2 - 2ab \cos \phi, \quad (479)$$

$$y^2 = b^2 + c^2 - 2bc \cos \psi. \quad (480)$$

As x is functionally (and thus statistically) independent of ϕ and η and y is independent of ψ and η , one can expect x and y to be independent of each other and thus the support of the marginal distribution over x and y is a rectangle:

$$(x, y) \in [|a - b|, a + b] \times [|b - c|, b + c] \quad (481)$$

Comparatively less obvious is the relationship between z and the variables a, b, c, x, y . To process, we begin by introducing two more planar angles, ω and ζ , at the vertex A :



The cosine laws for the planar angles incident to A are

$$\cos \phi = \frac{a^2 + b^2 - z^2}{2ab}, \quad \cos \omega = \frac{b^2 + y^2 - c^2}{2by}, \quad \cos \zeta = \frac{a^2 + y^2 - z^2}{2ay}. \quad (483)$$

To make progress, one needs to use a relationship holding between the dihedral angle η and the other specified variables. For instance [\[Ric02\]](#),

$$\cos \eta = \frac{\cos \zeta - \cos \phi \cos \omega}{\sin \phi \sin \omega}. \quad (484)$$

Combining the above identities, one obtains

$$z^2 = a^2 + y^2 - 2ay \cos \zeta, \quad (485)$$

$$\cos \zeta = \cos \eta \sin \phi \sin \omega + \cos \phi \cos \omega, \quad (486)$$

$$\cos \omega = \frac{b - c \cos \psi}{\sqrt{b^2 + c^2 - 2bc \cos \psi}}, \quad (487)$$

$$\sin \omega = \frac{c \sin \psi}{\sqrt{b^2 + c^2 - 2bc \cos \psi}}. \quad (488)$$

Thus, one can determine z as a function of the side lengths a, b, c and the angles η, ϕ, ψ :

$$z^2 = a^2 + y^2 - 2ay \cos \zeta, \quad (489)$$

$$= a^2 + y^2 - 2ay \cos \eta \sin \phi \sin \omega - 2ay \cos \phi \cos \omega, \quad (490)$$

$$= a^2 + b^2 + c^2 - 2bc \cos \psi - 2a \cos \eta \sin \phi (c \sin \psi) - 2a \cos \phi (b - c \cos \psi), \quad (491)$$

$$= a^2 + b^2 + c^2 - 2bc \cos \psi - 2ac \cos \eta \sin \phi \sin \psi + 2ac \cos \phi \cos \psi - 2ab \cos \phi, \quad (492)$$

$$= a^2 + b^2 + c^2 - 2bc \cos \psi - 2ab \cos \phi - 2ac(\cos \eta \sin \phi \sin \psi - \cos \phi \cos \psi), \quad (493)$$

or alternatively as function of a, b, c and x, y, η :

$$\begin{aligned} z^2 = & x^2 + y^2 - b^2 + \frac{(a^2 + b^2 - x^2)(b^2 + c^2 - y^2)}{2b^2} + \dots \\ & \dots - \cos \eta \frac{\sqrt{4a^2b^2 - (a^2 + b^2 - x^2)^2} \sqrt{4b^2c^2 - (b^2 + c^2 - y^2)^2}}{2b^2} \end{aligned} \quad (494)$$

Consequently, the admissible values of z (found by varying $\cos \eta \in [-1, +1]$) depend upon x, y according to $Z_-(x, y) \leq z \leq Z_+(x, y)$ where

$$\begin{aligned} Z_{\pm}(x, y) = & \left(x^2 + y^2 - b^2 + \frac{(a^2 + b^2 - x^2)(b^2 + c^2 - y^2)}{2b^2} + \dots \right. \\ & \left. \dots \pm \frac{\sqrt{4a^2b^2 - (a^2 + b^2 - x^2)^2} \sqrt{4b^2c^2 - (b^2 + c^2 - y^2)^2}}{2b^2} \right)^{\frac{1}{2}} \end{aligned} \quad (495)$$

The change of variables from (ϕ, ψ, η) to (x, y, z) (for constant a, b, c) is

$$\begin{pmatrix} x^2 \\ y^2 \\ z^2 \end{pmatrix} = \begin{pmatrix} a^2 + b^2 - 2ab \\ b^2 + c^2 - 2bc \\ a^2 + b^2 + c^2 - 2bc \cos \psi - 2ab \cos \phi - 2ac(\cos \eta \sin \phi \sin \psi - \cos \phi \cos \psi) \end{pmatrix}. \quad (496)$$

After taking square-roots, it has a Jacobian determinant equal to

$$\det J(\phi, \psi, \eta) = \frac{a^2 b^2 c^2 \sin \eta \sin^2 \phi \sin^2 \psi}{xyz} \quad (497)$$

If (ϕ, ψ, η) is sampled according to the measure

$$\frac{1}{4\pi} \sin \phi \sin \psi d\phi d\psi d\eta, \quad (498)$$

then the joint probability density for (x, y, z) can be shown to be equal to

$$p(x, y, z|a, b, c) = \left(\frac{1}{4\pi} \sin \phi \sin \psi \right) (\det J(\phi, \psi, \eta))^{-1}, \quad (499)$$

$$= \frac{1}{4\pi} \frac{xyz}{a^2 b^2 c^2 \sin \eta \sin \phi \sin \psi}, \quad (500)$$

$$= \frac{xyz}{2\pi abc} \frac{1}{\sqrt{Q(a, b, c, x, y, z)}} \quad (501)$$

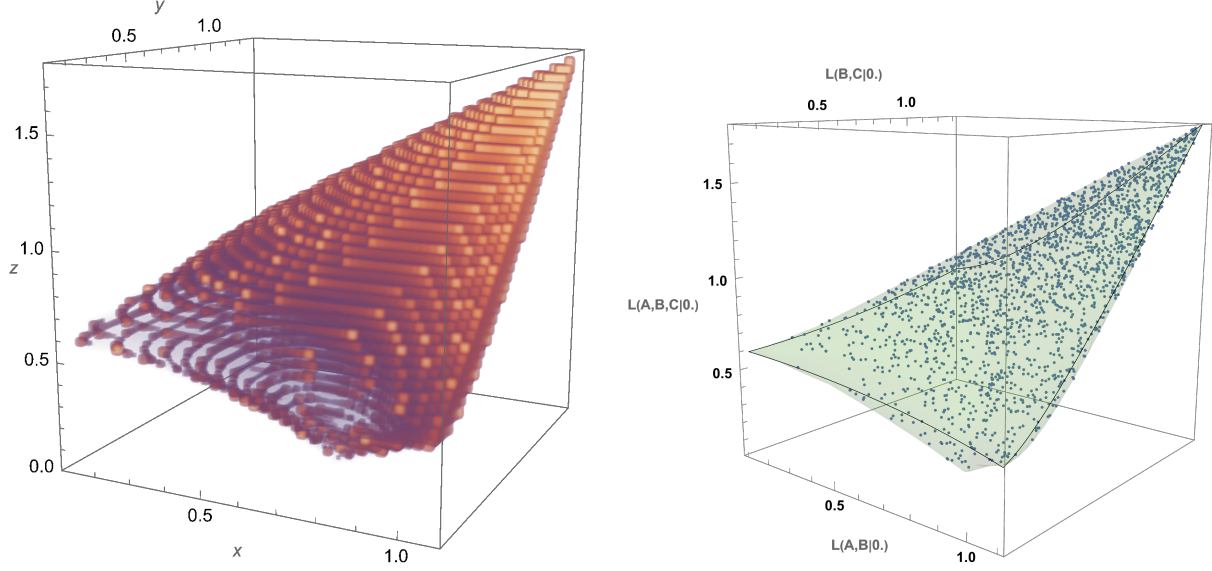


Figure 5: Admissible eigenvalues of partial sums ($A + B$, $B + C$ and $A + B + C$) of traceless 2×2 matrices (A, B, C) with spectra $(\alpha, \beta, \gamma) = (a, b, c) = (0.5, 0.6, 0.7)$. Left is a density plot of the analytic formula $p(x, y, z|a, b, c)$ restricted to the admissible region. Right is a numerical sample with admissible region boundary plotted in green.

where $Q(a, b, c, x, y, z)$ is the following homogeneous degree 6 polynomial:

$$Q(a, b, c, x, y, z) := -a^4c^2 + a^2b^2c^2 - a^2b^2x^2 + a^2b^2z^2 - a^2c^4 + a^2c^2x^2 \quad (502)$$

$$+ a^2c^2y^2 + a^2c^2z^2 + a^2x^2y^2 - a^2y^2z^2 - b^4z^2 - b^2c^2y^2 \quad (503)$$

$$+ b^2c^2z^2 + b^2x^2y^2 + b^2x^2z^2 + b^2y^2z^2 - b^2z^4 \quad (504)$$

$$+ c^2x^2y^2 - c^2x^2z^2 - x^4y^2 - x^2y^4 + x^2y^2z^2 \quad (505)$$

Further analysis [Ric02] (being careful to note the notational convention in [Ric02] is incompatible with the one used here $(a, b, c, x, y, z) \leftrightarrow (a, x, z, c, y, b)$) reveals that $Q(a, b, c, x, y, z)$ is equal to 114 times the volume squared of the tetrahedron with side lengths a, b, c, x, y, z , i.e.

$$Q(a, b, c, x, y, z) = 144 \text{Vol}(a, b, c, x, y, z)^2, \quad (506)$$

$$= \frac{1}{2} \det \begin{pmatrix} 2a^2 & a^2 + b^2 - x^2 & a^2 + y^2 - z^2 \\ a^2 + b^2 - x^2 & 2b^2 & b^2 - c^2 + y^2 \\ a^2 + y^2 - z^2 & b^2 - c^2 + y^2 & 2y^2 \end{pmatrix} \quad (507)$$

Therefore, the resulting probability density is simply

$$p(x, y, z|a, b, c) = \frac{xyz}{24\pi abc \text{Vol}(a, b, c, x, y, z)} \quad (508)$$

B.7 Challenges in generalizing the scaling approach

TC: The purpose of this section is to outline some of the challenges I faced when attempting to generalize the approach adopted by Roberts in proving the Ponzano-Regge asymptotic formula in [Rob99]. Roughly speaking, the stationary phase approach relies heavily on a rigorous understanding of the critical and their isolation from each other. In the case of $\text{SU}(2)$, this understanding is given essentially for free and without rigour – afterall, it is geometrically intuitive that given candidate edge lengths of a Euclidean tetrahedra,

there can only exist zero, one or two such tetrahedron corresponding to a tetrahedron and its mirror image.

TODO: Show how to generalize the construction due to Roberts [Rob99] by taking the Cartesian product of these manifolds over all cycles and then imposing additional group actions and moment-map conditions to stitch everything together.

Remark 94. Consider the following vector space which is twelve-fold tensor product of irreducible representations of $\text{GL}(n)$:

$$\mathcal{L} = \mathcal{L}_{\delta\epsilon\phi}^{\alpha\beta\gamma} := V_\alpha \otimes V_{\alpha^*} \otimes V_\beta \otimes V_{\beta^*} \otimes V_\gamma \otimes V_{\gamma^*} \otimes V_\delta \otimes V_{\delta^*} \otimes V_\epsilon \otimes V_{\epsilon^*} \otimes V_\phi \otimes V_{\phi^*}. \quad (509)$$

along with an associated symplectic manifold

$$\mathcal{M} = \mathcal{M}_{\delta\epsilon\phi}^{\alpha\beta\gamma} := \mathcal{O}_\alpha \times \mathcal{O}_{\alpha^*} \times \mathcal{O}_\beta \times \mathcal{O}_{\beta^*} \times \mathcal{O}_\gamma \times \mathcal{O}_{\gamma^*} \times \mathcal{O}_\delta \times \mathcal{O}_{\delta^*} \times \mathcal{O}_\epsilon \times \mathcal{O}_{\epsilon^*} \times \mathcal{O}_\phi \times \mathcal{O}_{\phi^*}. \quad (510)$$

While the vector space $\mathcal{L}$ naturally supports a representation of $\text{GL}(n)^{\times 12}$, for the purposes of the tetrahedral Horn problem, there are three additional representations acting on $\mathcal{L}$ that will be of interest; $\sigma_6 : \text{GL}(n)^{\times 6} \rightarrow \text{GL}(\mathcal{L})$, $\sigma_4 : \text{GL}(n)^{\times 4} \rightarrow \text{GL}(\mathcal{L})$ and $\sigma_1 : \text{GL}(n) \rightarrow \text{GL}(\mathcal{L})$:

$$\begin{aligned} \sigma_6(g_1, g_2, g_3, g_4, g_5, g_6) &= \pi_\alpha(g_1) \otimes \pi_{\alpha^*}(g_1) \otimes \pi_\beta(g_2) \otimes \pi_{\beta^*}(g_2) \otimes \pi_\gamma(g_3) \otimes \pi_{\gamma^*}(g_3) \otimes \pi_\delta(g_4) \otimes \pi_{\delta^*}(g_4) \otimes \pi_\epsilon(g_5) \otimes \pi_{\epsilon^*}(g_5) \otimes \pi_\phi(g_6) \otimes \pi_{\phi^*}(g_6) \\ &\quad \cdots \otimes \pi_\delta(g_4) \otimes \pi_{\delta^*}(g_4) \otimes \pi_\epsilon(g_5) \otimes \pi_{\epsilon^*}(g_5) \otimes \pi_\phi(g_6) \otimes \pi_{\phi^*}(g_6) \end{aligned} \quad (511)$$

$$\begin{aligned} \sigma_4(g_1, g_2, g_3, g_4) &= \pi_\alpha(g_1) \otimes \pi_{\alpha^*}(g_4) \otimes \pi_\beta(g_1) \otimes \pi_{\beta^*}(g_2) \otimes \pi_\gamma(g_3) \otimes \pi_{\gamma^*}(g_1) \otimes \pi_\delta(g_3) \otimes \pi_{\delta^*}(g_2) \otimes \pi_\epsilon(g_4) \otimes \pi_{\epsilon^*}(g_3) \otimes \pi_\phi(g_2) \otimes \pi_{\phi^*}(g_4) \\ &\quad \cdots \otimes \pi_\delta(g_3) \otimes \pi_{\delta^*}(g_2) \otimes \pi_\epsilon(g_4) \otimes \pi_{\epsilon^*}(g_3) \otimes \pi_\phi(g_2) \otimes \pi_{\phi^*}(g_4) \end{aligned} \quad (512)$$

$$\begin{aligned} \sigma_1(g) &= \pi_\alpha(g) \otimes \pi_{\alpha^*}(g) \otimes \pi_\beta(g) \otimes \pi_{\beta^*}(g) \otimes \pi_\gamma(g) \otimes \pi_{\gamma^*}(g) \otimes \pi_\delta(g) \otimes \pi_{\delta^*}(g) \otimes \pi_\epsilon(g) \otimes \pi_{\epsilon^*}(g) \otimes \pi_\phi(g) \otimes \pi_{\phi^*}(g) \\ &\quad \cdots \otimes \pi_\delta(g) \otimes \pi_{\delta^*}(g) \otimes \pi_\epsilon(g) \otimes \pi_{\epsilon^*}(g) \otimes \pi_\phi(g) \otimes \pi_{\phi^*}(g) \end{aligned} \quad (513)$$

At this stage, it is worth noticing that σ_1 is the representation of $\text{GL}(n)$ which is obtained by restricting σ_4 (respectively σ_6) to the diagonal subgroup $\text{GL}(n) \subseteq \text{GL}(n)^{\times 4}$ (respectively $\text{GL}(n) \subseteq \text{GL}(n)^{\times 6}$). Now consider the subspace of $\mathcal{L}$ consisting of vectors invariant under the action of $\text{GL}(n)^{\times 4}$ (via σ_4), i.e.,

$$\mathcal{V}_4 := (\mathcal{L})^{\text{GL}(n)^{\times 4}} = \{w \in \mathcal{L} \mid \forall g_1, g_2, g_3, g_4 \in \text{GL}(n) : \sigma_4(g_1, g_2, g_3, g_4)w = w\}. \quad (514)$$

Whenever it happens to be that $\mathcal{V}_4 \neq 0$, then one may conclude

$$\begin{aligned} (V_\alpha \otimes V_{\gamma^*} \otimes V_\beta)^{\text{GL}(n)} &\neq 0, \\ (V_{\beta^*} \otimes V_\phi \otimes V_{\delta^*})^{\text{GL}(n)} &\neq 0, \\ (V_\delta \otimes V_\gamma \otimes V_{\epsilon^*})^{\text{GL}(n)} &\neq 0, \\ (V_\epsilon \otimes V_{\phi^*} \otimes V_\alpha)^{\text{GL}(n)} &\neq 0, \end{aligned} \quad (515)$$

which, by Theorem 41, implies the existence of Hermitian operators in $\mathcal{M}$

$$A, A' \in \mathcal{O}_\alpha, \quad B, B' \in \mathcal{O}_\beta, \quad C, C' \in \mathcal{O}_\gamma, \quad D, D' \in \mathcal{O}_\delta, \quad E, E' \in \mathcal{O}_\epsilon, \quad F, F' \in \mathcal{O}_\phi, \quad (516)$$

satisfying the following four conditions

$$A - C' + B = 0, \quad -B' + F - D' = 0, \quad D + C - E' = 0, \quad E - F' - A' = 0. \quad (517)$$

From this perspective, it becomes clear that the existence of non-zero $\text{GL}(n)^{\times 4}$ -invariant vectors in $\mathcal{L}$ is a necessary condition for $\alpha, \beta, \gamma, \delta, \epsilon$, and ϕ to be tetrahedral spectra.

Remark 95. To address the insufficiency of Eq. (515), consider the subspace of $\mathcal{L}$ consisting of vectors invariant under the action of $\text{GL}(n)^{\times 6}$ (via σ_6), i.e.,

$$\mathcal{V}_6 := (\mathcal{L})^{\text{GL}(n)^{\times 6}} = \{w \in \mathcal{L} \mid \forall g_1, g_2, g_3, g_4, g_5, g_6 \in \text{GL}(n) : \sigma_6(g_1, g_2, g_3, g_4, g_5, g_6)w = w\}. \quad (518)$$

For all dominant weights λ , Schur's lemma implies

$$(V_\lambda \otimes V_\lambda^*)^{\text{GL}(n)} \simeq \text{Hom}_{\text{GL}(n)}(V_\lambda, V_\lambda) \simeq \mathbb{C} \text{id}_{V_\lambda}, \quad (519)$$

which in turn implies the subspace of $\text{GL}(n)^{\times 6}$ -invariant vectors in $\mathcal{L}$ is always one-dimensional, $\dim(\mathcal{V}_6) = 1$. In any case, if there exists a vector which is *both* $\text{GL}(n)^{\times 4}$ and $\text{GL}(n)^{\times 6}$, i.e., $w \in \mathcal{V}_6 \cap \mathcal{V}_4 \neq 0$, then it can be shown (by similar arguments to the proof of Theorem 41¹⁰) that there exists Hermitian operators

$$A \in \mathcal{O}_\alpha, \quad B \in \mathcal{O}_\beta, \quad C \in \mathcal{O}_\gamma, \quad D \in \mathcal{O}_\delta, \quad E \in \mathcal{O}_\epsilon, \quad F \in \mathcal{O}_\phi, \quad (520)$$

satisfying the following four conditions

$$A - C + B = 0, \quad -B + F - D = 0, \quad D + C - E = 0, \quad E - F - A = 0, \quad (521)$$

which are readily seen to be equivalent to tetrahedral constraints in Eq. (3). In other words, the existence of a $\text{GL}(n)^{\times 4}$ and $\text{GL}(n)^{\times 6}$ invariant vector in $\mathcal{L}$ is sufficient to conclude that the sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral spectra. However, using the results of [Rob99], it can be concluded for most tetrahedral $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ that no “doubly-invariant” vectors exist in $\mathcal{L}$, i.e. $\mathcal{V}_4 \cap \mathcal{V}_6 = 0$.

Remark 96. In [Rob99], Roberts originally used the stationary phase method to prove the Ponzano and Regge semiclassical formula [PR68] for the $6j$ coefficients (i.e., the $d = 2$ recoupling symbol). In order to apply the technique of stationary phases, it must be the case that exponent in the integrand has *isolated* critical points. For the case of the $6j$ symbol (with generic values for the six js), there were two isolated critical points corresponding to (i) the $\text{SO}(3)$ -orbit $[\tau]$ of a Euclidean tetrahedron τ with fixed side lengths and (ii) the $\text{SO}(3)$ -orbit $[\tau']$ of the Euclidean tetrahedron τ' which is the mirror-image of the tetrahedron τ with the same fixed side lengths.

TC: For the more general case of $n \geq 3$, it is unclear to me if the analogous critical points (whose union is identified with all Hermitian matrix-valued solutions to the tetrahedral equations) correspond to a finite number of disjoint coadjoint orbits under the subgroup $\text{GL}(n)$ (as in the $n = 2$ case). As a result, at the moment, it is unclear to me how to establish an asymptotic formula $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\}$ in the “scaling $\lambda \mapsto k\lambda$ ” regime. Instead, it seems the next best thing is to use the “completeness-type” relations between various recoupling symbols to prove the existence of a convergent sequence of weights such that the corresponding sequence of recoupling symbols admits of a power-law type lower bound.

Even if we were to stipulate/conjecture/assume that $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are such that $(\mu_E^{-1}(0) \cap \mu_V^{-1}(0)) / \text{GL}(n)$ consists of a finite number of isolated fixed points (as in the $n = 2$, non-degenerate case considered by Roberts, it would be a great deal of work to compute the Hessians and phases at each of the critical points in order to formally check that the phases are *NOT* “misaligned” in the sense that the complex contributions destructively interfere leading to a trivial lower-bound on the norm of $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\}$ of zero.

Even worse, even in the $n = 2$ case, one generically expects non-isolated critical points even after integrating over a global $\text{GL}(n)$ because of the existence of non-rigid polyhedra with fixed edge lengths (the tetrahedron is locally rigid, while a cube is *not* locally rigid in this sense).

B.8 An alternative asymptotic characterization

There are a number of technical challenges, namely the difficulty in performing a stationary phase analysis, which prevent one from making concrete statements about the *scaling* $\lambda \mapsto k\lambda$ limit of the squared norm of the $\text{SU}(n)$ recoupling operator. However, there is another quantity, denoted by $Q_{\delta, \epsilon, \varphi}^{\alpha, \beta, \gamma}$ in Eq. (524) below, for which it is comparatively easier to prove asymptotic bounds. Moreover, while the “recoupling theory” approach concerned with $R_{\delta \epsilon \phi}^{\alpha \beta \gamma}$ is seemingly limited by the commutativity of the projection operators involved (the approach cannot handle more than two mutual non-commuting projection operators), the approach which produces $Q_{\delta, \epsilon, \varphi}^{\alpha, \beta, \gamma}$ does *not* share this apparent limitation.

¹⁰Specifically, one begins with a generic twelve-fold coherent product vector $v_0 \in \mathcal{L}$ and iteratively flowing along gradients to minimize the norm of $\sigma_6(g'_i)\sigma_4(g_j)v_{n-1} \in \mathcal{L}$ with respect to g'_i and g_j to obtain $v_n \in \mathcal{L}$. The resulting sequence of points exhibits a limit point $\tilde{v}$ which is (i) non-zero (because $\langle w, \tilde{v} \rangle = \langle w, v_n \rangle = \langle w, v_0 \rangle \neq 0$), and (ii) minimal in norm with respect to both orbits, i.e. both moment maps vanish: $\mu_4([\tilde{v}]) = 0$ and $\mu_6([\tilde{v}]) = 0$. Thanks to Harold for sketching out this argument for me.

The quantity is built from reconsidering the $\text{GL}(n)^{\times 4}$ and $\text{GL}(n)^{\times 6}$ fixed subspaces of $\mathcal{L} = \mathcal{L}_{\delta\epsilon\phi}^{\alpha\beta\gamma}$ defined above with projection operators Γ_4 and Γ_6 ; one can simply consider their tensor product and the associated $\text{GL}(n)^{\times (4+6)}$ fixed subspace of $\mathcal{L}^{\otimes 2}$ with projection operator $\Gamma_4 \otimes \Gamma_6$. With these subspaces in mind, the quantity $Q_{\delta,\epsilon,\phi}^{\alpha,\beta,\gamma}$ is defined as

$$Q_{\delta,\epsilon,\phi}^{\alpha,\beta,\gamma} := \text{Tr}_{\mathcal{L}^{\otimes 2}}[(\Gamma_4 \otimes \Gamma_6)\Pi_{\text{double}}] \quad (522)$$

where Π_{double} is the projection operator onto the “doubled” subspace inside $\mathcal{L}^{\otimes 2}$:

$$\mathcal{L}_{2\delta,2\epsilon,2\phi}^{2\alpha,2\beta,2\gamma} \subseteq \mathcal{L}_{\delta,\epsilon,\phi}^{\alpha,\beta,\gamma} \otimes \mathcal{L}_{\delta,\epsilon,\phi}^{\alpha,\beta,\gamma}. \quad (523)$$

Diagrammatically, the quantity $Q_{\delta,\epsilon,\phi}^{\alpha,\beta,\gamma}$ equals

{eq:workable-recoupling-like-quantity} (524)

Using the strong-duality theorem and central-limit theorem results [FW20, Theorem 1 & Proposition 3.1], one can prove that if $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral, then $Q_{k\delta,k\epsilon,k\phi}^{k\alpha,k\beta,k\gamma}$ exhibits a power-law in k lower bound, otherwise they are not tetrahedral and then $Q_{k\delta,k\epsilon,k\phi}^{k\alpha,k\beta,k\gamma}$ exhibits exponential decay in k .

The primary difference between $Q_{\delta,\epsilon,\phi}^{\alpha,\beta,\gamma}$ and $R_{\delta,\epsilon,\phi}^{\alpha,\beta,\gamma}$ is diagrammatically expressed in the following figure:

$Q_{\delta,\epsilon,\phi}^{\alpha,\beta,\gamma}$	$R_{\delta,\epsilon,\phi}^{\alpha,\beta,\gamma}$

TC: Developing a better understand of the relationship between these two operators would be beneficial for proving cleaner and tighter results about the recoupling operator. While the operator on the left is a quasiprojection operator onto the invariant subspace of $V_{\lambda}^* \otimes V_{\lambda}$, the operator on the right is a non-negative operator with support containing the invariant subspace in $V_{\lambda}^* \otimes V_{\lambda}$ (by virtue of the fact that it contains the invariant subspace in $V_{2\lambda}^* \otimes V_{2\lambda}$).

C [Research Notes]

TC: This section contains a number of notes I've collected about some of the paper's I've been researching over the course of this project. It will, of course, be removed for the final draft. TC: At some point I made an effort to start being consistent regarding various conventions. In particular, dimensions of complex vector spaces should always be n , many copy asymptotics should be indexed by k , and the edges of the tetrahedra are always labelled according to the main figure.

C.1 Horn's problem & generalizations

- Horn's original paper where they conjectured the form of a necessary and sufficient set of inequalities via an inductive scheme for increasing dimension [Hor62]
- An excellent expository overview of the big picture by Knutson [Knu00]
- An excellent introduction to these ideas is the recent masters thesis on various aspects of additive and multiplicative Horn problems in connection to Vinnikov curves and the Lax conjecture due to Hess and supervised by S. Kuhlmann and M. Infusino [Hes20]
- Quantum-info style proof of Littlewood-Richardson coefficient connection due to Christandl [Chr08]
- Coquereaux and Zuber explicitly outline the link between the orbital measures (i.e. the induced distributions in the sense of Klyachko's random walk approach) and the magnitude of the stretched Littlewood-Richardson coefficients [CZ18]. See also the papers [CMZ20; CMZ19; Zub18].
- A recent article by McSwiggen is an excellent introduction to the Harish-Chandra integral [McS21]. The proof of the Harish-Chandra formula provide by McSwiggen using the heat equation methods from Fegan's 1978 paper [Feg78] seems very promising for deriving a connection to the character formula for the norm of the recoupling operator!
- Speyer [Spe05]
- Generalizations of Horn's problem to the product of unitary matrices [AW97; Bel08; BK16]
- Bhatia [Bha01]
- History account offered by Daftuar and Hayden [DH05], and technical review article by Fulton [Ful00].
- Berline, Vergne, and Walter a pedagogical account of the triple role played by Horn's inequalities in characterizing the existence of invariants vectors in tensor products, the eigenvalues of sums of Hermitian matrices, and intersections of Schubert varieties [BVW18]
- Fulton provides a comprehensive treatment of the connection between representation theory of $GL(n)$ and the geometry and (co)homology of the complex Grassmannians [Ful97]
- Prakash Belkale has contributed greatly to the theory surrounding Horn-type problems: for example, in [Bel04], Belkale provides a *geometric* proof of the equivalence between (i) the structure coefficients of the representation theory of $GL(n)$ (i.e. the LR-coefficients $c_{\mu\nu}^\lambda$), and (ii) the structure coefficients of the cohomology ring of the Grassmannian $Gr(d, n)$ of n -dimensional subspaces in $\mathbb{C}^n$. This proof is now complementary to the previously known algebraic proofs which rely on knowledge of the algebra of symmetric functions. This geometric approach taken by Belkale, which emphasizes the tangent spaces of the involved Schubert varieties, is later used by Belkale to provide a geometric proof of the saturation conjecture [Bel02].

C.2 Thompson’s conjecture

- Original (indirect) proof through the lense of random walks Klyachko [Kly00]
- The proof of Thompson’s conjecture in the style of Alekseev, Meinrenken, and Woodward is the “linearization paper” due to [AMW01].
- See also the related article due to Lamb which summarizes the ideas of [AMW01] while also providing more technical background [Lam05]
- Lu and Evens extend the proof technique of [AMW01] to a larger class of Lie groups [LE03]
- Alekseev, Podkopaeva, and Szenes [APS17a]
- Alekseev, Podkopaeva, and Szenes [APS17b]
- Alekseev and Meinrenken [AM07]
- Deforming the symplectic volume by the s parameter Alekseev et al. [Ale+18]
- **TODO:** track down pdfs for the original triple of papers by Thompson and Therianos titled “On the singular values of matrix products” I, II, and III from 1973
- **TODO:** track down pdfs for the original triple of papers by Thompson and Freede titled “On the eigenvalues of sums of Hermitian matrices” III from 1973 (I and II have been found)
- **TODO:** track down a pdf for Thompson’s 1988 “Matrix Spectral Inequalities” referenced by Klyachko
- Thompson and Therianos [TT73] present a number of interesting counter-examples to the question of whether the inequalities pertaining to the spectra of general matrices can be deduced from looking at the spectra of diagonal matrices (both in the additive and multiplicative cases).

C.3 $6j$ symbols

- For a comprehensive reference textbook for the $6j$ coefficients, see the textbook by Varshalovich, Moskalev, and Khersonskii [VMK88], specifically Chapter 9.
- A great summary by Roberts [Rob02]; also the original result for the asymptotics of the $6j$ symbols due to Roberts [Rob99]
- The masters essay/thesis of Backens (**TODO: check if the pdf Matthias sent to me is the final version**) [Bac10]
- The spectral quantum marginal problem and the $6j$ symbol for the symmetric group Christandl, Şahinoğlu, and Walter [CSW18]
- Usage of Wigner’s $6j$ symbol to compute asymptotic error rates for the promise problem of distinguishing between two possible “change points” for a sequence of unknown pure states to transition from one unknown type to another [AH11]
- See also Coquereaux’s later paper which promotes the alternative, albeit equivalent, usage of the integer-valued gon and tet functions in place of the rational-valued theta (triangular) and irrational-valued $6j$ symbols [Coq23]. Most relevant for this project is [Coq23, Section IV D] which discusses the apparent lack of substantial generalization from $SU(2)$ to $SU(N)$ (“*apart from very few special cases (low dimension of irreps and no multiplicity) worked out by physicists at the time of the eightfold way, we are not aware of any published general result*”).
- The recent paper by Alekseev, Morozov, and Sleptsov (different Alekseev!) covers the case of some multiplicity free formulas for the quantum $6j$ symbols for $U_q(\mathfrak{sl}_N)$ [AMS20]

- Keppeler, Plätzer, and Sjodahl recently calculated $SU(n)$ $6j$ symbols with adjoint representations [KPS24] and a bit earlier [Alc+23] calculated $SU(n)$ $6j$ symbols with at least two fundamental representations [Alc+23].
- Representation theoretic proof of the Regge symmetries due to Boalch [Boa07]
- Asymptotics of quantum $6j$ symbols Taylor and Woodward [TW06]
- The integral formula for the square of the $SU(2)$ $6j$ symbol can be performed by an appropriate change of variables [FL03]. See also other papers by Laurent Freidel and collaborators that are relevant [FK00; FNR07; FKL10].
- Alcock-Zeilinger consider the case of $6j$ symbols for $SU(N)$ for wherein two opposing edges are labelled by fundamental irreps of $SU(N)$ so that all vertices are multiplicity one (or zero) [Alc18] **TC: They also freely use the birdtrack notation which I enjoy.**
- Only the tetrahedral symmetry of the $U_q(SL(N))$ $6j$ -symbol is known, the counterpart of the Regge symmetry is still unknown, although Morozov and Sleptsov conjectures the generalized Regge symmetry may be related to the eigenvalue conjecture, just as it is in the case when $N = 2$ [MS18].
- The graphical notion, specifically the use of an additional edge for multiplicity spaces can has been used for a long time, as early as [AB68; Mou84].
- The original Ponzano-Regge paper [PR68]
- Gurau derives the Ponzano-Regge asymptotic formula directly using Stirling’s approximation applied to Racah’s single-sum formula [Gur08]
- Bartlett and Nzaganya work with so-called *coherent loop states* or *Bohr-Sommerfeld states* to develop some semiclassical results concerning the inner products between such states (see also the talk on Youtube) [BN23]
- Aquilanti et al. provides an excellent introduction to subject of classical $6j$ symbols, their history, diagrammatics, presentations and asymptotics [Aqu+12]
- Marzuoli and Rasetti connects quantum computing models to the recoupling theory of $SU(2)$ angular momentum and also considers the semiclassical limit [MR05]
- Artamonov derives an explicit formula for the $n = 3$ case [Art23], and later to arbitrary n [Art24].
- There is a series of papers involving Alexey Sleptsov, starting with a computation of quantum $6j$ symbols for generic dimension n but with symmetric representations in terms of certain hypergeometric polynomials [MMS17], then with a discussion of the eigenvalue hypothesis and its potential connection to generalized Regge symmetries [MS18], then with a calculation of all multiplicity free quantum $6j$ symbols for arbitrary dimension [AMS20], and then finally with the discovery of novel “tug-the-hook” symmetries for quantum $6j$ symbols for all n [LS23].

C.4 Minkowski’s theorem and tetrahedron geometry

- Connecting group-valued moment maps to spherical, flat, and hyperbolic tetrahedron via Minkowski’s theorem from the reconstruction of flat tetrahedron Haggard, Han, and Riello. See also the comments in [HHR16, Section 11] on the utility of *quasi-poisson* (controlled failure of the Jacobi identity) manifolds and group-valued moment maps.
- Foundational results concerning the relationships between the edge lengths, face areas, dihedral and solid angles of a tetrahedron by Richardson [Ric02]

- Bianchi, Dona, and Speziale uses Minkowski’s theorem to parametrize convex polyhedra in $\mathbb{R}^3$ with specified face areas and the associated Kapovich-Millson phase space (i.e., the *space of shapes of polyhedra with fixed areas* and then considers its connection to loop quantum gravity via the usual Guillemin-Sternberg quantization commutes with reduction philosophy which associates invariants of the quantization of the unreduced phase space to the reduced phase space [BDS11] (notice the connection with [HHR16]).
- Regarding an algorithmic construction for the polyhedra appearing in Minkowski’s theorem and their complexity, see this post: [constructive-algorithm-for-minkowskis-theorem](#)
- Charles quantization of polygon spaces, tetrahedron and semiclassical limits [Cha10]

C.5 Quantization, Poisson geometry & quantum groups

- A popular textbook on quantum groups with an emphasis on the connection to Poisson manifolds, Lie bialgebras and their quantizations by Chari and Pressley [CP95]
- Course notes on Poisson geometry with an eye toward quantization by Ciccoli and Witkowski [CW06]
- The PhD thesis of Esposito (from the maths department here in Copenhagen) serves as a introduction to the world of Poisson momentum maps and the notion of a “quantum” momentum map as a deformation of the ordinary “classical” momentum map (where deformation is in the sense of deformation quantization) [Esp12].
- Alekseev [Ale97]
- Papers contained results pertaining to the generalization of Duistermaat-Heckman measures for Hamiltonian group actions with group-valued moment maps by Alekseev, Meinrenken, and Woodward [AMW99; AMW00]
- Kirillov and Reshetikhin [KR90]
- Octahedron Recurrence/Rule Robbins and Rumsey Jr [RR86] applied to the “Hive” ring Knutson, Tao, and Woodward [KTW03], and also Henriques and Kamnitzer [HK06]
- Henriques and Kamnitzer [HK05]
- A reference which contains the formula for the quantum dimension of a representation of $U_q(\mathfrak{sl}(n))$ labelled by λ , thanks Rinat! [Ais97]. See also [AM98] which clarifies the relation between various parameters. Apparently, the x, v, s parameters are apparently natural parameters to consider in the context of invariants of knots and links. The substitution mentioned in [AM98] is the following

$$s = e^{h/2}, \quad v = e^{-dh/2} = s^{-d}, \quad x = e^{-h/2d} = s^{-\frac{1}{d}}, \quad q = e^h = s^2.$$

This is consistent with [Res87] in the sense that

$$[k] := \frac{s^{+k} - s^{-k}}{s^{+1} - s^{-1}} = \frac{q^{+k/2} - q^{-k/2}}{q^{+1/2} - q^{-1/2}}. \quad (525)$$

- Reshetikhin develops the diagrammatics of the category of representations of $U_q(\mathfrak{g})$ motivated by connections to the developing theory of polynomial invariants of knots and links [Res87].
- Moak investigates a q -Stirling’s formula for the logarithm of the q -factorial [Moa84].

C.6 Eigenvalue inequalities

- The general subject of multivariate trace inequalities involving exponentials of sums of Hermitian and products of their exponentials is, of course, intimately related. See for instance the paper by Sutter, Berta, and Tomamichel [SBT17].
- Also the article [FT14] by Forrester and Thompson which describes the connection between Thompson’s inequality and hyperbolic geometry quite explicitly.
- Friedland and Gaubert considers submodular functions associated to operator monotones evaluated on different principal submatrices of a single Hermitian matrix [FG13]. While it is not clear to be yet whether or not one can relate partial sums of matrices to principal submatrices, [FG13] references a paper by Bouhtou, Gaubert, and Sagnol [BGS08] which proves the following family of inequalities for all positive semidefinite $A, B, C \geq 0$ (and $p \in [0, 1]$) [BGS08, Lemma 5.3]:

$$\sum_j \lambda_j(A+B)^p + \sum_j \lambda_j(B+C)^p \geq \sum_j \lambda_j(B)^p + \sum_j \lambda_j(A+B+C)^p. \quad (526)$$

- Audenaert, Hiai, and Petz consider a similar form of “strongly subadditive” functions [AHP10]
- For multivariate inequalities based on idempotents of the symmetric group algebra see Huber’s paper [Hub20].
- For spectral inequalities derived using semidefinite programming, see Huber and Wyderka’s paper [HW22].
- Paksoy, Turkmen, and Zhang produce a large family of polynomial inequalities for positive semidefinite $A, B, C \geq 0$ using the notion of a “generalized matrix function” [PTZ14] (see also [HM21; MM65])

C.7 Uncategorized

- Elementary introduction to toy models of three-dimensional quantum space [Bar03]
- Regge and Williams [RW00] a relatively recent review of discrete approaches to gravity. The essential idea, in contrast to the differential geometry of general relativity, is to model curvature discretely instead of continuously. In particular, one can think of a triangulation of the surface of a sphere; the curvature of the sphere is encoded in the triangulation when a configuration of three (or more) triangles meet at a common vertex with sum of interior angles deviating from π . It is in this sense that Regge and Williams refers to these types of models as “piecewise-flat”. Contained is an overview of various attempts to generalize discrete theories of three-dimensional gravity to four dimensions [RW00, Section III].

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